ROBOBALL VOLUME I: DESIGN, MODELING, AND NUTATIONAL INSTABILITIES OF SOFT PENDULUM DRIVEN SPHERICAL ROBOTS

A Dissertation

by

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Submitted to the Graduate and Professional School of
Texas A&M University
in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY

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December 2025

Major Subject: Mechanical Engineering

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ABSTRACT

RoboBall is a family of spherical, inflatable robots designed for mobility across diverse environments. While hard-shell spherical robots are well represented in the literature, few alternatives exist that use inflatable designs rugged enough for long-term deployment.

This dissertation presents the design and development of three major subsystems: an internal pendulum, an inflatable outer shell, and a pressure regulation system. The shell and pressure control architecture represents a state-of-the-art contribution to spherical robotic hardware, with prototypes that have endured more than three years of continuous testing. Their design, maintenance, and repair are described in detail, along with the packaging and modeling of the internal compressor–tank system that regulates shell pressure.

With the hardware matured to a stable operating state, the dissertation then examines performance and modeling. A set of baseline mobility experiments highlights limiting design factors and informs improvements for operation on slopes, in water, and during a wobble-induced instability. In particular, the wobble tests empirically reveal a nutation instability in spherical robots—a phenomenon present in existing prototypes but previously absent or overlooked in prior models.

Addressing this modeling gap, the final chapters focus on dynamics. A forward-dynamics model of RoboBall is implemented using Featherstone's Articulated Body Algorithm in the py-Drake framework and is shown to reproduce the nutational instability in the presence of relevant contact dynamics. Building on this result, a theoretical link is investigated between the spherical robot nutation instability and a special case of the Intermediate Axis Theorem, which states that an oblate body constrained to roll about its minimum inertia axis is inherently unstable. An analytical model for an oblate sphere rolling on a plane is derived and validated against both experimental results and the Drake simulation. Finally, the interaction between the pendulum mechanism and the observed nutation is analyzed in the context of mobility and dynamic performance.

DEDICATION

To my siblings, parents, and anyone who works with rolling things

ACKNOWLEDGMENTS

Robotics is a team sport. I have had the privilege of working alongside talented graduate students, research engineers, undergraduates, and faculty during my time in the RAD Lab. I have been with the RoboBall II from the first idea sketches to the most recent tests, and I can no longer distinguish the fingerprints on each part of the robot. To everyone who has been on this project for a summer or multiple years, thank you. It will be wild to see where the robot rolls next.

CONTRIBUTIONS AND FUNDING SOURCES

Contributions

This work was supported by a variety of graduate students, summer undergraduates, and research engineers.

Work presented in this thesis either led directly to, or was a significant part of, the following publications.

1. **Micah J. Oevermann**, Meghali Prashant Dravid, Garrett Jibrail, Jared Janak, Rishi Jangale, David McDougall, David Dugas and Robert O. Ambrose. "A Soft Spherical Robot for Lunar Crater Exploration," AIAA 2024-1961. AIAA SCITECH 2024 Forum. January 2024.

2. **M. Oevermann**, D. Pravecek, G. Jibrail, R. Jangale and R. O. Ambrose, "RoboBall: An All-Terrain Spherical Robot with a Pressurized Shell," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024.

3. M. P. Dravid, **M. Oevermann**, D. McDougall, D. Dugas and R. Ambrose, "Design of a Soft Shell for a Spherical Exploration Robot Traversing Varying Terrain," 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Abu Dhabi, United Arab Emirates, 2024.

4. D. J. Pravecek, **M. J. Oevermann**, G. C. Thomas and R. O. Ambrose, "Empirically Compensated Setpoint Tracking for Spherical Robots With Pressurized Soft-Shells," in IEEE Robotics and Automation Letters, vol. 10, no. 3, pp. 2136-2143, March 2025. MJO

5. R.V. Jangale, A. Villanueva, G. Jibrail, **M.J. Oevermann**, D.J. Pravecek, M.P. Dravid, and R.O. Ambrose, "Scaling of RoboBall: A Parametric Robot Family for Crater Exploration," 2025 IEEE Aerospace Conference, Big Sky, MT, USA, 2025.

6. **Micah J. Oevermann**, Meghali P. Dravid, Derek J. Pravecek, Will Olejnik, and Robert O. Ambrose, A Pressure Model and Control System for a Pressurized Pendulum Driven

Spherical Robot," 2025 22nd International Conference on Ubiquitous Robots (UR), College Station, TX, USA, 2025

7. **Micah J. Oevermann** and Robert O. Ambrose "Empirical Contact Models for Soft Spherical Robots in Drake", in IEEE Robotics and Automation Letters (*Accepted 2025*)

8. Material from Chapter 7 is under review for contribution to the 2026 IEEE International Conference on Robotics and Automation (ICRA) at the time of this document's preparation.

Funding Sources

Research supported by the Texas A&M Chancellor's Research Initiative and matching funds from the Texas Governor's University Research Initiative.

TABLE OF CONTENTS

Page

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ABSTRACT ii

DEDICATION iii

ACKNOWLEDGMENTS iv

CONTRIBUTIONS AND FUNDING SOURCES v

TABLE OF CONTENTS vii

LIST OF FIGURES x

LIST OF TABLES xiv

1. Introduction 1
   - 1.1 Background 2
   - 1.2 Organization of Dissertation 4

2. Internal Pendulum Design, Instrumentation, and Modeling 6
   - 2.1 Introduction 6
   - 2.2 Pendulum Instrumentation for State Measurements 8
     - 2.2.1 Vectornav IMUs 8
     - 2.2.2 Absolute Encoders 8
   - 2.3 Isolated Controls Testbeds 9
     - 2.3.1 Hanging Static Stand 9
     - 2.3.2 Gyroscopic Stand 10
     - 2.3.3 Steering Stand 10
     - 2.3.4 Drive Stand 11
   - 2.4 Decoupled Dynamic Model and Control Design 12
     - 2.4.1 Steering Dynamic Model 13
     - 2.4.2 Linear Quadratic Control of Steering 15

3. Inflatable Soft Shell Design 18
     - 3.0.1 Mounting Interface 18
     - 3.0.2 Exercise Ball Based Shell 19

3.0.3 Molded Shells ... 24
3.0.4 Shell Repair ... 27

3.1 Models of Soft Shell ... 29
3.1.1 Shape and Stability ... 29
3.2 Characterization of Shell Prototypes ... 32
3.2.1 Flat Model ... 33
3.2.2 Effects of Soft Shell on Control Design ... 36

4. Internal Pressure Control System ... 39

4.1 Background ... 39
4.2 Introduction ... 41
4.2.1 Pressure Control Hardware ... 41
4.2.1.1 Pressure Control Range ... 42
4.3 Dynamic Pressure Model and Control ... 43
4.3.1 Measurement of Constants ... 45
4.3.2 Bang Bang Control of Pressure System ... 46
4.3.3 Empirical Response of Pressure Control Scheme and Dynamic Model ... 46
4.4 Design and Maintenance of Ring and Hubcap ... 48
4.4.1 Hubcap Counterbores ... 48
4.4.2 Ring Threads ... 49

5. RoboBall II Mobility Baselines ... 51

5.1 Water Mobility ... 51
5.1.1 Methods ... 51
5.1.2 Water Mobility Results ... 53
5.2 Slope Climbing Ability ... 55
5.2.1 Description of Parametric Models ... 55
5.2.2 Slope Climbing Experiments ... 56
5.2.3 Comparing Slope Climbing Models ... 57
5.2.4 Slope Climbing Results ... 58
5.3 Rolling with a Locked Pendulum ... 60
5.3.1 Locked Steering Tests ... 61
5.3.2 Locked Steering Results ... 64
5.4 Chapter Conclusions ... 64

6. Numerical Modeling of the Spherical Robot in Drake ... 65

6.1 Introduction ... 65
6.2 Modeling Approach ... 67
6.2.1 URDF Model ... 67
6.2.2 Joint Friction Models ... 69
6.3 Outer Shell Contact Models ... 72
6.3.1 Experimental Shell Dynamics in Steering Direction ... 73

6.3.2 Injected Contact Dynamics ... 75
6.3.3 Hydroelastic Contact ... 76
6.3.4 Hydroelastic Parameter Sensitivity ... 77
6.4 Modeling the Full Dynamics Through Uncontrolled Descent ... 78
6.5 Discussion and Model Limitations ... 80
6.6 Application to Wobble Experiments ... 81
6.7 Chapter Conclusions ... 83

7. Nutation Instabilities in Rolling Objects ... 84

7.1 Chapter Introduction ... 84
7.1.1 Wobbling Bicycles ... 84
7.1.2 Relaxation of Rotating Asteroids and Satellites ... 86
7.1.3 Inertial Profiles of Robots in Literature ... 86
7.2 Dynamic Analysis of Rotating Bodies Supplemented with Relaxation Dynamics ... 88
7.2.1 Euler Dynamics of a Rotating Body ... 89
7.2.2 Relaxation Dynamics of a Damped Oblate Body ... 92
7.2.3 Comparing Responses of Rotation and Relaxation Dynamics ... 94
7.3 Relaxation Dynamics for Constrained Bodies ... 96
7.4 Shell Rolling Experiments ... 101
7.5 Simulation Studies ... 104
7.6 Rolling Dynamics with a Pendulum ... 107
7.6.1 Revisiting the Mobility Studies ... 108
7.7 Comments On The Relaxation Angle as a Feedback Parameter ... 110
7.8 Chapter Conclusions ... 110

REFERENCES ... 112

LIST OF FIGURES

FIGURE Page

1.1 Assembled RoboBall I and its pendulum 2

1.2 RoboBall II and III pendulums and shells unassembled 3

2.1 Depiction of Pendulum Construction and Mounting Interface of Soft Shell 6

2.2 Kinematic Model of Drive and Steering Axes 7

2.3 Photos of Pendulum rotating about its main axes 7

2.4 Pendulum in hanging static stand. The hex clamps could be swapped for rotating bearings if desired. 9

2.5 Pendulum in 3-DOF Gyro Stand 10

2.6 Pendulum in steering stand 11

2.7 Pendulum in drivestand on a slope descent test 12

2.8 Schematic of Robot States 13

2.9 Simulated response of optimal controller on ideal nonlinear system 17

3.1 Cross-sectional view of the shell mounting interface 19

3.2 The bare yoga ball prototype with excessive expansion 20

3.3 Nylon shell prototype resting and during a test 20

3.4 Assembled bedliner constrained prototype 21

3.5 Illustration of RoboBall Bedliner Manufacturing Process 22

3.6 Examples of mis-formed bedliner-coated shells. One undersized (left) and one oversized (right) 23

3.7 A sheared interface of the ball-clamp interface. The black material is Flex Seal, applied in a vain attempt to fortify the joint. 23

3.8 A summer's term worth of testing 24

3.9 Isometric sketch of the core mold assembly and disassembly process (steps 1 and 2) 25
3.10 Vertical rotary mold stand 26
3.11 Horizontal Spray Stand and Mold 27
3.12 Illustration of Molded Shell Manufacturing Process 27
3.13 Attempt to repair thin walls at edges 28
3.14 Taping method to repair small holes in the shell 29
3.15 2D Schematic of a Soft Spherical Shell Deforming with Pressure 30
3.16 Potential Energy of the System with Changing Scale Parameters 32
3.17 Measured Shape Parameters for Shell Prototypes 33
3.18 Schematic Relating Geometry of a Flat of Pressurized System (rolling out of page).. 34
3.19 Measurement of the Flat from below 35
3.20 Comparison of Flat Model with Experimental Data 36
3.21 LQR Stabilization of the System at Different Pressures 37
3.22 Steady State Steering Error Plotted against Flat Diameter Model 37
4.1 Packaging of Pneumatic System on the Robotic Pendulum 42
4.2 Pressure Control System Diagram 45
4.3 Pressure Model and Robot Following the Same Setpoint 47
4.4 Modeled Tank Pressure with Measured Tank Pressure 47
4.5 Picture of the worn-down counterbore, note the lack of gold anodization 49
4.6 Split Ring Design 50
5.1 Overview of IPG fording feature 52
5.2 Schematic of Testing Setup for Aqueous Mobility 53
5.3 Relationship between the rotational and forward velocities of the robot in water with a stock shell 54
5.4 Rooster tails produced during tests against the wind. 54
5.5 Diagram Relating the Physical Parameters for RoboBall's slope climbing ability 56

5.6 Ramp Testing Experimental Setup 56

5.7 RoboBall with the copper ballast installed 57

5.8 Model comparisons between experimental data 58

5.9 Testing Results with model using both aluminum and copper ballasts 59

5.10 Image of a pendulum in a shell with a clamp applied to the steering axis 61

5.11 Single test with locked steering at a target speed of 3.11rad/s 62

5.12 Average Wobbling Amplitude and Frequency of multiple locked steering tests at 2.5psi 63

5.13 Robot experiencing a nutation instability at a target speed of 6.22rad/s 63

6.1 A soft spherical robot and respective experiment using Drake's contact model 66

6.2 Flowchart of the modular Drake modeling framework, showing integration of rigid-body dynamics with friction losses and soft-shell contact models. 68

6.3 URDF kinematic structure of RoboBall, showing pendulum, pitch center, and shell components. 69

6.4 Robot State Diagram 70

6.5 Comparison of modeled torque responses from viscous, Coulomb+viscous, and Majd friction models. 71

6.6 Experimental setup isolating the steer and drive joints, alongside its digital twin in MeshCat. 71

6.7 Experimental data of the robot's actuation axes overlaid with model output with different friction models 72

6.8 Reinforced 3D printed clamp to lock the steering. 73

6.9 Comparison of measured damping ratio and frequency with point contact, injected dynamics, and hydroelastic models. 77

6.10 Effect of $\pm 50\%$variations in hydroelastic modulus and dissipation on damping ratio predictions. 78

6.11 Experimental ramp descent (left) and digital twin simulation in MeshCat (right). 79

6.12 Drive speed errors at each pressure 80

6.13 Simulated Wobble tests under different contact models 82

7.1 Side by side of RoboBall and Bicycle Steering Axis Trail 85

7.2 Diagram of the shell's body frame with angular velocity vector, applied moment components and relaxation angle between the velocity and minimum inertia axis $\hat{x}_2$. 89

7.3 RoboBall shell in two edge case relaxation states 92

7.4 Free rotations of an oblate body about $\hat{x}_2$ 95

7.5 Damped rotations of the shell about the drive axis 95

7.6 The robot shell located in the world frame.. 96

7.7 World frame aligned with object rolling down slope 100

7.8 Instrumentation of Robot Shell 102

7.9 Screenshots of the shell experiencing a nutation instability............................. 102

7.10 Data from a single run of the shell rolling down a slope 103

7.11 Relaxation Rates with R2 Shell at different Pressures 104

7.12 Simulated Response and Relaxation Angle of a Robot Shell Rolling down a slope with perfect initial conditions... 105

7.13 Simulated Response and Relaxation angle of Robot Shell Rolling down a Slope Modeled with a Point contact model 106

7.14 Simulated Response and Relaxation Angle of a Robot Shell Rolling down a slope modeled with a hydroelastic contact model.. 106

7.15 Simulated Response and Relaxation Angle of a Robot Shell Rolling down a slope modeled with a flipped inertial profile 107

7.16 Locked Steering Tests Revisited with Relaxation Angles............................... 108

7.17 Relaxation Parameters in Water Mobility Tests 109

LIST OF TABLES

TABLE Page

2.1 Parameter Values for FIRST Pendulum in 2023 15

4.1 System Volume Parameters 43

4.2 Measured Pressure Leak Rates 46

5.1 Slope Climbing Experiment Parameters 58

6.1 Parameters for Joint Friction Models (steer and drive axes). 72

6.2 Physical parameters measured for URDF model and used in dynamic simulations. .. 81

7.1 Literature Model Comparison and Inertial Profiles of Outer Shells 88

1. Introduction

NASA launched Artemis I in late 2022. The capsule and rocket launched as the start of a massive United States effort to return humans to the moon to explore the lunar south pole. One of the potential landing sites is on the lip of Shackleton crater, whose walls permanently shield its depths from sunlight. Deep and dark, these shadowed regions at the South Pole harbor frozen water deposited over eons. Exploration and utilization of Shackleton's depths will be paramount to the success of Artemis and future programs. It is this theater that the use case for the RoboBall family of robots was initially inspired. Current extraterrestrial rover design paradigms follow a well-known and predictable pattern of a series of wheels attached to a central chassis. Due to their over-actuated nature, these rovers move slowly, following carefully planned routes through highly studied terrain, guided by onboard sensors or orbital cameras.

RoboBall is an alternative approach to traditional wheeled rovers. It consists of an internal pendulum surrounded by a soft spherical outer shell. State-of-the-art rovers, such as VIPER and Chariot, employ a slow, methodical, and controlled approach to descend slopes and traverse terrain. A sphere, however, can utilize its natural locomotion mode of rolling and bouncing to descend a crater's internal walls in a fast, semi-chaotic, yet semi-controlled motion. Its symmetric geometry ensures that there is no orientation in which it could be trapped on its back, for it has no back. For this reason, it is ideal for rolling into inhospitable craters.

The benefits of this type of system are only just becoming viable. The Robotics and Automation Design Lab has made strides in producing multiple generations of these types of robots. It has been a privilege to be part of a team that designs, builds, controls, and studies state-of-the-art prototypes of these systems. Robotics is a team sport; the robots discussed in this dissertation would not be possible without the team as a whole. Yet dissertations are often seen as a record of individual contributions to the field. As a result, this document is the oxymoronic task of writing a robotics dissertation. I will present my contributions to the project, untangling them while acknowledging their place within the work being carried out by the team as a whole.

1.1 Background

In this thesis, the term "RoboBall" refers explicitly to the iterations of soft pendulum-driven robots developed by the RAD lab. At the time of this dissertation, three different iterations of these robots have been published in the literature. To differentiate between the prototypes, they have been dubbed "RoboBall I", "RoboBall II", and "RoboBall III" or in shorthand R1, R2, and R3.

RoboBall I was initially built as a senior project over a semester program at the Johnson Space Center in Houston. It consisted of a green inflated exercise ball shell with an internal pendulum. This was a rough proof of concept by L. Bridgwater and my chair, Dr. Robert Ambrose, in 2003 [1]. The prototype has since been excessed, but it was challenging to assemble, uninstrumented, and its outer shell was prone to tearing.

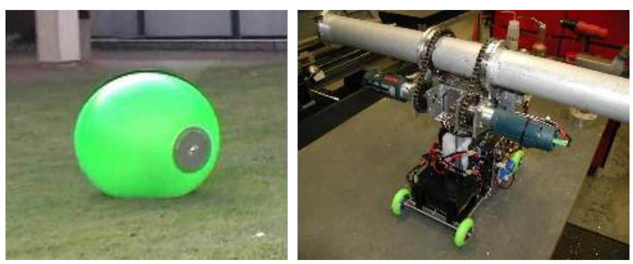


Figure 1.1: Assembled RoboBall I and its pendulum

RoboBall II is a revisit to the R1 concept with modern sensors and hardware. The dramatic decrease in the cost of Inertial Measurement Units, combined with a serendipitous connection to the materials industry, allowed us to instrument R2 and achieve a robust inflatable shell. This dissertation will primarily focus on the RoboBall II unit. References to an unnumbered "RoboBall" in the following chapters can be assumed to be a reference to R2.

RoboBall III is the newest member of the RoboBall lineup. It is a larger, more powerful version of R2. I did not study it directly and will not discuss it in depth in this dissertation, but it should not be left out in the discussion of the current RoboBall lineage.

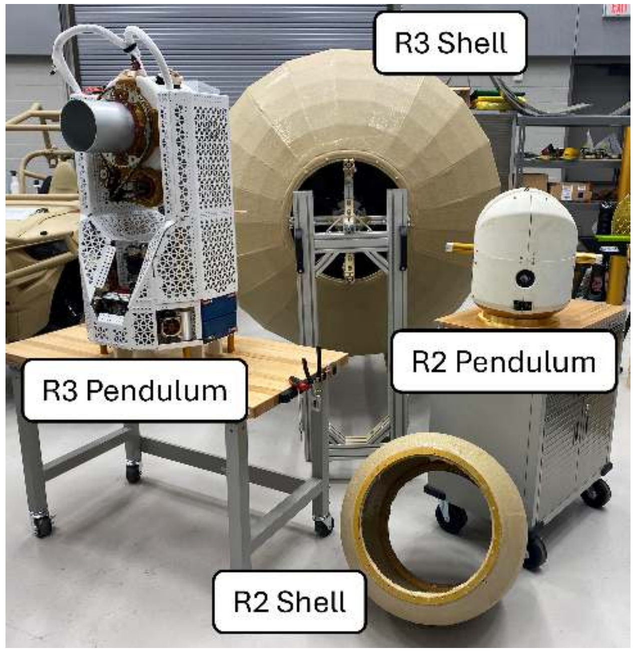


Figure 1.2: RoboBall II and III pendulums and shells unassembled

1.2 Organization of Dissertation

A high-level overview of this dissertation's chapters is given below:

Chapter 2: Internal Pendulum Design and Instrumentation. This chapter outlines the first significant component of RoboBall: its pendulum. My work on instrumenting, dynamic and performance study, and initial control work will be presented. Terminology such as the pendulum, pitch center, pipe assembly, hubcaps, and shell will be defined for use in the rest of this report. The choice of IMU and absolute encoder components, along with their performance, is presented. Along with initial control and dynamics studies through four testing stands and a simplified drive-steer model. Chapters 2 and 3 are heavily adapted from references [2, 3].

Chapter 3: Inflatable Soft Shell Design. This chapter presents my continuation of the work initiated by Dravid in [4], with a primary focus on Dravid's iterations of the soft shell design and my enhancements to them. I will describe previous shell prototypes and my observations on their effects on the early control scheme. Based on these observations, I will present shell design metrics, such as sphericality and flatness models, that describe desirable properties for shell production. I will then refine the previous prototypes by introducing a new molding method and incorporating the design iterations to meet the metrics. The chapter will conclude with a discussion on ring longevity and the repair methods employed over the three-year testing period with the prototype.

Chapter 4: Pressure Control System. This chapter presents the pressure subsystem packaged within the internal pendulum. The system is designed for the internal air management of the robot's pressure. A novel dynamic model based on exponential change laws will be introduced to continuously track the change in pressure over time resulting from leakage. The model will be compared against empirical data from the robot's leaks. Due to the slow dynamic nature, a bang-bang controller will be described and implemented.

Chapter 5: RoboBall Mobility Baselines This chapter will address the next step in RoboBall's development after a stable hardware state was reached. Baseline studies of the robot's motion in water, on slopes, and rolling with simplified dynamics will be studied. Each scenario will show different factors that affect the robot's performance. These factors range from environmental impacts, design parameters, and largely ignored factors in the robot's dynamics if modeled according to the decoupled approach presented in Chapter 2. These ignored dynamics manifest themselves as *nutation instabilities* where oscillations in the steering axis will grow with increasing drive speed. This effect is missed in some models, and its causing mechanisms are not fully understood.

Chapter 6: Empirical Contact Models for Soft Spherical Robots in Drake. This chapter presented tools developed to study the robot numerically. The models rely on the Articulated-Body Algorithm from Featherstone [5] implemented in the Drake modeling library [6]. The library will parse a Unified Robot Description Format (URDF) file to yield a 3D numerical model of the system. This model alone does not accurately account for the contact interface of the outer shell. Drake's hydroelastic model will be tuned with an injected linear dynamic contact model for better model performance. These models were tuned using experimental data and validated against a dynamic test of the robot rolling down a ramp. This chapter will close out by recreating the nutation tests from Chapter 5 to highlight the effect that the chosen contact model has on the nutation frequency.

Chapter 7: Nutation Instabilities in Spherical Rolling Robots. This chapter describes another potential cause of the nutation instability. The chapter will draw parallels between Euler's equations for rigid body rotation and the wobbling motion of rolling bodies. The theory will then be extended by applying rolling constraints to the free-floating system, incorporating a new translational energy term into the relaxation dynamics. This new term is the cause of relaxation in rolling bodies. Experiments with the robot's shell will support this, and the pendulum will be added and analyzed experimentally alongside the tools developed in Chapter 6, within the experimental context established in Chapter 5.

2. Internal Pendulum Design, Instrumentation, and Modeling

2.1 Introduction

At a high level, the pendulum itself contains three main sub-assemblies: a driveshaft or "pipe", a pendulum, and a pitch module linking them. These three sections are constructed in series to form a 2-DOF pendulum with intersecting axes of rotation. The robot can move by shifting the center of mass with the pendulum and pitch center to roll and pitch.

The first-generation electro-mechanical components were primarily sourced from a FIRST Robotics kit, including four Rev NEO BLDC motors (two on each axis) featuring built-in encoders, a RoboRio computer, and a Wi-Fi router for wireless communication with the host computer. Software and control algorithms were implemented in the WPILib software library for compatibility and data logging. Each axis's input is commanded from a PlayStation joystick connected to the host computer. A single VN-100 IMU sensor is rigidly attached to the pitch module to measure system angles and shell pressure. An overview of the assembled robot pendulum and components is shown in Figure 2.1.

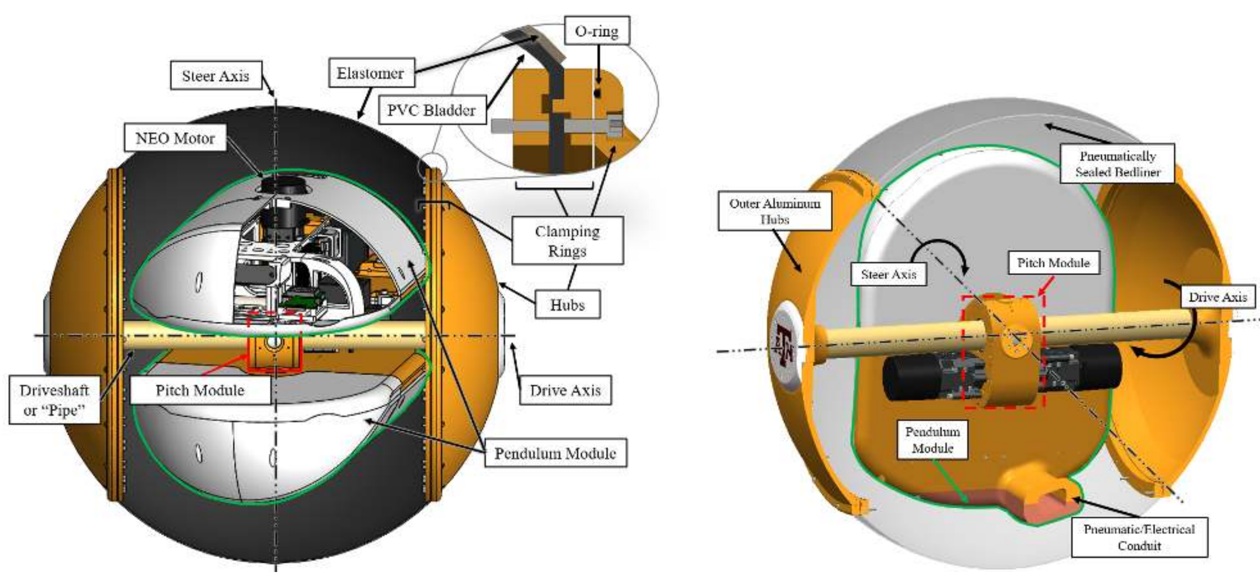


Figure 2.1: Depiction of Pendulum Construction and Mounting Interface of Soft Shell

The axes of the 2-degree-of-freedom pendulum are kinematically shown in Figure 2.2. The two axes of the pendulum intersect at the center of the robot. Applying a torque via a motor will lift the pendulum, changing the system's center of mass and initiating rolling motion. The drive axis of the robot is unconstrained for continuous motion, while the steering axis can tilt the robot from side to side and is limited by the drive axle. Tilting while driving causes the robot to steer in that direction. Video stills of the pendulum executing basic rotations about each axis in a testing stand are shown in Figure 2.3. Here, from left to right, the pendulum will demonstrate basic steering and driving motion.

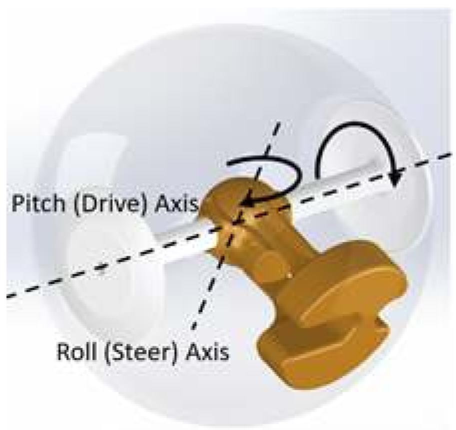


Figure 2.2: Kinematic Model of Drive and Steering Axes

Figure 2.3: Photos of Pendulum rotating about its main axes

2.2 Pendulum Instrumentation for State Measurements

This section will discuss the iterations of sensors used to measure important states of the robot.

2.2.1 Vectornav IMUs

Knowledge of the attitude of the pitch and pendulum links in space is critical for dynamic insight. IMUs have a long history in aircraft and satellite sensor arrays; however, the dramatic drop in their cost and size over the last decade has also enabled their deployment on small, mobile robots.

The VN-100 IMU fuses nine sensors (equal accelerometers, gyroscopes, and magnetometers about each axis) with its proprietary onboard sensor fusion and filtering algorithms to provide an extremely reliable attitude vector [7]. Its onboard barometric pressure sensor, commonly used for sensing altitude, is perfect for sensing the internal pressure of the ball. The raw acceleration and gyroscopic data can be read directly for different analyses.

Initial sensor configurations placed an IMU on the pendulum's steer axis just above the steer axis motors. Since there is extra magnetic interference from the motors, it is recommended to mount them as far away from the active motor as possible.

2.2.2 Absolute Encoders

Initially, an FRC Lamprey large-bore absolute encoder was mounted directly between the pendulum and pitch center links. However, the mounting bolt pattern also doubled as the structural coupling which sees some minor flexing between the links. So the added forces would crack the magnet ring. After a few runs with this setup, we eventually switched to the native encoders built into the Neo brushless motors. Ideally, a better encoder would be used; however, integrating non-FRC standard components with the RoboRio computer is challenging if an existing library is not available.

2.3 Isolated Controls Testbeds

The 2-DOF pendulum finished construction relatively quickly, within 9 months from its first design review. In contrast, RoboBall's novel soft outer shell development underwent many iterations before we settled on the initial design presented in Chapter 3. To make headway on controls and software components before the soft shell was completed, a series of dynamic testbeds were built to mimic the actual rolling dynamics of the robot. The following sections will discuss each iteration and the merits and drawbacks of each.

2.3.1 Hanging Static Stand

This stand hung the pendulum between two static beams, shown in Figure 2.4. The driveshaft hex could be rigidly clamped to disable relative motion between the pendulum and the ground. These clamps could be swapped for bearings, but the lack of inertia on the output meant that the pendulum would not pitch upward at normal speeds. This stand was perfect for the initial operations tests and assembly, but its lack of reflection of a sphere's natural dynamics did not accurately model the necessary rolling dynamics.

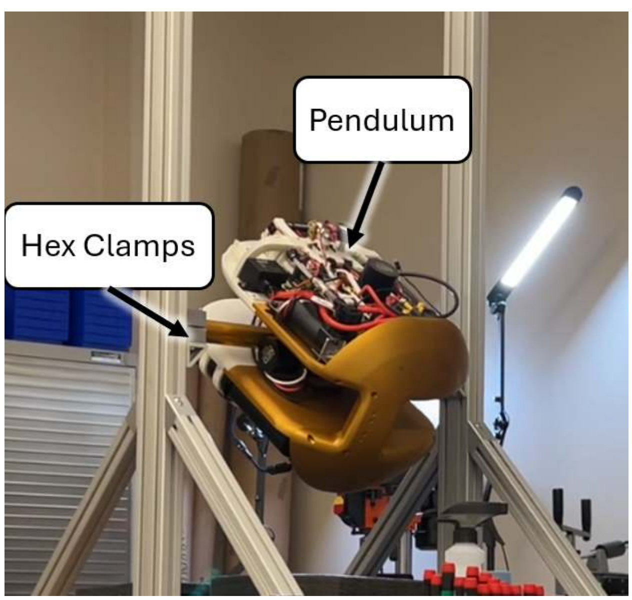


Figure 2.4: Pendulum in hanging static stand. The hex clamps could be swapped for rotating bearings if desired.

2.3.2 Gyroscopic Stand

The next generation of stands implemented a set of flywheels to compensate for the lack of inertia with the pure bearings of the hanging stand. These flywheels would mimic the shell's reflected inertia to verify that the pendulum pitches upward with forward velocity, Figure 2.5. However, the belts would slip after achieving final velocity and lose active flywheel braking. This issue, in tandem with the safety risks of unsheathed flywheels, led us to scrap the stand after its functional test.

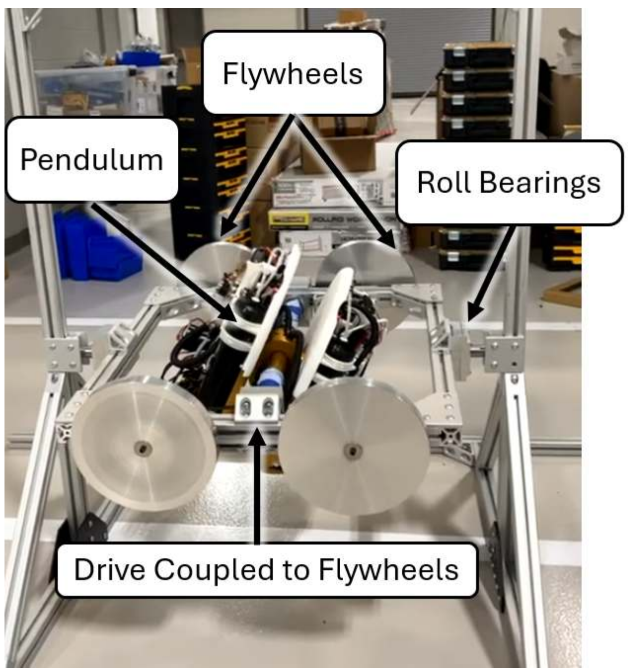


Figure 2.5: Pendulum in 3-DOF Gyro Stand

2.3.3 Steering Stand

Hanging concepts of testing stands were abandoned after the gyro stand, since they did not capture the linear coupling between the rolling and translational velocities. A stand that mimicked the outer roll with the steering joint was constructed, Figure 2.6. This steering stand isolated the pendulum's continuous drive axis from the limited steering axis. Initial balancing tests with an

LQR controller were promising. Still, the difference in inertias and contact dynamics between this stand and the shell prototypes made transitioning gains from the testbed to the robot unrealistic.

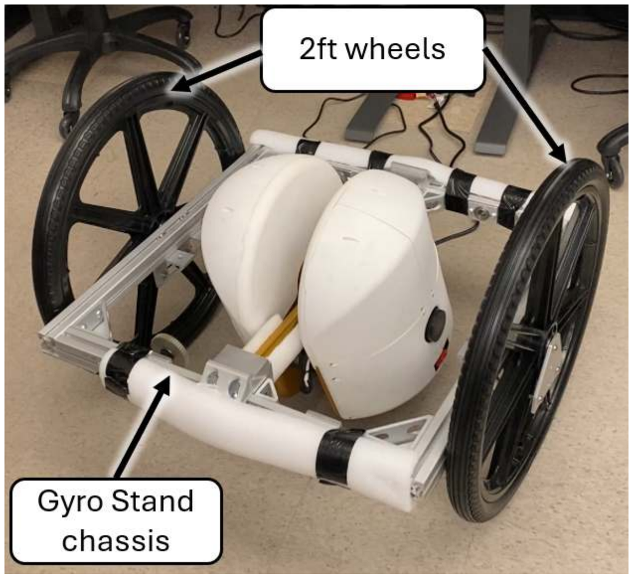


Figure 2.6: Pendulum in steering stand

2.3.4 Drive Stand

The final stand was simply a set of wheels with hex adapters mounted to the outer ends of the driveshaft. This stand proved the most useful, as it served the purpose of the hanging stand to check alignments and calibrations. Simple rolling tests with the pendulum could be conducted without assembling the shell. The stand even appears in early field tests, Figure 2.7.

Figure 2.7: Pendulum in drivestand on a slope descent test

2.4 Decoupled Dynamic Model and Control Design

The angles that describe the pendulum and motion of the outer shell are presented in Figure 2.8. For the Steering direction, a positive rotation about the center axis of the outer shell, ϕ, requires a negative rotation of the pendulum, θ_s. Both ϕ and θ_s are measured with the mounted IMU to the ground frame. The relative angle between them, θ, is measured with a motor encoder and relates to the others through equation (2.1). For the Driving direction, a positive lifting of the pendulum, θ_{dr}, results in a negative shell rotation, ϕ_d. The drive angle is measured with an IMU, but the robot's rolled angle, ϕ_d, is measured with encoders on the drive axis motor.

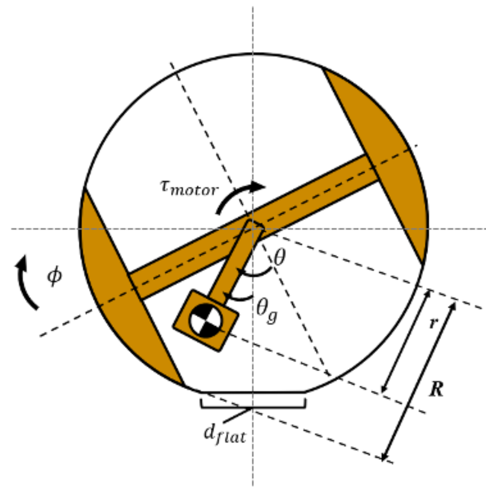


Figure 2.8: Schematic of Robot States

$$\theta_g = \phi + \theta \tag{2.1}$$

2.4.1 Steering Dynamic Model

To derive an initial model, assume that the outer shell does not deform significantly and is a circle. Additionally, assume the robot can be decoupled as a planar system in a hoop with a 1-DOF pendulum, then twisted 90° about the vertical to represent the other axis. The system states are shown in Figure 2.8, where ϕ is the absolute tilt angle of the driveshaft with respect to the ground, θ_g is the angle the pendulum makes with the ground, and θ is the relative angle between the pendulum and the shaft. This relative angle is measured by an encoder and actuated with a motor, τ_m. Both ϕ and θ_g are measured by an IMU rigidly attached to the pitch center frame. The angles of ϕ, θ, and θ_g are related through Equation (2.2).

$$\theta_g = \phi + \theta \tag{2.2}$$

Similarly, R is the nominal outer radius of the ball without deformation and r is the distance to the center of gravity of the pendulum from the axis of rotation. d_{flat} is the diameter of the flat section of the shell resting on the ground and is assumed to be zero so that the outer shell can be approximated as a sphere.

We can then define the system energy with respect to absolute angles: ϕ and θ_g. Potential and kinetic energies are shown in Equations (2.3) – (2.5).

$$V = m_p g \left(R - r\cos(\theta_g)\right) \tag{2.3}$$

$$T_{rot} = \frac{1}{2} I_b \dot{\phi}^2 + \frac{1}{2} I_p \dot{\theta}_g^2 \tag{2.4}$$

$$T_{trans} = \frac{1}{2} m_b (R\dot{\phi})^2 + \frac{1}{2} m_p (\dot{x}_p^2 + \dot{y}_p^2) \tag{2.5}$$

Where m_p, m_b, I_p, and I_b are the masses and rotational inertias of the pendulum and ball shell, respectively. The pendulum translational energies can be expressed with $x_p = R\phi - r\sin(\theta_g)$ and $y_p = (R - r\cos(\theta_g)$. To find the planar equations of motion, we use Lagrange's method for $L = T_{rot} + T_{trans} - V$.

$$\frac{\partial}{\partial t}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \tau_q \tag{2.6}$$

Applying equation (2.6) with $q = \phi$ and θ_g yields the following EOMs:

$$\tau_m = I_{eq}\ddot{\phi} - m_p r R \cos{(\theta_g)}\ddot{\theta}_g + m_p r R \sin{(\theta_g)}\dot{\theta}_g^2 \tag{2.7}$$

$$-\tau_m = - m_p r R \cos{(\theta_g)}\ddot{\phi} + (I_p + m_p r^2)\ddot{\theta}_g + m_p r g \sin{(\theta_g)} \tag{2.8}$$

where $I_{eq} = (m_b + m_p)R^2 + I_b$

which can be simplified to the standard robotic form

$$M(\theta_g)\begin{pmatrix}\ddot{\phi}\\ \ddot{\theta}_g\end{pmatrix} + C(\dot{\theta}_g, \theta_g)\begin{pmatrix}\dot{\phi}\\ \dot{\theta}_g\end{pmatrix} + V(\theta_g) = F\tau_m \tag{2.9}$$

where

$$M(\theta_g) = \begin{bmatrix} I_{eq} & -m_p r R \cos(\theta_g) \\ -m_p r R \cos(\theta_g) & I_p + m_p r^2 \end{bmatrix}$$

$$C(\dot{\theta}_g, \theta_g) = \begin{bmatrix} 0 & m_p r R \sin(\theta_g)\dot{\theta}_g \\ 0 & 0 \end{bmatrix}$$

$$V(\theta_g) = \begin{bmatrix} 0 \\ m_p r g \sin(\theta_g) \end{bmatrix} \quad F = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The physical parameters used in the above model were measured from a CAD model of the robot and shown in Table 2.1. Individual masses were verified with a scale.

Table 2.1: Parameter Values for FIRST Pendulum in 2023

Symbol	Description	Value [units]
m_p	mass of the pendulum	22.65 [kg]
m_b	mass of the ball or shell	14.49 [kg]
I_p	the inertia of the pendulum	0.4279 [kgm^2]
I_b	the inertia of the shell	0.902 [kgm^2]
R	outer radius of shell	0.305 [m]
r	radius to c.g. of pendulum	0.0975 [m]
g	gravitational constant	9.81 [m/s^2]
τ_m	input motor torque	[Nm]

2.4.2 Linear Quadratic Control of Steering

In Equation (2.9), the matrix F has a rank less than two, meaning this system is under-actuated. Linear Quadratic Regulators (LQRs) are a popular choice for regulating similar systems in previous works [8], so we present a similar derivation here. We can simplify (2.7) and (2.8) by assuming low speeds and small angles such that $\dot{\theta}_g^2 \approx 0$, $\sin(\theta_g) \approx \theta_g$ and $\cos(\theta_g) \approx 1$. These simplified equations become (2.10) and (2.11).

$$\tau = I_{eq}\ddot{\phi} - m_p r R \ddot{\theta}_g \tag{2.10}$$

$$-\tau = - m_p r R \ddot{\phi} + (I_p + m_p r^2)\ddot{\theta}_g + m_p r g \theta_g \tag{2.11}$$

these equations can be manipulated into the form of (2.12) where $x = (\phi, \theta_g, \dot{\phi}, \dot{\theta}_g)$.

$$\dot{x} = Ax + B\tau \tag{2.12}$$

By substituting values from Table 2.1, (2.12) yields the following A and B matrices:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -6.22 & 0 & 0 \\ 0 & -40.2 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0.0128 \\ 1.57 \end{bmatrix} \tag{2.13}$$

Solving the optimal control problem requires minimizing a cost function (2.14) according to chosen weighted matrices Q and R.

$$J = \int \left(x^T Q x + \tau_m^T R \tau_m \right) dt \tag{2.14}$$

The values of Q and R were refined based on experiments with the robot in its shell to achieve acceptable performance and stability.

$$Q = \begin{bmatrix} 1000 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 5 \end{bmatrix} \tag{2.15}$$

These matrices can then be used to solve the optimal Linear Quadratic Regulator problem to yield a linear feedback control law (2.16) which stabilizes the linear system (2.10) and (2.11).

$$\tau_m = -Kx \tag{2.16}$$

For our system, the K matrix was obtained offline and loaded into the robot on startup. Encoders and onboard IMUs measure all states.

A simulated response of the derived controller, shown in Figure 2.9, shows that it stabilizes the underactuated system with a rigid shell, as modeled by (2.9). This model would be accurate if the outer shell behaved as a rigid body with a point contact. these results support the viability of the small angle assumption to simplify the equation for the form in (2.10) and (2.11).

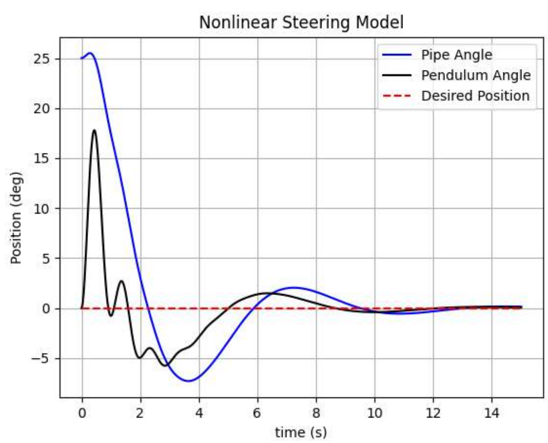


Figure 2.9: Simulated response of optimal controller on ideal nonlinear system

Any shell that deviates from this assumption would affect the performance of the control scheme. Design iterations of the outer soft shell are discussed in the next chapter, along with basic models to quantify their behavior. For the sake of continuity, their effects on the controller in (2.16) will be discussed in the next section after they have been introduced.

3. Inflatable Soft Shell Design

The outer shell design is the most unique and challenging aspect of creating RoboBall. The shell must be both soft and airtight to function as desired, yet rugged enough to withstand repetitive testing. The shell design consists of two main parts: the soft membrane, which comprises most of its contact area, and a mounting interface that grips the membrane, allowing torque to be transferred from the pendulum to the membrane. This chapter will discuss two methods of manufacturing a soft outer shell, one based on commercially available exercise balls and the second on a molded spray polymer. Both types of shells utilized the same mounting hardware. The chapter will conclude by comparing the designs and providing comments on the longevity of the mounting hardware over a three-year testing period.

3.0.1 Mounting Interface

A large-bore circular ring clamp attaches all prototype membranes to the rigid driveshaft. Figure 3.1 a shows a cutaway of the rings and attached hub. From the bottom to the top of Figure 3.1, the Lower Clamp and Upper Clamp are bolted together with fasteners from the underside (not shown). The hub is bolted on from the outside through clearance holes into the bottom clamp. This allows the shell material to be firmly held between the rings with screws from either direction. To ensure an airtight seal between the rings and hubs, an O-ring dovetail groove is cut around the outer radius of the hubcap.

When the robot needs to be removed from the shell for maintenance or charging, only the upper layer of bolts and the hub need to be removed. This keeps the clamps and shell membrane intact between tests.

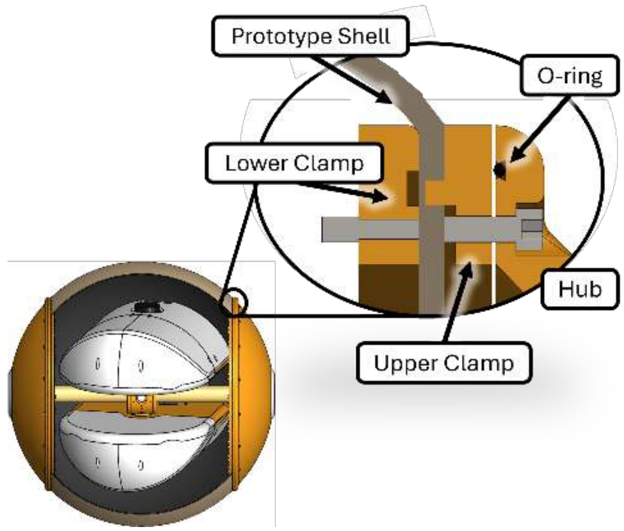


Figure 3.1: Cross-sectional view of the shell mounting interface

3.0.2 Exercise Ball Based Shell

The first type uses an off-the-shelf exercise ball, commonly referred to as a yoga ball. This design was based on the original prototype for RoboBall I, built by L. Bridgwater in [1], which used a similar shell and is shown in Figure 3.2. This shell could not contain the high pressures needed to hold the pendulum off the ground without excessively expanding. This required the pendulum to be outfitted with wheels to keep from tearing through the outer shell, see Figure 1.1. Subsequent efforts focused on trying to constrain this undesired expansion.

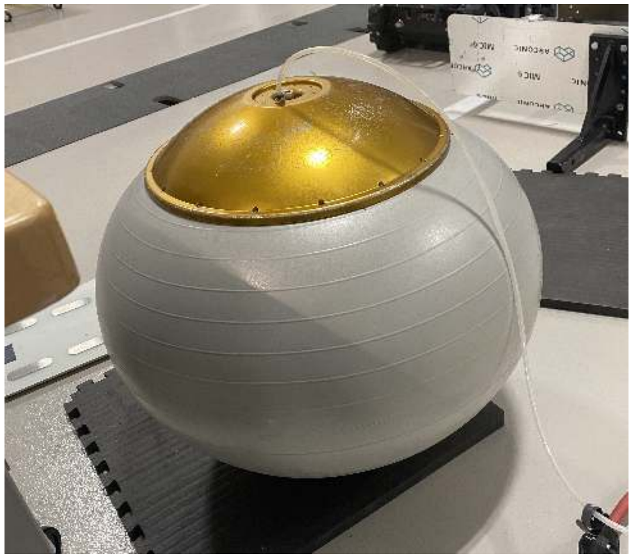


Figure 3.2: The bare yoga ball prototype with excessive expansion

An initial approach to constrain the inner ball was to sew a nylon jacket, constructed with the help of a local tailor, to provide a barrier against the outer shell's expansion. This iteration was stable enough to enable preliminary control studies, as shown on the right side of Figure 3.3.

Figure 3.3: Nylon shell prototype resting and during a test

The outer jacket was sufficient for short periods. However, it had to wrap around the aluminum hubs. During tests, natural instabilities in the robot caused the shell to grind the nylon between the ground and the aluminum. Holes would inevitably form, ruining the shell's structural integrity. As a result, the outer layer would wear out and need to be replaced frequently. The difficulty of sewing the shell to expand to a perfect sphere at the desired pressure also meant that the nylon prototypes would become ellipsoidal or "egg-like" at pressures necessary to keep the pendulum off the ground. This pushed us to our next design.

Using the same exercise ball as a base for the inner bladder, we took some prototypes to Rhino Linings in Dallas to experiment with various sprayed-on polymers. We settled on an outer layer that is a military-grade spray-on elastomer [9]. This outer layer was durable yet flexible enough to constrain the inner bladder to the appropriate spherical shape at operating pressures; a final prototype is shown in Figure 3.4.

Figure 3.4: Assembled bedliner constrained prototype

While the bedliner-coated exercise balls were strong and able to withstand initial dynamics and control studies, they were difficult to produce consistently. The process to construct a bedliner-coated exercise ball is outlined in Figure 3.5. First, a ball would be blown to size in a pre-made rig to be marked 3.5. These marks would serve as guides to clamp and cut the clamping rings on the inner shell, 3.5B. Another slightly smaller ball could then be inflated inside of it to give the outer ball a semi-final state to spray 3.5C-D.

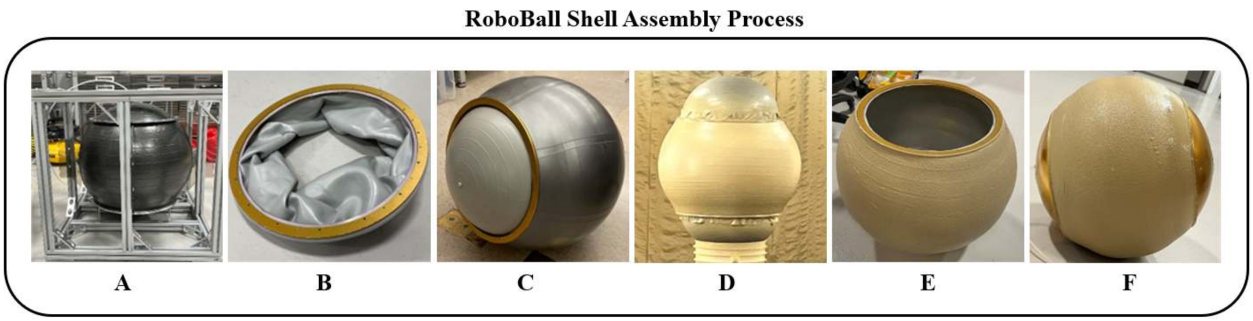


Figure 3.5: Illustration of RoboBall Bedliner Manufacturing Process

However, the bedliner coating would be sprayed at elevated temperatures, causing undesirable expansion of the inner ball during step D, resulting in some outer shells that were not spherical. Figure 3.6 shows two such prototypes where the resulting shell was no longer spherical. The process was eventually streamlined after learning the average rate of expansion from previous attempts. However, even then, the process was anything but reliable.

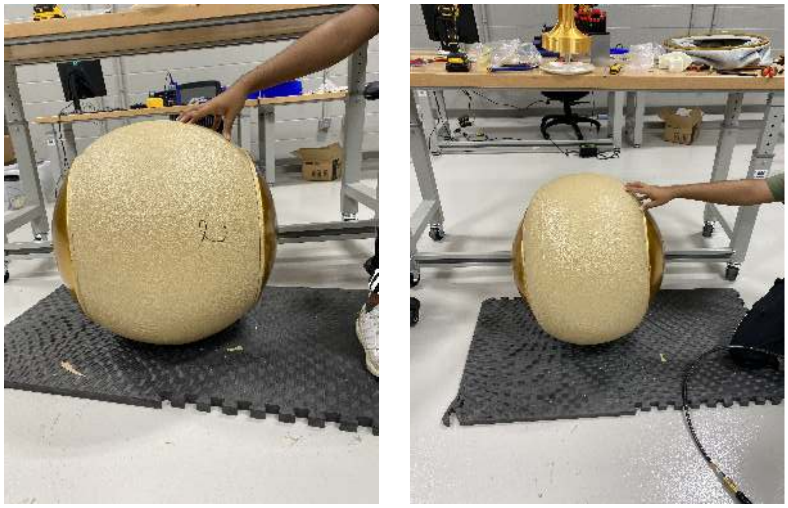


Figure 3.6: Examples of mis-formed bedliner-coated shells. One undersized (left) and one oversized (right)

This method was stable enough to carry out testing primarily discussed in [2]. However, as the control system evolved, the outer shell was subjected to increasingly demanding environments and dynamic scenarios. In most of these scenarios, shells would shear along the clamp-ball interface, shown in Figure 3.7.

Figure 3.7: A sheared interface of the ball-clamp interface. The black material is Flex Seal, applied in a vain attempt to fortify the joint.

This would happen to almost all prototypes after a few days of testing, as seen in Figure 3.8, resulting in constant supply runs to build more. This practice was neither sustainable nor desirable, as it was impossible to study a prototype that continually changed one of its core components.

Figure 3.8: A summer's term worth of testing

3.0.3 Molded Shells

Replacing the exercise ball as the main torsional interface was a key hurdle to overcome in finalizing a stable robot design. The bedliner coating has proven robust in testing across various harsh environments, such as gravel and water environments. Some experiments caused the pendulum to strike the inner layer of the ball, resulting in a tear in the rubber. In these scenarios, the bedliner still maintained the airtight seal, allowing it to provide a complete airtight membrane. The ball was initially used as a shape reference for spraying, and the bedliner would hold its shape even after some parts of it peeled off. See some of the prototypes in Figure 3.8. Replacing it with something that had a more predictable shape would allow the bedliner to be the entire shell.

To solve this, a 3D printable mold was developed that could be sprayed and disassembled from the inside, leaving a complete shell behind. The mold consists of three distinct parts apart from the mounting hardware: an upper panel, a middle panel, and the inner clamping ring from the original ball design. The upper panels would be attached to the inner ring for structural support, and the inner panels are positioned between them, as shown in Figure 3.9. This structure could then be placed on a rotary stand, allowing a human sprayer to apply an even coat of the polymer.

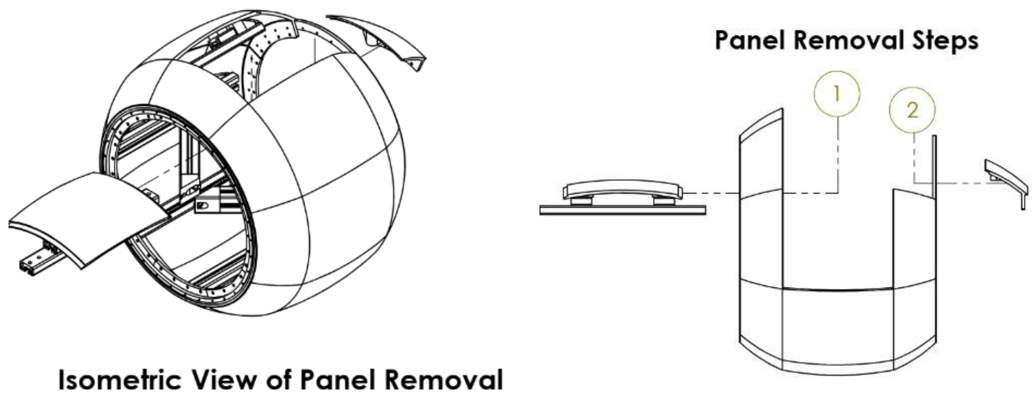


Figure 3.9: Isometric sketch of the core mold assembly and disassembly process (steps 1 and 2)

This system went through two rotary stand prototypes. The first mounted and rotated the mold vertically, shown in Figure 3.10.

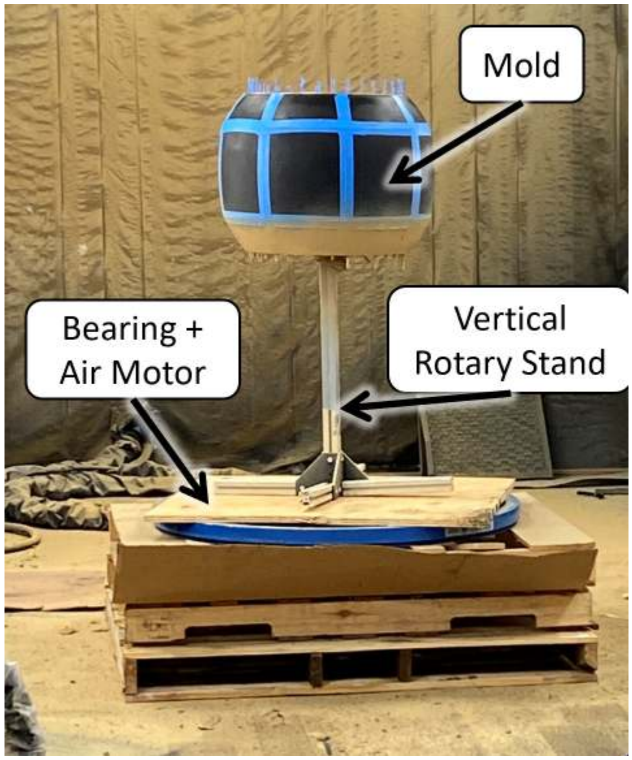


Figure 3.10: Vertical rotary mold stand

This was derived from the existing architecture at the RhinoLinings warehouse. The first few shells came out lopsided, as the overspray from the nozzle would disproportionally collect on the lower hemisphere. So we redesigned the stand to roll horizontally for a more uniform application, shown in Figure 3.11.

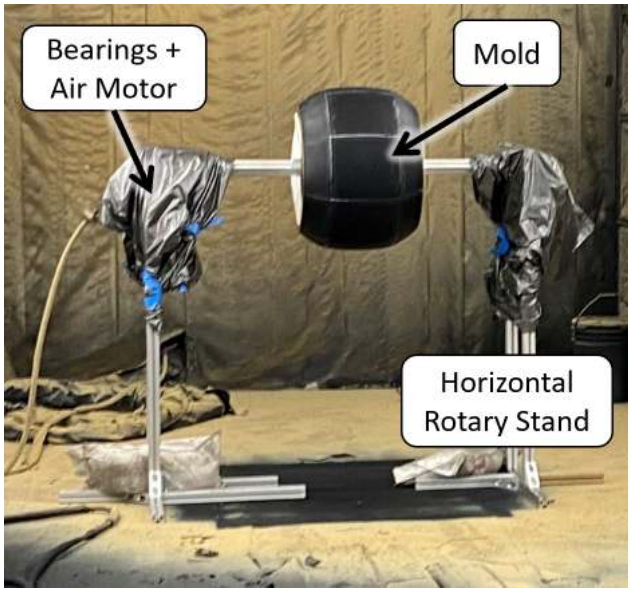


Figure 3.11: Horizontal Spray Stand and Mold

This led to a standard workflow for completing molded shells. Similar to the process in Figure 3.5, shown in Figure 3.12

Figure 3.12: Illustration of Molded Shell Manufacturing Process

3.0.4 Shell Repair

Two significant issues arise from the aerosolized spraying process. These issues need to be corrected when initially spraying or post-processing the shell.

In the first issue, inadequate coverage of the rings leaves a thin wall where the clamps apply pressure; this will eventually tear through, as shown on the left side of Figure 3.13. Attempts to fix this in post-production with silicone, polyurethane concrete glue, or a combination of polyurethane and a mesh tape matrix all would fail after a few tests. Shown on the right of Figure 3.13. The shells could also be resprayed, but that would throw off their balance. It is best to ensure the rings are evenly coated on the first spray attempt.

Figure 3.13: Attempt to repair thin walls at edges

The second issue arises when air bubbles become trapped as the polymer sets. These air bubbles compromise the shell's pneumatic integrity and must be plugged. Filling them with polyurethane or silicone and covering them with tape from the inside is usually enough. The leftover mold-releasing compound on the inside must be sanded off before it adheres. An example of this style of repair is shown in figure 3.14.

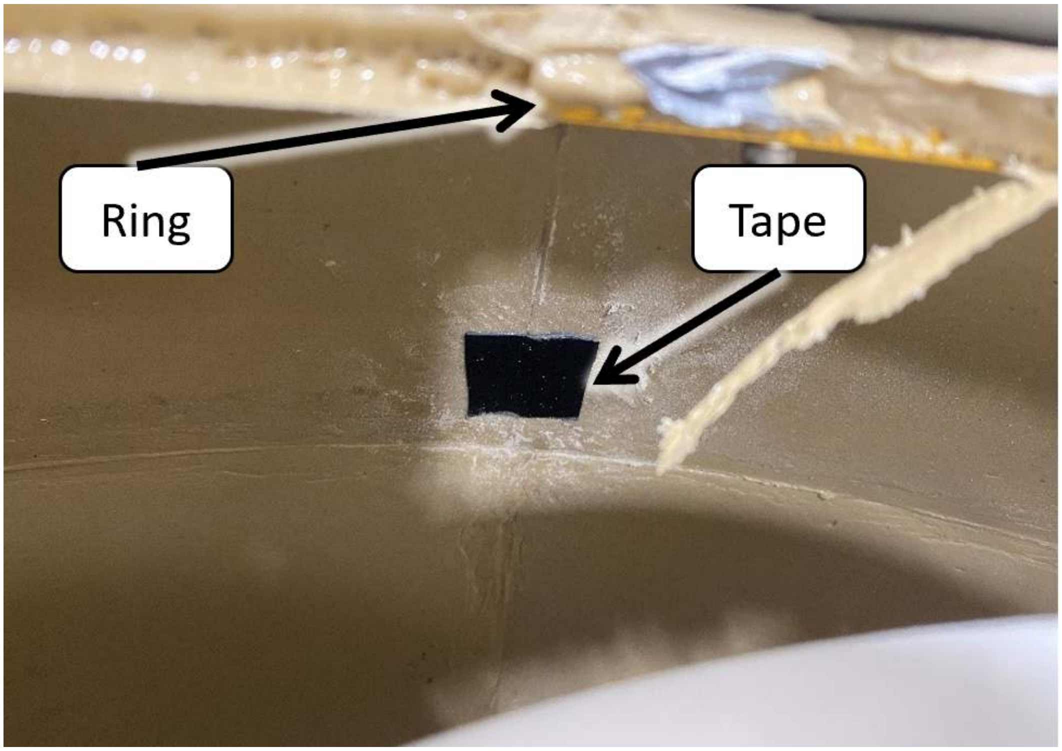


Figure 3.14: Taping method to repair small holes in the shell

3.1 Models of Soft Shell

Introducing a robot with a compliant shell nulls one of the most common assumptions made when working with these robots—the assumption of a rigid, perfectly spherical shell. Through our initial testing with the bedliner-exercise ball shell, two basic models were proposed and validated. One that tied the shape of the robot to its natural rolling stability and yielded design requirements on the sphericality of the shells. The second dealt with predicting the size of the flat that appears on the bottom of the robot at different pressures on a hard surface.

3.1.1 Shape and Stability

The rolling motion of the robot is relatively simple as long as the outer shell stays spherical. Most spherical robots use a hard outer shell for its ease of procurement and machining. By doing this, spherical robot designers have not harnessed the potential benefits a soft outer layer would provide. In addition to providing more damped protection for the inner mechanism and payload environment, a soft shell would also offer additional damping, resulting in a smoother ride. A soft, pliable surface, in conjunction with a pressure control system, would enable greater flexibility in varying contact forces, thereby enhancing traction.

Most models and controllers assume the robot keeps a spherical shape throughout its operation. Since the shell is flexible and can be pressurized, it will naturally change its shape in response to pressure. For the robot, its horizontal axis is held at a rigid length by the drive shaft. As the robot is pressurized, it expands along its vertical axis, resembling an ellipse. This is illustrated in Figure 3.15. When this happens during rolling experiments, the system shifts from the pendulum hanging down at zero degrees and rolling straight to the side of the robot. This causes it to turn and wobble.

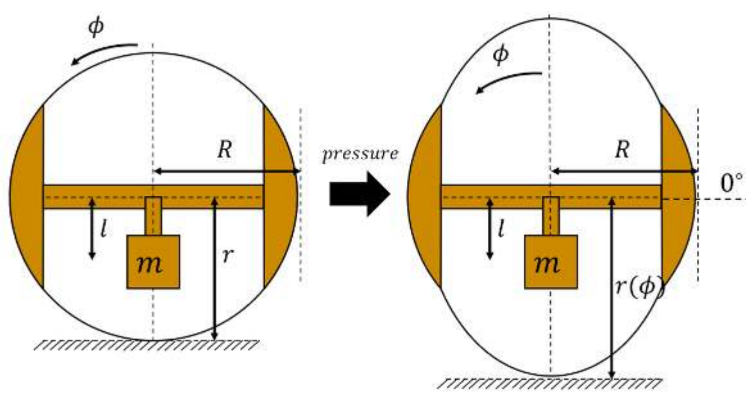


Figure 3.15: 2D Schematic of a Soft Spherical Shell Deforming with Pressure

To design a shell that mitigates this expansion, we seek to understand the underlying process. The gravitational potential energy of a system is related to its geometry, so we develop a relationship to track this change in relation to pressure. For a similar 2D circular system with a pendulum and a chosen datum on the ground, its potential is given by equation (3.1) below. Note that in the spherical case, the distance from the ball's center to the ground, r, is a constant value of the nominal ball radius, R (or $r(\phi) = R$), and ϕ is the global ball tilting angle. Here it is assumed that the roll, or steering, motors are locked in place so that the pendulum will stay perpendicular to the drive axle. where m is the mass, l is the distance to the pendulum's center of gravity.

$$V = mg\left(r(\phi) - l\cos\phi\right) \tag{3.1}$$

We assume that the deformation can be modeled well enough by scaling an ellipse. The standard equation of the ellipse is repeated below. If $a = b = 1$, then (3.2) describes a circle with radius R. Whereas shifting a and b will scale the circle into an ellipse about the respective horizontal and vertical axis.

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = R^2 \tag{3.2}$$

Because the drive axle holds the horizontal axis constant, we can assume that $a = 1$ for any pressure. However, b will change based on the system pressure and shell design. Ideally, $b = 1$, and the robot will be a perfect sphere. In practice, irregularities in construction and over-inflating or under-inflating the system will change its value.

To study how the change in b affects the rolling stability of the system, (3.2) is transformed into polar coordinates and rotated 90° with $x = r\cos(\phi + \pi/2)$ and $y = r\sin(\phi + \pi/2)$, then substituted into (3.1). This yields (3.3), or the potential of the robot as a function of roll (steer) angle, and the pressure dependent scale parameter, b.

$$V = mg\left(\sqrt{\frac{R^2b^2}{b^2\sin^2(\phi) + cos^2(\phi)}} - lcos(\phi)\right) \tag{3.3}$$

By plotting this equation, we can graphically study the effect this scaling parameter will have on the system's dynamics. Equation (3.3) is plotted in Figure 3.16, with various values of b. The graph is normalized to mass and gravity, mg, to focus on the geometric parameters: R, l, and b. Were R is 12in and l is 3in. The corresponding ellipse shape is on the right for reference. This is analogous to changes in the system pressure deform the outer shell. From the theory of potential wells, minima in the graph correspond to stable equilibrium points that the robot will tend toward.

By inspection of Figure 3.16, we can study what happens to regions of minimal potential energy as the ball is inflated and b changes. To start, when $b = 1$, the system is a perfect sphere. This is the ideal case for our robot operation. With this geometry, the potential minimal is at a roll angle of 0°.

Indicating that the ball will naturally roll to where the pendulum is hanging down. Additionally, if $b < 1$ the system retains this inherently stable position, but the pull towards the middle position is more extreme. When $b > 1$ the old equilibrium point flips to an unstable equilibrium point with two minima on either side. This bifurcation of the stability point will cause the robot to move without actuation from its motors, such an effect would appear as a wobbling motion in the robot, but a different source than in described in Chapter 5 Section 3. Therefore it is vital that the shell be designed to maintain a shape close to, or less than, a sphere at its operating pressure.

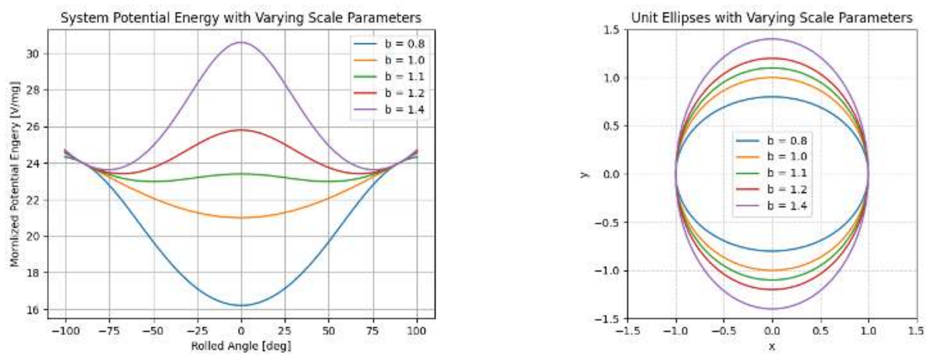


Figure 3.16: Potential Energy of the System with Changing Scale Parameters

3.2 Characterization of Shell Prototypes

Knowledge of each shell's scale parameter, b, will allow us to study the inherent stability and predict the system's performance. Each prototype was placed on a hub and inflated to various pressures to measure the scale parameters. At each pressure, the major circumference of the shell was measured. The results of each test is overlaid in Figure 3.17.

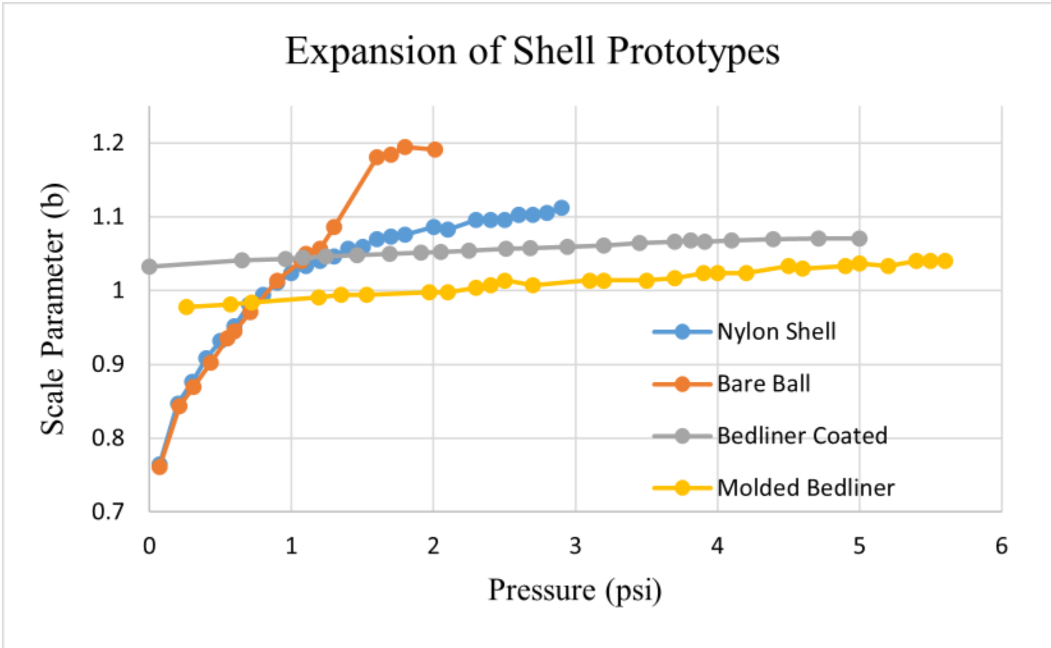


Figure 3.17: Measured Shape Parameters for Shell Prototypes

Of the shells tested, the ball reinforces bedliner and nylon shells had scale parameters close to 1. However, the nylon had more variation along the tested pressure range. Since the robot is prone to leaking, the nylon would have the most variation in its shape over a test. The bedliner shell was the best at retaining its shape and is used for most types of testing. Additionally, the nylon-reinforced and bedliner shells were tested with the assembled pendulum in basic rolling tests. The nylon shell, shown in Figure 3.3, exhibited profound side-to-side rocking as it rolled forward due to its elliptical shape. It was difficult to coax it to roll straight without a more robust balancing controller. Both variations of the shell with bedliner had limited expansion and stayed closer to its ideal spherical shape.

3.2.1 Flat Model

A rigid sphere will contact a flat surface at exactly one point. However, inflatable bodies conform to the surface on which they rest. This deformation also depends on pressure, as objects at higher pressure will deform less. For RoboBall, this deformed patch or "flat" will be our primary traction surface and significantly affects the system dynamics. In this section, we propose a model for the flat. A sketch of the deformed surface is shown in Figure 3.18. Where δ represents how much the ball has deformed from a perfect sphere, W_b is the system's total weight, and P is the internal system pressure.

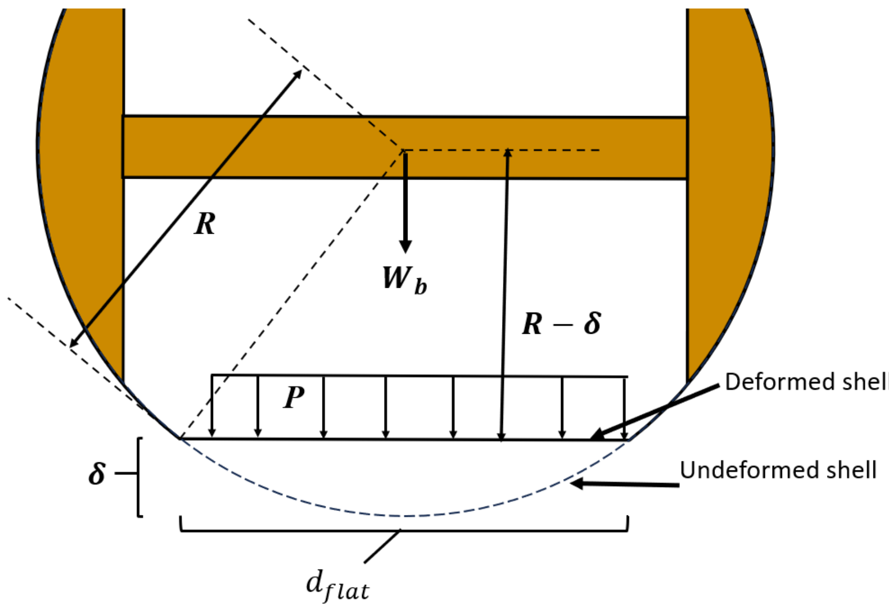


Figure 3.18: Schematic Relating Geometry of a Flat of Pressurized System (rolling out of page)

Then relate R, δ, and d_{flat} through the Pythagorean Theorem shown in Equation (3.4). A constraint must be placed on δ so the compressed shell does not interfere with the pendulum. For our system, which was to keep a pressure such that $\delta < 2$ in. This constraint is based on the design's radial clearance between the bottom of the pendulum and the shell.

$$R^2 = (\frac{1}{2}d_{flat})^2 + (R - \delta)^2 \tag{3.4}$$

The flat that supports the system's weight through pneumatic pressure, P, over the flat area, must equal the system's weight to be in equilibrium, given in Equation (3.5).

$$\sum F = W_{ball} = A_{flat}P = P\frac{\pi}{4}d_{flat}^2 \tag{3.5}$$

Rearranging Equation (3.5) to Equation (3.6) yields a model relating the flat to system pressure which is plotted in Figure 3.20. Where $W_b = (m_b + m_p)g$ from Table 2.1.

$$d_{flat} = \sqrt{\frac{4W_b}{\pi * P}} \tag{3.6}$$

To verify Equation (3.6), the robot was placed on a clear polycarbonate table with a tape measure beneath it. A light was shown to add contrast to the deformed and undeformed regions. The robot was then manually inflated to a pressure, and the diameter of the flat was recorded. Figure 3.19 shows a picture of the test setup. The results were then plotted against the model, shown in Figure 3.20. The data fits in the middle but becomes a poor fit on the extremes. For low pressures, the stiffness of the polymer adds more structure and prevents the flat from growing. At higher pressures, the change in diameter is less pronounced, and the shell stiffness would also work against a change in shape.

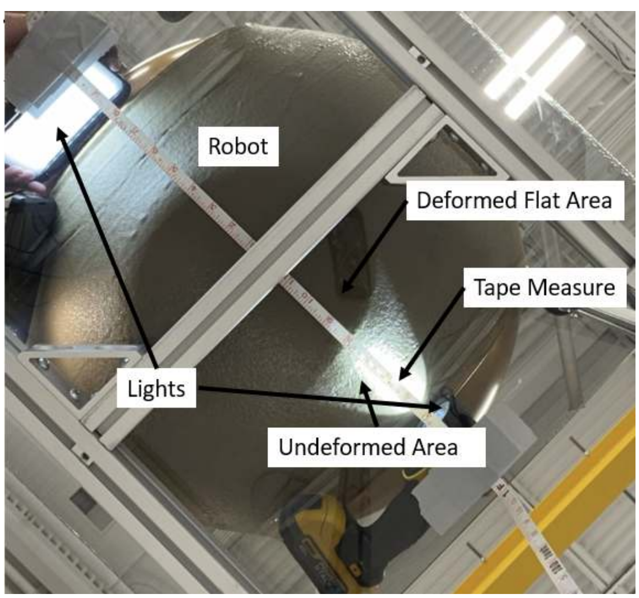


Figure 3.19: Measurement of the Flat from below

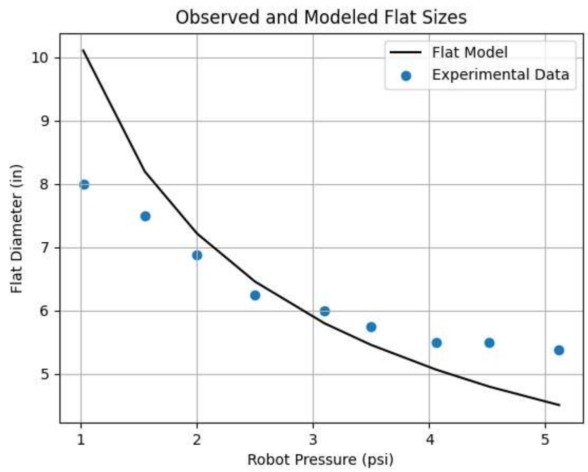


Figure 3.20: Comparison of Flat Model with Experimental Data

3.2.2 Effects of Soft Shell on Control Design

The pure full-state feedback control law that was simulated on the system is shown in Figure 2.9. In the figure, the system eventually stabilizes to a zero pipe angle; the model and control law assume a rigid outer shell through point contact.

The results of running the controller on RoboBall II, at various pressures, while not driving, are shown in Figure 3.21. The system exhibits a more critically damped response and significant steady-state error. This suggests that the shell stiffness, damping, and flat size may be affecting its performance, as these factors are not considered in the controller derivation.

To further investigate the effect pressure has on the system, the same balancing experiment was conducted at different system pressures from 0.5 to 5.5 psi. The compiled results are shown in Figure 3.21. To simplify comparison, the data is normalized with respect to its initial position. Low pressures are colored blue, while higher pressures are red, to illustrate the spectrum of responses. As the pressure increases, both the steady state error and flat diameter (presented in the next chapter as Figure 3.20) decrease.

To establish a correlation, the steady-state error from Figure 3.21 was plotted against the model for the flat sizes, derived in the next chapter as equation (3.6), evaluated at the measured shell pressure. This is shown in Figure 3.22 with colors representing increasing pressure. The obvious positive correlation between the steady state error and the flat size points to its effects on the steering controller's performance.

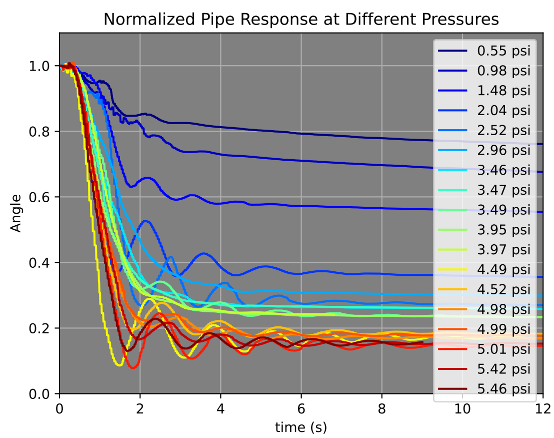


Figure 3.21: LQR Stabilization of the System at Different Pressures

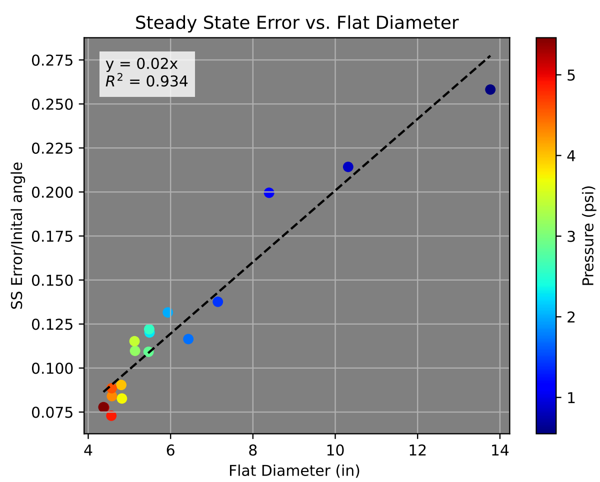


Figure 3.22: Steady State Steering Error Plotted against Flat Diameter Model

These tests were carried out in the bedliner-constrained exercise ball-based shell. These results were replicated and appropriately compensated for by Pravecek et al. in [10] using the molded shell.

4. Internal Pressure Control System

4.1 Background

Spherical rovers are prime candidates for investigating hostile terrain. A sphere lacks crevices for dirt and debris to accumulate and damage the system. If the system were deployed in cold Arctic or space environments, the shell would provide an enclosed environment that keeps internal avionics warm. Such rovers have proven effective even without active control. A notable example is the Tumbleweed robot, which traversed unassisted across Greenland's tundra, relying solely on arctic winds. The robot relayed valuable data back to the base for many days [11]. If multiple systems are deployed with an appropriate supporting communications network, they could form a versatile charting system to map planetary areas [12]. During surveying missions, these robots will encounter various terrain features, potentially causing them to bump, skid, and slam. Thus, if this class of robot is to be ubiquitous in the field, its outer shells must be soft enough to withstand these scenarios.

A subclass of these spherical robots uses an internal pendulum to control the rolling motion of these systems. Lifting the pendulum changes the system's center of gravity and allows the system to roll. They have been a popular class of robots to build, and numerous papers have been devoted to their design and control strategies [13]. However, most robots that are built use a hard outer shell. While simple to manufacture, they are poorly suited to rough conditions unless some padding is added outside, as in [14]. However, it would be more useful if the spherical system were inherently designed with a compliant outer shell. Bruhn first introduced a preliminary design for a soft outer shell, and their design sought to incorporate solar cells, micro instrumentation, and thermal management schemes in the flexible membrane [15]. Unfortunately, to my knowledge, Bruhn's prototype was never fully realized. Ylikorpi et al. constructed a 1-degree-of-freedom pendulum-driven spherical robot with an inflatable exercise ball as the shell. Although the main contributions of their inflatable robot focused on such a system's ability to overcome obstacles rather than the dynamic benefit of having a soft shell [16]. Robots with soft outer shells do not need to be inflatable. Kim et al. implemented a spherical robot that used a spherical elastic cage. While they acknowledged a flexible shell's benefits on a system, they do not specify how it might affect its dynamics [17]. An open cage design would also allow dirt and debris to enter and harm the inner electronics. Another group published a series of papers showing that changing the system pressure has some effects on applied dynamics and control schemes [2]. The significant effects must be compensated for in the control scheme [10].

Another take on soft spherical rovers forgoes the inner pendulum and instead relies on deforming the robot's outer shell. By segmenting the outer layer into multiple inflatable sacs, inflating one will cause the robot to tip and roll off it. Coordinating these patches' inflate and deflate sequences will allow for robot motion control [18]. This process is unique and capable; however, when implemented, most of the robot's internal hardware becomes dedicated to supporting the inflatable structures. Internal solenoids and air sources take up valuable real estate that could go to communications or scientific packages [19]. In addition, implementing these types of pneumatic control schemes usually requires using a specialized novel actuator system such as [20], which could be unreliable to send on long-duration missions far from repair. However, there might be a way to combine the pneumatic control systems of these deformable robots with off-the-shelf parts in tandem with the simplicity of a pendulum-driven robot.

In typical vehicles with pneumatic tires, proper inflation pressure is fundamental for the safe and efficient operation of the vehicle. Central Tire Inflation Systems have been used in agricultural vehicles to improve traction and performance [21]. With a system dedicated to adapting tire pressure, the handling of the vehicles can improve handling for the operators [22] and can reduce undesirable vibrations when rolling over uneven terrain [23]. Dravid showed that varying the pressure in spherical systems influences performance in [4]. For example, a spherical robot released at the top of a slope could "air down" its shell to better absorb impacts or prevent unnecessary bouncing. Likewise, increasing the internal pressure could make for better handling at higher speeds on level ground, driving a need for an internal air management system through explicit control of bounce frequency and damping by controlling pressure.

4.2 Introduction

Dynamic modeling is an essential aspect of robotic systems. Accurate dynamic models make predicting and tuning the performance of control systems easier. The rolling performance of an inflatable sphere is strongly related to its internal pressure, as shown in previous work by Pravecek et al. [10]. Since pressure is significant, it becomes a noteworthy state variable and should be accounted for when modeling. So, developing an accurate dynamic model of the time-varying pressure response of the system would make for more accurate dynamic models.

This chapter focuses on the fusion of RoboBall II with an internal air control system. The following sections will present the design of the inflatable outer shell and outline the packaging of an onboard compressor and tank system in the system's internal pendulum. A novel dynamics model will be developed to study how the internal pressure responds to leaks and inputs from the compressor and solenoid. The model reveals that the underlying process is slow, so a simple bang-bang control scheme is selected for control. Additionally, since most work so far has focused on the rolling dynamics of the robot, a brief empirical study of the robot's bouncing dynamics as a function of pressure will be presented.

4.2.1 Pressure Control Hardware

The pressure control system hardware was designed to fit within the inner pendulum of the robot alongside the system's motors and batteries and consists of off-the-shelf parts. These parts are three 35cu. in. plastic Clippard air reservoirs, an electronically controlled 12V solenoid, and a Viair 90C Air Compressor. All of these components are fitted together with 1/4in pneumatic fittings. Air is moved from the ball to the tanks to lower the ball pressure via the air compressor. Venting the solenoid moves air back to the ball to increase its pressure. The electrical components were sourced from the FIRST robotics kit for quick implementation. The tanks and compressor are fitted vertically on the pendulum to maximize space for other components, shown in Figure 4.1. The tank pressure is read from an analog pressure sensor attached to the pneumatic lines.

An onboard Vectornav 100 IMU system records ball pressure and angular states. All sensors, compressors, and solenoids are controlled, and data is logged with a RoboRio 2.0 computer.

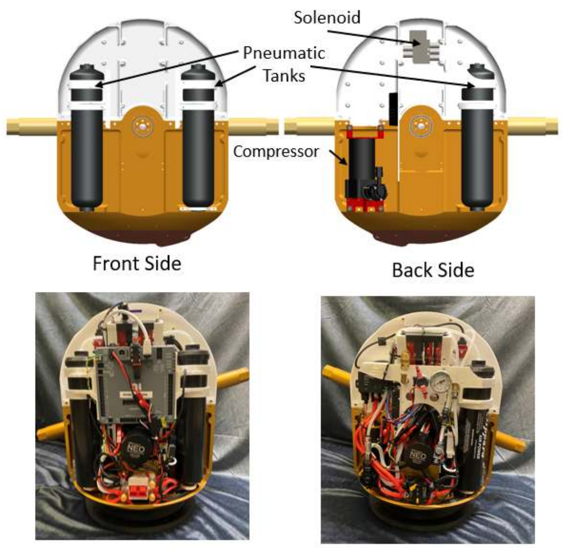


Figure 4.1: Packaging of Pneumatic System on the Robotic Pendulum

4.2.1.1 Pressure Control Range

The primary use of the pneumatic control system system is to transfer air stored in the tanks into the ball with a solenoid valve, and back with the compressor, allowing the ball shell pressure to be controlled. The maximum change in ball pressure the system can produce can be estimated from a modified ideal gas law in (4.1). Where P_i and V_i are the pressure and volume of the tanks and ball, respectively, and γ is the percentage of V_b taken up by the pendulum.

$$\Delta P_t \frac{V_t}{(1-\gamma)V_b} = \Delta P_b \tag{4.1}$$

Table 4.1 lists the volume parameters for the above equation. The pendulum volume was estimated from a SolidWorks model of the system, and V_b was calculated assuming the system was a perfect sphere with a radius of 12.5in. The volume change due to the compression of the shell during normal operation was assumed to be insignificant.

Table 4.1: System Volume Parameters

Parameter	Value
V_t	105 (in^3)
V_b	8181 (in^3)
γ	0.15

Under this model, complete discharge of the rated tank capacity of 120psi to a perfectly sealed ball will result in a pressure change of about 1.8psi in the ball. To affect the system's dynamics, only properties significantly affected within this limited range would be of use. Upgrading to larger tanks or tanks with a higher-rated pressure would allow the robot to work within a larger pressure range or counteract leaks for longer.

4.3 Dynamic Pressure Model and Control

Three main factors influence the robot's internal pressure: the compressor, the solenoid, and the slow leaking of various components. This section will develop a dynamic model that captures all three pneumatic aspects, measures the required leaking constants from the robot, and implements a bang-bang control scheme.

We assume that leaking pressure is similar enough to a cooling thermal system that the exponential change laws can represent the ball and tanks. We present the law in (4.2) with appropriate variable name substitutions. The leak rate of the vessel $\dot{P}$ depends on the vessel's pressure and the pressure of its environment, P_{env}. The speed of this relationship is governed by k or a measurable leak rate.

$$\dot{P} = \frac{dP}{dt} = -k\left(P - P_{env}\right) \tag{4.2}$$

For the tanks, (4.2) can be modified with the appropriate state variables. This is shown in equation (4.3) where P_t is the tank pressure, P_b is the ball pressure, k_t α and β are respective tank, compressor, and solenoid leak rates, and u_c and u_s are the control inputs for the compressor and solenoid respectively.

$$\dot{P}_t = -k_t\left(P_t - P_b\right) + \alpha_{t_c} u_c - \beta_t u_s \tag{4.3}$$

We take a similar approach with the ball in equation (4.4). Where the first term is the rate of air leaking out of the ball, the second is the air that's leaked out of the tanks into the ball, and the final terms are the control terms. The difference in volume between the tanks and the ball is accounted for in (4.4) by a volume ratio $V_r = V_t/V_b$. For both (4.3) and (4.4), the control inputs, u, are allowed to be either 1 or 0 to signify *on/off* or *open/closed*.

$$\dot{P}_b = -k_b P_b - V_r k_t\left(P_t - P_b\right) - \alpha_{b_c} u_c + \beta_b u_s \tag{4.4}$$

Since the solenoid is not forcing air out like the compressor, the coefficients β can be thought of as additional leakage of the tanks. They can then be replaced by another instance of (4.2).

$$\beta_t = -\alpha_{t_s}(P_t - P_b) \tag{4.5}$$

$$\beta_b = -\alpha_{b_s}(P_b - P_t) \tag{4.6}$$

Combining (4.3)-(4.6) and placing into state space form yields (4.7)

$$\dot{P} = A\begin{pmatrix} P_t \\ P_b \end{pmatrix} + B(P)\begin{pmatrix} u_c \\ u_s \end{pmatrix} \tag{4.7}$$

where

$$A = \begin{bmatrix} -k_t & k_t \\ V_r k_t & -k_b - V_r k_t \end{bmatrix}$$

$$B = \begin{bmatrix} \alpha_{t_c} & -\alpha_{t_s}(P_t - P_b) \\ -\alpha_{b_c} & \alpha_{b_s}(P_t - P_b) \end{bmatrix}$$

Figure 4.2 contains a graphical representation of the locations of the six leaking rate constants in the system's air flow, along with the two pressure states, measurement layout, and control logic. The control logic is discussed later in section 4.3.2.

4.3.1 Measurement of Constants

Equation (4.7) introduces six pressure change rates that must be measured. This equation is a collection of multiple instances of (4.2) with the ball and tank pressures (P_b and P_t) represented by environmental pressure (P_{env}) or state pressure (P) accordingly. By assuming different parts of (4.7) are zero we can isolate the desired measurable leak rate in a form of (4.2). The solution of which is given generally by (4.8).

$$P(t) = P_{env} + (P_0 - P_{env})e^{-kt} \tag{4.8}$$

If the system is open to atmosphere, then $P_{env} = 0$, and the system can be linearized as in (4.9).

$$\ln(P) = \ln(P_0) - kt \tag{4.9}$$

To measure k in (4.7), the ball and tanks were independently filled with air and allowed to leak

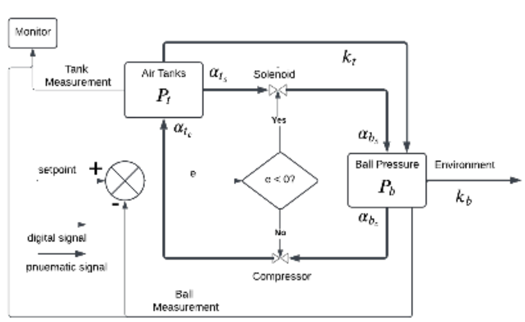


Figure 4.2: Pressure Control System Diagram

into the atmosphere over time. The resulting trace was plotted over a log axis to measure k as in (4.9). The solenoid and compressor leaking coefficients, α_{i_j}, were also measured similarly. Table 4.2 summarizes these measured values.

Table 4.2: Measured Pressure Leak Rates

Parameter	Measured Value (psi/s)
k_t	0.900e-3
k_b	1.36e-4
α_{b_c}	0.0446
α_{t_c}	2.35
α_{b_s}	1.51e-3
α_{t_s}	9.78e-2

4.3.2 Bang Bang Control of Pressure System

The measured constants presented in the previous section show that the system dynamics are slow. As a result, we selected a bang-bang controller for the system. The controller was implemented by sampling the ball's pressure and comparing it with the current pressure setpoint. Then, the compressor or the solenoid is activated, depending on the direction of the error. The chosen action is continued until the error is reduced to a specified tolerance. This deadband was set at 0.1 psi to prevent excessive chattering. Figure 4.2 shows the controller flow graphically.

4.3.3 Empirical Response of Pressure Control Scheme and Dynamic Model

A graph of the robot tracking a predefined ball pressure set point is shown in Figure 4.3. Figure 4.4 shows the measured and modeled pressure in the tanks. The system would continue tracking pressure until the tank's air ran out.

During the initial 10 seconds of the response, the tank pressure drops by about 80 psi, increasing

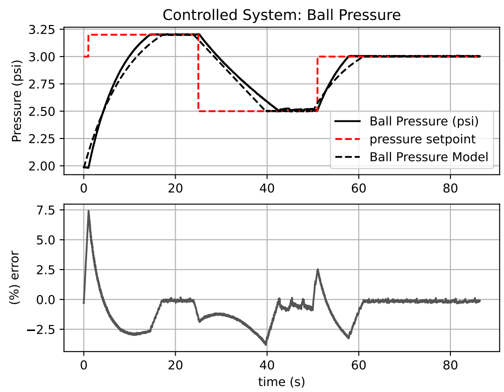


Figure 4.3: Pressure Model and Robot Following the Same Setpoint

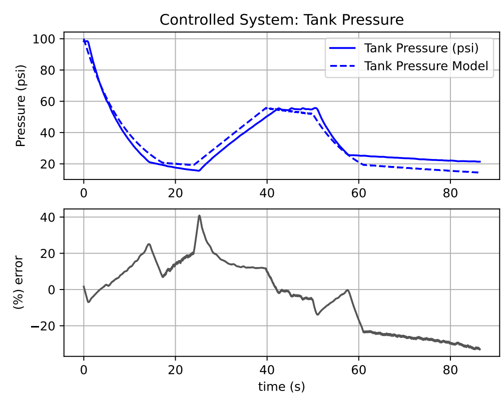


Figure 4.4: Modeled Tank Pressure with Measured Tank Pressure

the ball pressure by 1.25 psi. Equation (4.1) predicts a change of 1.20psi for the same ΔP_t.

The ball pressure model specified by equation (4.4) follows the robot's response reasonably well, plotted in Figure 4.3. However, the model lags the response when the compressor is enabled but leads when the solenoid is open. This error could result from slightly changing model constants between the robot's disassembly and reassembly for parameter measurement and tracking tests. Additionally, the percent error of the models when the solenoid or compressor is active takes on a unique curvature until they are shut off. This could arise from the assumption that the solenoid and compressor were approximated with an exponential law. As more air is added to the tanks, this would help or hinder the solenoid and compressor, respectively, as either would be aided by or work against the stored potential in the tanks. This phenomenon may not be wholly captured by (4.2), but for our system, these approximations work reasonably well.

4.4 Design and Maintenance of Ring and Hubcap

The leaking models presented in this chapter are heavily dependent on the measured leak rates. These rates are heavily reliant on the health of the numerous sealing surfaces on the robot's shell. Over the three years of testing with the robot's hardware, these surfaces began to wear down. Since the overall ability of the robot to maintain pressure is dependent on every single bolt along the hubcap, I thought it relevant to review some modes of failure in this section. I will also review methods employed to prolong the life of the parts.

4.4.1 Hubcap Counterbores

The initial designs of the hubcap called for 6-32 socket head sealing screws. The screws were a custom and expensive part due to a custom groove and O-ring on the underside of the bolt head. These sealed well and remained the standard for nearly a year and a half on the first generation of hubcaps. However, over time, the geometry of the O-ring groove in the stainless steel bolts ate away at the soft aluminum hubcap's counterbores, see Figure 4.5.

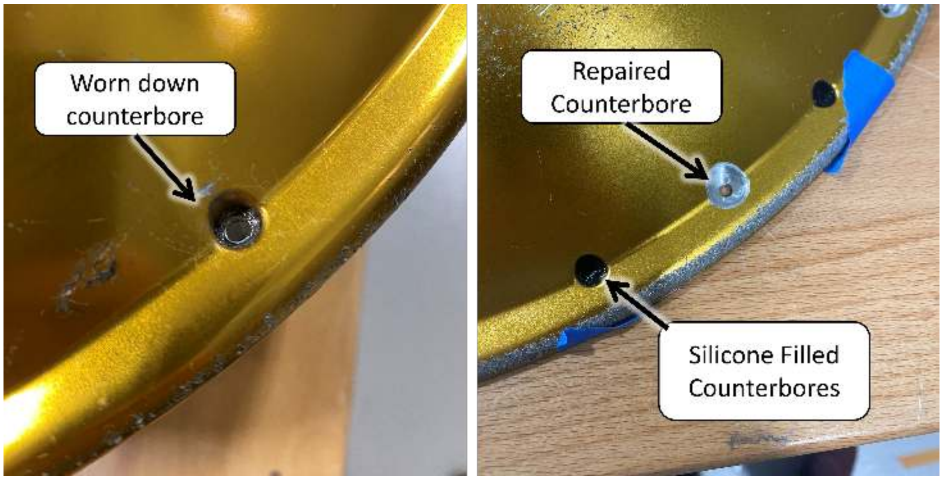


Figure 4.5: Picture of the worn-down counterbore, note the lack of gold anodization

Eventually, one of them wore thin enough to blow through completely, and the hubcap could no longer seal. To remedy this and prevent it from happening to the other hubcaps, new, wider counterbores were added to allow for the use of sealing washers. These were flat gaskets used in tandem with standard 6-32 washers to seal the hubs. The original hubcaps were repaired with silicone on the old counterbores.

4.4.2 Ring Threads

The removal of sixteen screws during robot assembly eventually wore down the threads. Sometimes the holes would be stripped clean or torn threads would cause the bolt to catch and shear. In both cases, the inner clamping ring would need to be removed and repaired or replaced. The lead time and cost of an individual ring lent itself to repair first. The two primary methods for repairing broken threads are helicoils and keyed inserts.

The unsatisfactory performance of the thread repair kit prompted us to redesign the ring. Recall that the only reason the ring is a solid piece is to clamp the yoga ball into place for spraying, Figure 3.5B. By moving to the molded shell, which could hold its shape on its own, it is not entirely necessary to have a complete ring. Instead, the threads are made easier to repair by splitting the inner ring into eight equal parts. Then, when a threaded hole eventually fails, only a section of the ring would need to be replaced. This since the segments are all inside the ball, and the segment seams do not pass across the pressure seal to increase leaking.

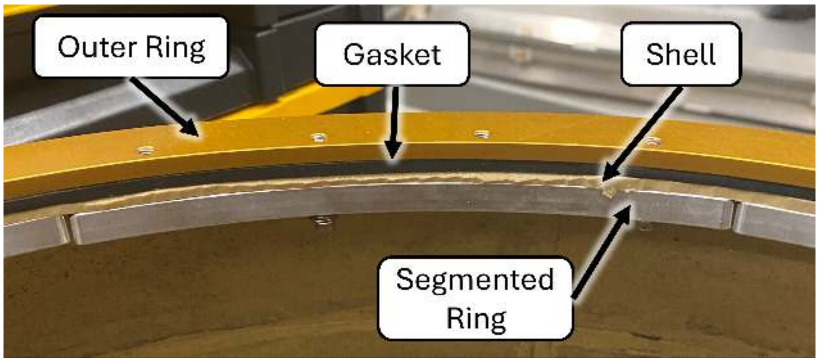


Figure 4.6: Split Ring Design

5. RoboBall II Mobility Baselines

With the robot now equipped with stable hardware, a series of capability studies was conducted to get a baseline understanding of its typical movement in various environments. This chapter will discuss three elementary maneuvers: the ball rolling through water, its fundamental ability to climb slopes, and its natural rolling behavior when moving with a restricted pendulum.

5.1 Water Mobility

The motivation for conducting water mobility tests was to establish baseline aquatic locomotion performance. RoboBall is pressurized, so it needs no additional modifications to float. The natural finish roughness of the molded bedliner provides it with just enough traction against the water to move. Previous studies on spherical robots included limited water performance testing. Arif et al inclded a single test run of their prototype to compare rotational speed against forward sailing speed [24], but included no comments on the limits of this locomotion mode outside of a single test.

This section outlines an experiment of RoboBall in the fording feature at the RELLIS IPG-MCC testing facility. It will describe the experimental methods and video-based data collection techniques to characterize the relationship between the robot's rolling velocity and maximum forward velocity. The results will be presented with a discussion on the primary mechanism that limit forward speed in water.

5.1.1 Methods

A picture of the fording feature is shown in Figure 5.1. The feature is a rectangular pond with entrance and exit ramps designed for vehicles. The water level in the feature was adjusted to ensure that RoboBall would be free-floating along its length with a water depth deep enough to prevent any ground contact.

Figure 5.1: Overview of IPG fording feature

A camera was positioned in the water to capture video of the robot as it traversed north to south and south to north, left of Figure 5.2. The 2ft-wide concrete blocks on the side of the trench were used in conjunction with the video timestamps to estimate the linear speed of the robot, as shown in Figure 5.2. The rotational speed of the shell is measured with the onboard NEO motor encoders and averaged during the entire testing run. The robot was tested at four commanded rotational speeds (up to 1000 deg/s, battery-limited), with each trial conducted in both directions relative to the prevailing wind. A control case with no rotation was included. The wind from the testing day was from SE at 13.5 mph (119.8 fps), estimated from nearby airport data. The robot was allowed to drift with the wind with no rotational speed to establish a control point.

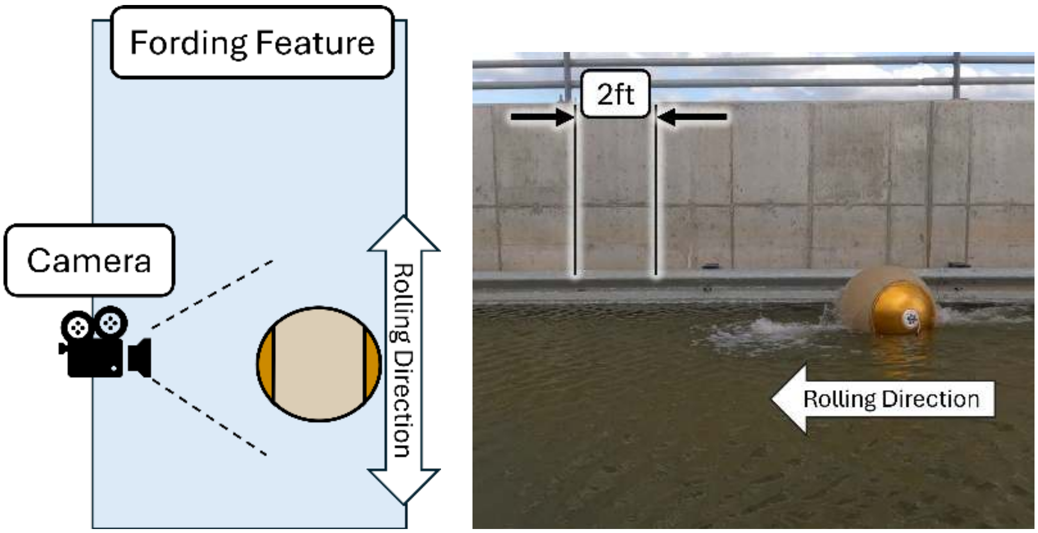


Figure 5.2: Schematic of Testing Setup for Aqueous Mobility

5.1.2 Water Mobility Results

The linear speed and rotational speeds and their directions are shown in Figure 5.3. All testing runs with the wind were faster than the control drifting speed so the robot was able to move forward under its power and not only be pushed by the wind. The large gap between tests 3 and 4 could be due to a momentary sustained wind gust.

From the figure, the max forward sailing speed is about 2.5-3fps (1.7-2mph) at 10 rad/s; however, the two tests at higher speeds have significantly reduced forward speeds. The fording feature is below ground level, and both directions are reasonably close in speed, so wind is not a major factor in the forward speed reduction past 10 rad/s. Review of video footage revealed another potential culprit: at around 10 rad/s, the robot picks up water from behind and throws it forward. This hydrodynamic phenomenon is often referred to as a "rooster tail". Figure 5.4 compares video stills of tests with the wind direction. From the stills, once the rooster tail breaks over the top of the robot, its forward speed is reduced. Once this happens, the robot starts throwing mass in the wrong direction, which results in lost efficiency and speed.

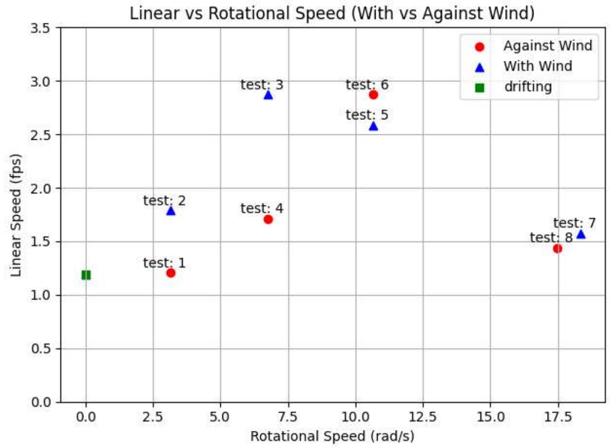


Figure 5.3: Relationship between the rotational and forward velocities of the robot in water with a stock shell

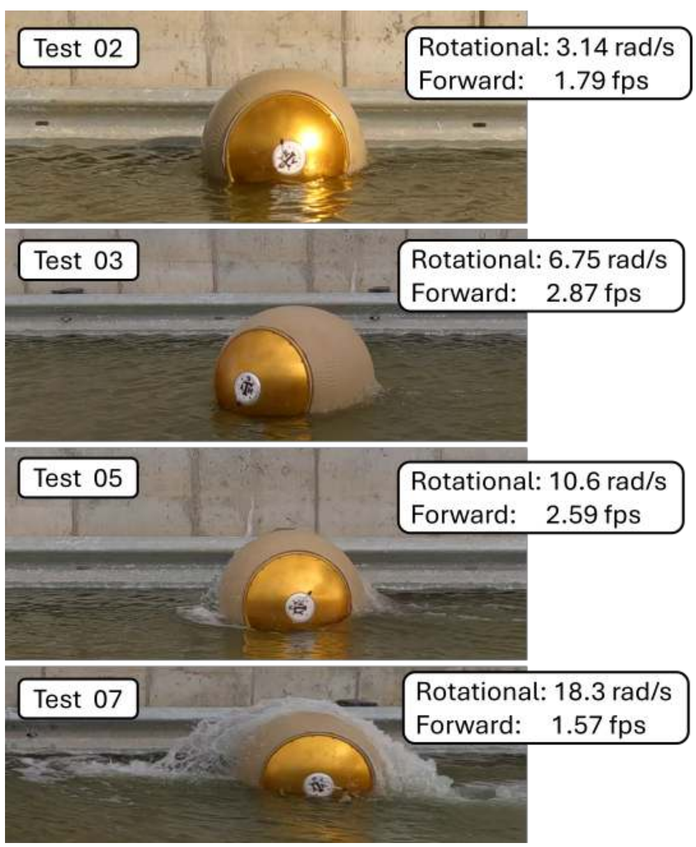


Figure 5.4: Rooster tails produced during tests against the wind.

5.2 Slope Climbing Ability

Although RoboBall was primarily designed for crater descent, its limited ability to ascend slopes was also investigated. Spherical robots with pendulums have been known to possess this ability, and a variety of literature exists that derives relationships between the robot's design parameters. However, they are often presented without empirical evidence to support them. This section compares two parametric models: one from the RoboBall I paper by Bridgewater [1], and another from Roozegar et al. [25].

5.2.1 Description of Parametric Models

Bridgewater proposed an initial model for predicting slope climbing ability in his paper, repeated below as equation 5.1. There, r_p is the distance to the robot's center of gravity, R is the nominal outer radius of the shell, θ is the angle the pendulum makes with the ground, and α is the slope incline.

$$\frac{r_p}{R}\sin(\theta) > \sin(\alpha) \tag{5.1}$$

Roozegar's relationship differs in an additional offset that the drive angle climbing on a ramp will produce, α. Bridgewater only accounts for the slope to climb straight from the ground. The relationship is shown below in our terminology.

$$\frac{r_p}{R}\sin(\theta + \alpha) > \sin(\alpha) \tag{5.2}$$

To isolate the independent variable, recast equations 5.1 and 5.2 into equations 5.3 and 5.4, respectively.

$$\theta_{brid} = \sin^{-1}\left(\frac{R}{r_p}\sin(\alpha)\right) \tag{5.3}$$

$$\theta_{rooz} = \sin^{-1}\left(\frac{R}{r_p}\sin(\alpha)\right) - \alpha \tag{5.4}$$

A diagram illustrating the different parameters is shown in Figure 5.5.

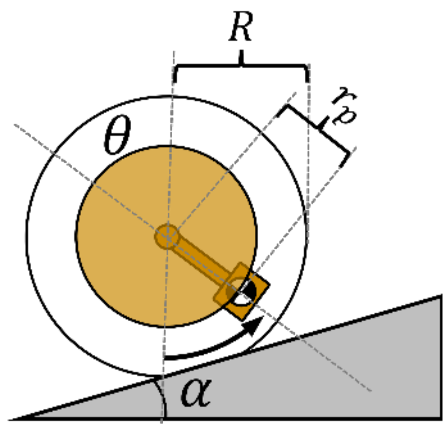


Figure 5.5: Diagram Relating the Physical Parameters for RoboBall's slope climbing ability

5.2.2 Slope Climbing Experiments

To study the relationship between R/r_p, we constructed a wooden ramp resting on stacked weights. The ramp's angle, α, varied by stacking plate weights at one end. Two foam mats at the start of the ramp ensured a smooth transition between flat and sloped ground. A complete setup with four stacked plates is shown in Figure 5.6.

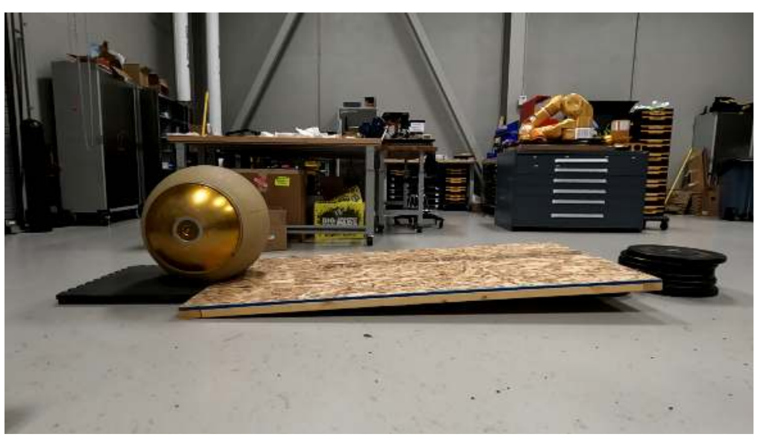


Figure 5.6: Ramp Testing Experimental Setup

To vary the R/r_p ratio, two copies of the robot's bottom plate were constructed (often referred to as a "ballast")—one of aluminum and the second of solid copper. Copper is denser than steel, but less toxic than lead. The ballasts are shown side by side in Figure 5.7.

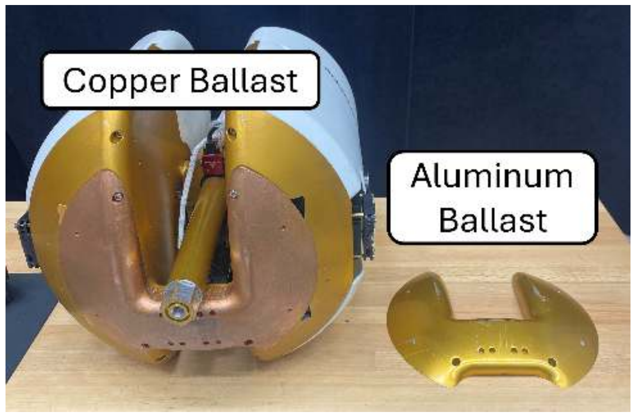


Figure 5.7: RoboBall with the copper ballast installed

To climb the ramp, a PI control law on the drive speed gradually increased the pendulum angle, θ. The pendulum angle at which the robot initiated ramp ascent was recorded as θ, and the ramp angle increased. The test concluded once the pendulum flipped over and backwards at its highest slope climb angle.

5.2.3 Comparing Slope Climbing Models

Plotting equations 5.3 and 5.4 with appropriate parameters for R and r_p, along with collected data from the copper ballast yield Figure 5.8. Both models initially poorly fit the data; however, Roozegar was much closer in slope. The experimental data appear to have an intercept not at the origin, contrary to what the experimental models predict. Intuitively, the robot generates rolling torque by lifting its pendulum. If it initially does not roll, then it can lift further. RoboBall is equipped with a soft outer shell, which provides some rolling resistance before the pendulum moves the robot.

This aspect is incorporated into our slope model by adding an arbitrary offset term, δ, to Roozegar's model, as shown in Equation 5.5 and plotted in Figure 5.8, which yields a good fit. Note that a smaller δ makes it easier for the robot to climb a slope.

$$\theta_{offset} = \sin^{-1}\left(\frac{R}{r_p}\sin(\alpha)\right) + \alpha + \delta \tag{5.5}$$

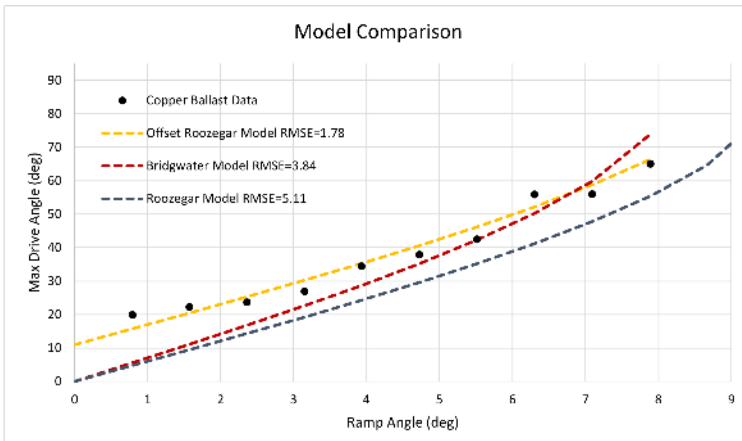


Figure 5.8: Model comparisons between experimental data

5.2.4 Slope Climbing Results

Equation 5.5 with experimental data from the aluminum and copper ballasts is shown in Figure 5.9. The parameters in Table 5.1 were arbitrarily chosen in excel for a good fit.

Table 5.1: Slope Climbing Experiment Parameters

Parameter	Value [Units]
$(R/r_p)_{Cu}$	7 [-]
$(R/r_p)_{Al}$	9 [-]
δ_{Cu}	11 [rad]
δ_{Al}	20 [rad]

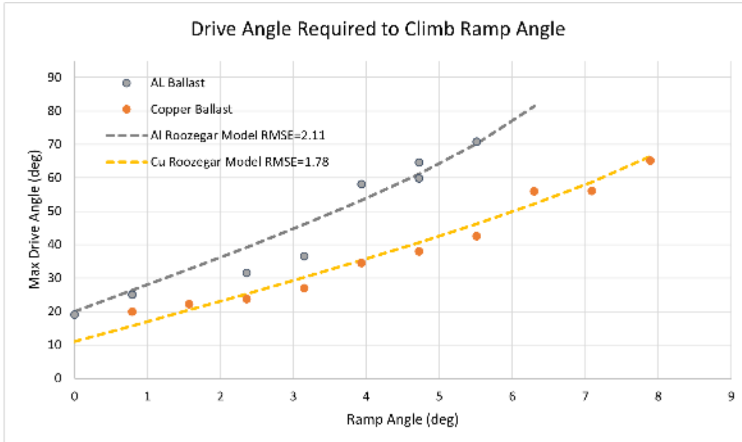


Figure 5.9: Testing Results with model using both aluminum and copper ballasts

The need for this offset suggests that shell compliance plays a non-negligible role in slope climbing dynamics, an effect absent from existing models. Now, to improve slope climbing ability, δ must also be minimized along with the R/r_p ratio. R. Jangale used the methods developed in this section to show RoboBall III's improved slope climbing performance by scaling and improving R/r_p in [26] along with a larger shell based on the techniques in Chapter 3.

5.3 Rolling with a Locked Pendulum

Building a dynamic model for a spherical robot typically involves one of two approaches: decoupling the drive and steering axes into planar cases or employing a complete three-dimensional modeling scheme that is subsequently simplified. This section outlines the motivation for and experiments to determine whether the decoupling approach is sufficient or if there is a correlation between the two directions.

The decoupled dynamics approach is typically implemented for special cases or to facilitate control design. In the previous section, Roozegar used a decoupled model to study the robot's performance on a slope [25]. Modeling off-axis effects was unnecessary since the robot was only rolling in a straight line up the slope, so it was sufficient for this scenario. Wang et al also used this approach to design a similar LQR-style controller that was presented earlier in Chapter 2 [27]. The controller stabilized the robot at lower speeds, but they cited "coupling effects" between the drive and steer directions as the reason their scheme broke down at higher speeds. Mao and Arif [28, 24] also acknowledged this coupling effect. While they did not try to model the system dynamically, they successfully designed a flywheel about the drive axis to counteract its effects.

Several articles also attempt to study the full dynamics of the system, yielding conflicting results. Schroll worked the EOM symbolically without simplifications using MATLAB's symbolic toolbox [14]. He was unable to instrument his prototype but qualitatively verified the symbolic model by constructing a 3D visualizer and replicating observed behaviors on a constructed prototype. Schroll assigned the term "nutation instability" to the robot, which spontaneously oscillated in its steering axis while attempting to drive straight. Eventually, at higher speeds, the nutations would grow until the system controller could no longer counteract them. Ivanova also attempted to study the 3D dynamics of a pendulum-driven robot; however, they stated that the robot exhibited oscillations that were not present in the numerical simulations [29]. Li also presented a 3D dynamics model, but it was adjacent to their hopping design [30]. Li acknowledged a wobbling behavior in the model but did not study it further. Singhal focused on a dynamic analysis of the uncontrolled system's wobble, and claimed the undesirable wobbling amplitude in the driveshaft will decrease at increased speeds (Figure 9 in [31]). However, they presented no data from the prototype to back up the claim. This is contrary to Schroll's claim, where the nutations would eventually overcome the robot.

The cause of these coupled effects has yet to be seen in any serious empirical study. To fully unlock the potential of spherical pendulum-driven mobility platforms, investigating the fundamental physics of these wobbles is needed before considering ways to control the system. The following section presents an experiment to determine whether a natural coupling effect exists that induces nutations when rolling straight, as claimed by Schroll, or if they eventually die out, as theorized by Singhal.

5.3.1 Locked Steering Tests

Singhal and Schroll disagreed on whether the amplitude of the robot's nutation was proportional to its speed. To test this, the steering axis of the pendulum was clamped, shown in Figure 5.10. This clamp physically locks out any steering motion from the pendulum, keeping it perpendicular to the drive axis in the straight-ahead steering angle and simplifying the dynamics.

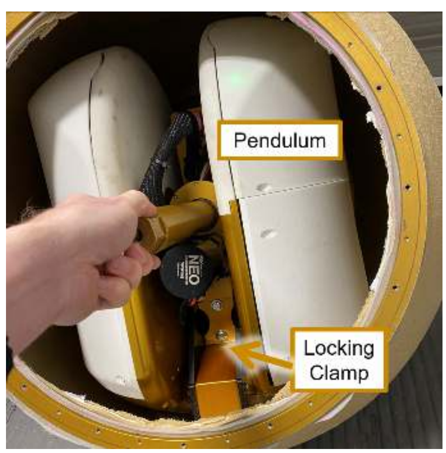


Figure 5.10: Image of a pendulum in a shell with a clamp applied to the steering axis

The robot was then inflated and placed on a flat concrete pad. The drive axis was ramped to a set speed and held there, shown in Figure 5.11. The pipe angle was measured, and an offline analysis was conducted to identify the peaks in the response. These peaks are marked with a blue "X" in the figure. The average frequency and magnitude for the steady-state period, as indicated by the red box, were calculated and recorded. These tests were conducted at various speeds with robot pressures of approximately 2.5 psi. A collection of the average amplitudes and frequencies at set speeds is shown in Figures 5.12A and 5.12B, respectively.

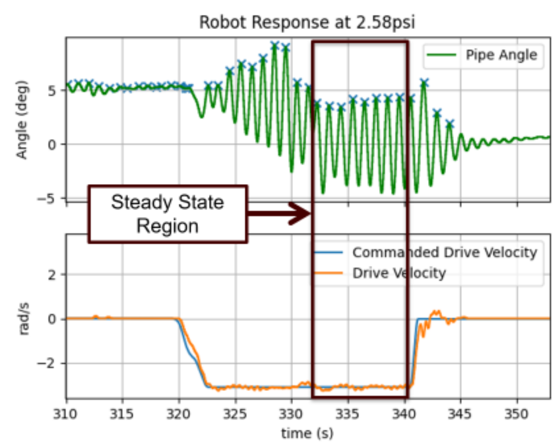
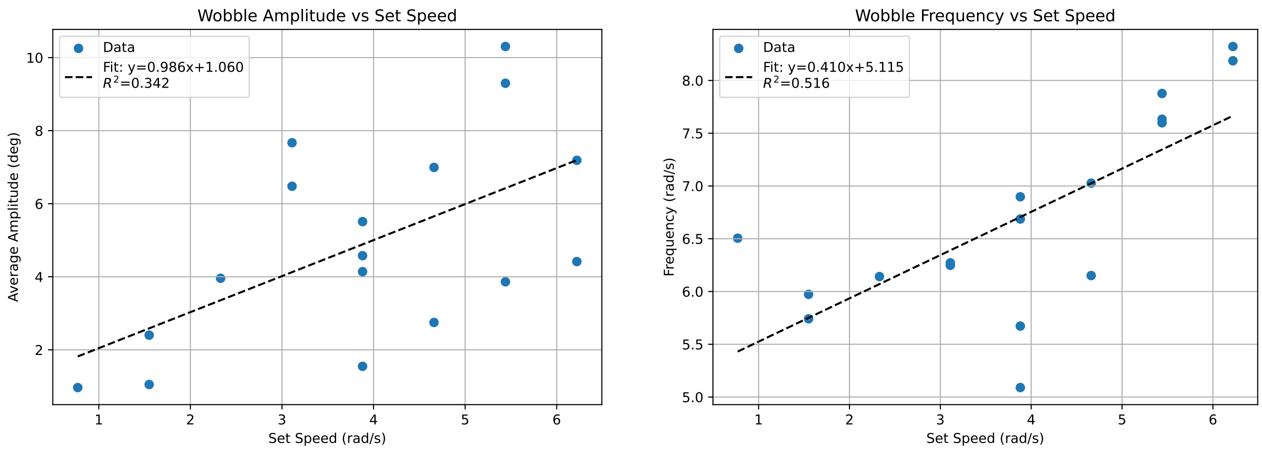


Figure 5.11: Single test with locked steering at a target speed of 3.11rad/s

The oscillatory behavior shown in the pipe angle for the locked steering tests is referred to as wobble. This motion spontaneously occurs even when the steering is locked, indicating some coupling between the dynamics, rather than the control laws for the drive and steering axes. According to Figure 5.12A, as the commanded speed increased, so did the average wobble amplitude. This is reasonable to expect as more energy is injected into the system through the motors and batteries. This is in direct opposition to what Singhal reported.



A B

Figure 5.12: Average Wobbling Amplitude and Frequency of multiple locked steering tests at 2.5psi

In addition, for speeds less than a critical speed. The amplitude of each test was bounded. However, after reaching speeds of more than 6 rad/s, the amplitude of the pipe angle would increase unbounded, even at constant speed, as shown in 5.13 before the robot must be slowed to preserve the hardware. If allowed to persist, the instability would result in a catastrophic end-over-end (hubcap-to-hubcap) tumbling motion. This effect is the *nutation instability* that Schroll described.

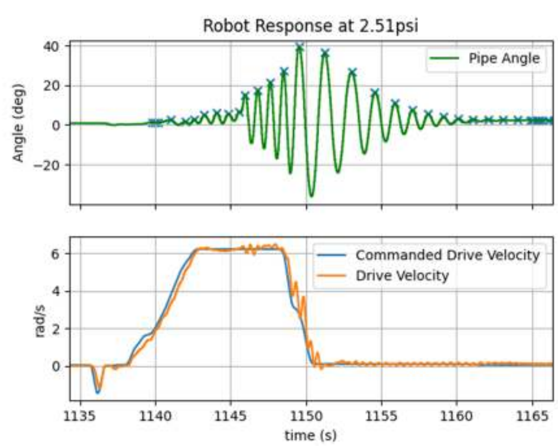


Figure 5.13: Robot experiencing a nutation instability at a target speed of 6.22rad/s

5.3.2 Locked Steering Results

Figures 5.12A and B show a correlation between the driven speed of the robot and a corresponding response in the wobbling pipe angle amplitude. This is the effect that Schroll discussed and was observed in [29], [30], and Roboball, but not currently captured in the system dynamic models.

5.4 Chapter Conclusions

This chapter addressed the next step in RoboBall's development after a stable hardware state was reached. Baseline studies of the robot's motion in various mobility environments were presented and empirically studied, with appropriate modifications made to the robot or the environment for each experiment. Each scenario presented its own unique limiting factor in the robot's performance. These limiting factors include the rooster tail effect in water, the mass ratio on slopes, and the coupling force between the drive and steering states when driving on flat ground.

6. Numerical Modeling of the Spherical Robot in Drake

6.1 Introduction

Spherical robots provide a unique exploration platform, enclosing avionics within a protective outer shell that shields against debris and environmental hazards. Multiple highly functioning prototypes have been developed and studied over the years. Ren et al. developed a system that operates in various environments using a custom sensor array [32]. RoboBall and Moball are two other prototypes that rely on soft outer shells. RoboBall uses a custom pneumatic spherical shell propelled by a pendulum [26]. Moball deployed a sewn outer frame stretched between spars; it relied on wind to drive it [33].

A large amount of literature is devoted to describing the complete dynamics of spherical robots. Most spherical robots are modeled from first principles [13], but these approaches are often limited by the complexity of the full analytical equations [14]. Researchers have invested time in exploring different modeling strategies to avoid explicitly stating the entire system dynamics, either by exploiting Lie bracket symmetries [34] or system identification methods [35]. Montenegro et al. used a Coppelia simulator to study the dynamics and prototype various machine learning controllers numerically [36]. While the rigid body components are an essential foundation to modeling, it is often the non-conservative forces that arise through actuation friction and contact that affect a modeling scheme the most. Few studies incorporate these forces directly; instead, they simplify with a point-contact assumption.

The effect of a soft outer shell on a robot has a significant impact on its dynamics. A soft outer shell of the robot can be used to absorb impacts when rolling across unknown terrain, allowing for a smoother ride [12]. Dravid studied a similar robot and found that inflation pressure has a noticeable effect on the overall power output of the system, but did not give an explicit model of its exact effects [37]. Other researchers have designed specialized compensation terms to account for any soft shell effects, but do not explicitly attempt to model the outer shells' rolling dynamics

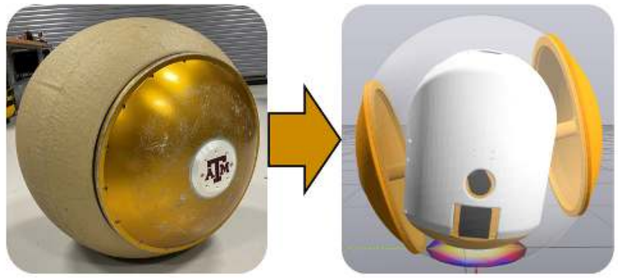


Figure 6.1: A soft spherical robot and respective experiment using Drake's contact model

[10]. In developing a modeling scheme, the impacts of the soft outer shell cannot be ignored.

Featherstone published a landmark work on methods for numerically acquiring a system's rigid body dynamics through a set of algorithms [5]. Such techniques have been incorporated into many popular robotics toolboxes. Most share the common feature of parsing the rigid body dynamics from a supplied URDF file, allowing researchers to study the numerical response of the robot's equation of motion. Drake and SOFA are two such implementations of Featherstone's work, yet they include additional functionality to model soft contact dynamics beyond rigid body properties. Drake's hydroelastic contact model uses pressure fields to approximate the contact area and resulting forces [6]. SOFA is geared toward modeling the whole bodies of soft robots of all shapes through Finite Element methods [38]. Since the outer shell is typically the only compliant component, its geometry makes Drake's lightweight hydroelastic model well-suited compared to full finite-element approaches. A side-by-side comparison of our physical robot and its digital twin in Drake is shown in Figure 6.1.

This chapter outlines a systematic modeling approach for RoboBall, combining rigid-body dynamics from Drake with friction and soft-shell contact models. The robot's rigid body dynamics will be numerically generated using Drake's implementation of Featherstone's rigid body dynamics algorithms through a custom URDF file for a spherical robot. Methods for characterizing the frictional losses in the two actuated degrees of freedom of the robot will be shown for model completeness. Three different models for the soft outer shell will be introduced, one based on Drake's native hydroelastic contact model, another for linear injected dynamics, and a regular point contact case for control. These models will be tuned against empirical data to match the dynamic frequencies and damping ratios in the steering direction. A simple parameter sensitivity study for the hydroelastic parameters will also be introduced. Finally, the tuned models will be tested in a dynamic experiment of the robot rolling down a ramp to assess the models' ability to follow the driving direction.

6.2 Modeling Approach

Based on work introduced in Featherstone [5], the forward dynamics of a rigid body system are given in the form of equation (6.1), Here, τ represents applied joint torques, $H(q)$ is the configuration-dependent inertia matrix, and $C(q, q, f^x)$ accounts for Coriolis, damping, and external forces including gravity. These equations are typically found and integrated by hand, or Drake can generate the numerical system by parsing a properly designed URDF.

$$\tau = H(q)\ddot{q} + C(q, \dot{q}, f^x) \tag{6.1}$$

Figure 6.2 presents a flowchart for our modeling framework. In the figure, a rigid body dynamic model is connected to joint friction losses and external forces. The friction losses are in series with the control effort on the actuated joints, and additional dynamics from the soft shell models are implemented through external spatial forces, f^x. The simulated states are then fed to a graph, data file, or Drake's Meshcat visualizer for analysis or comparison to experimental data.

6.2.1 URDF Model

Drake uses a URDF as a starting point for deriving the system dynamics. Obtaining an accurate file is trivial for common robots thanks to a recent database compiled by Tola and Corke [39]. Their database includes many robot arms, dogs, and humanoids; however, it lacks a URDF for any spherical robot, so a bespoke one was developed. The URDF's fidelity directly determines the accuracy of the dynamic model.

A SolidWorks assembly already existed for designing and manufacturing parts for our hardware testbed. To create an accurate URDF, the mass values of all individual components were updated,

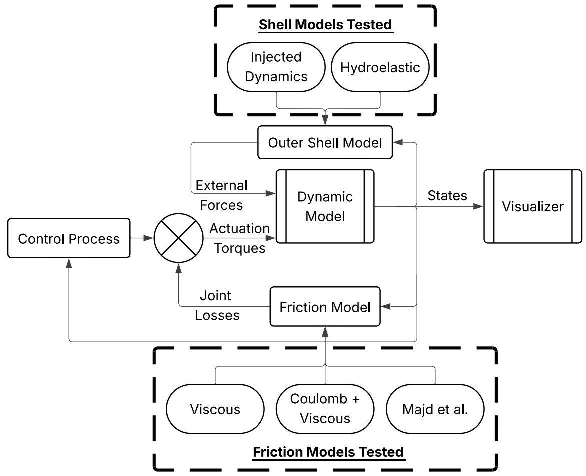


Figure 6.2: Flowchart of the modular Drake modeling framework, showing integration of rigid-body dynamics with friction losses and soft-shell contact models.

and corresponding colors were applied. With this information, the SolidWorks' Mass Analyzer tool was used to approximate the inertia and the center of mass of the various links based on the components' relative positions and masses. This process also revealed slight off-axis inertial imbalances, which are naturally captured in a URDF-based model. The export of frames, masses, and meshes was handled with the Solidworks2URDF ROS tool; a selection of these values can be found in Table 6.2 at the end of this document.

The robot consists of three key parts: the Pendulum, the Pitch Center, and the outer shell. The outer shell consists of both soft and rigid components. The Bedliner link represents the soft outer shell and primary soft contact point between the robot and the world. The Pipe Assembly contains the robot's driveshaft and connects the bedliner to the rest of the robot. The Pitch Center and Pendulum links make up the 2-degree-of-freedom pendulum, the primary control mechanism. These links can be visualized in the kinematic model on the left of Figure 6.3.

The exported URDF consists of a two-branch floating body structure. The Pitch Center is the kinematic tree's root and sits between the pendulum and two outer shell links. A 6-DOF floating joint connects it to the simulation inertial or world frame. The shell is split into the Pipe Assembly and Bedliner links, which are fixed together. Drake-specific tags are later applied to the Bedliner collision properties to register it as a soft object. This topology is shown on the right of Figure 6.3, next to the kinematic visual model.

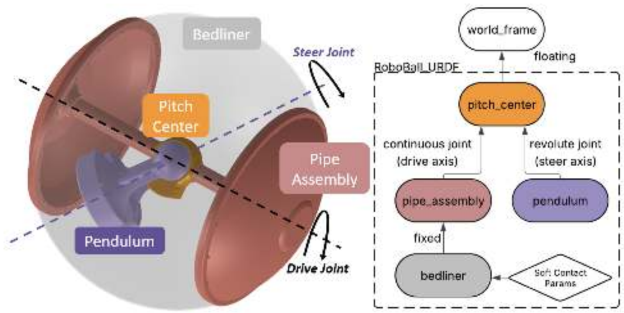


Figure 6.3: URDF kinematic structure of RoboBall, showing pendulum, pitch center, and shell components.

State Names: To validate the proposed models in the following sections, measured robot state data will be compared against simulated data. These states are illustrated in Figure 6.4 and will be called the pendulum angle, θ, and the pipe angle, ϕ, respectively. The pendulum angle is the angle between the pendulum and driveshaft in the steer direction or the revolute joint (steer axis) value in the URDF. The pipe angle refers to the angle the pipe assembly forms with the ground. This is found from the roll angle in an Euler angle interpretation of the base floating joint. τ_s is the net torque acting on the steer axis; this is the same Actuation Torque called out in Figure 6.2. The symbol τ_{app} represents the additional contact forces a spherical soft shell has on the outer Bedliner and Pipe Assembly links. The drive velocity of the robot is the angular speed of the pipe assembly about its $\hat{x}$ axis, denoted by ω_d.

6.2.2 Joint Friction Models

URDF format only natively supports proportional damping. However, more detailed models can be implemented through Drake's LeafSystem abstraction layer. This abstraction is shown in Figure 6.2, where the dynamics are wired into the system through input torques at the simulated actuators (τ_s). This will mimic frictional losses in the gear reductions on the robot's motors.

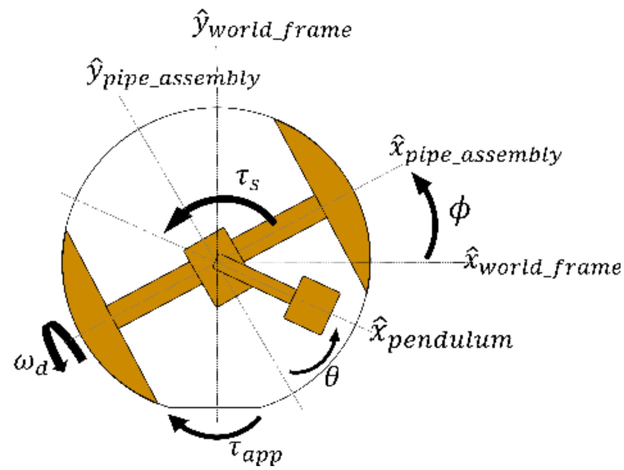


Figure 6.4: Robot State Diagram

Three different models were implemented and compared against experimental data from the robot. All three models follow a structure that takes in the joint's velocity, v, and yields some resisting torque as an output, $\tau_{friction}$. Equation (6.2) represents standard viscous damping, extended with Coulomb friction terms. The static and dynamic coulomb parameters μ_s, μ_d with a viscous coefficient c. The outcome of this is shown in Figure 6.7 as the Viscous. Tuning μ_s, μ_d to non-zero values is the Coulomb+Viscous model.

$$\tau_{friction} = \begin{cases} \mu_s \text{sign}(v), & \text{if } v < tol \\ -\mu_d \text{sign}(v) - cv & \text{otherwise} \end{cases} \tag{6.2}$$

The third model was derived by Majd et al. to handle stiction in servos and is presented at equation (6) from [40] but repeated in (6.3) below. This model includes additional terms for tuning the modeled response, which are $[f_\omega, f_c, \sigma, \omega_c, n]$. All three models were tuned to the steer response of the robot with a 15:1 reduction, and the output torques are shown in Figure 6.5.

To collect data to tune the friction models, our physical robot was placed in a testing stand that locked its outer rolling mode in the steer direction. This isolated the inner steer joint of the pendulum to study the decaying effect of the pendulum's swing due to the gearing's friction. Figure 6.6 shows the pendulum in the test stand and its rendering in Drake's Meshcat.

$$\tau_{friction} = f_\omega v[f_c + \sigma e^{-|v/\omega_c|} - (\sigma + f_c)e^{-|nv/\omega_c|}]\text{sign}(v) \quad (6.3)$$

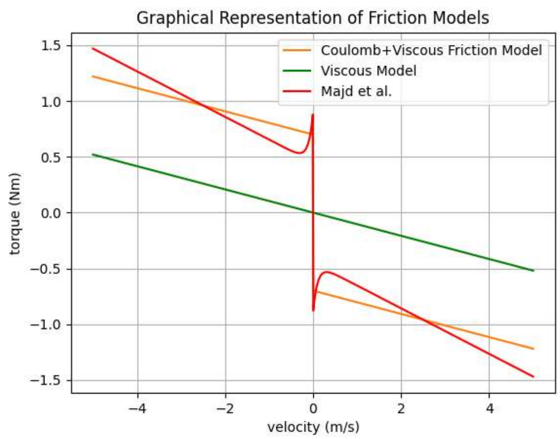


Figure 6.5: Comparison of modeled torque responses from viscous, Coulomb+viscous, and Majd friction models.

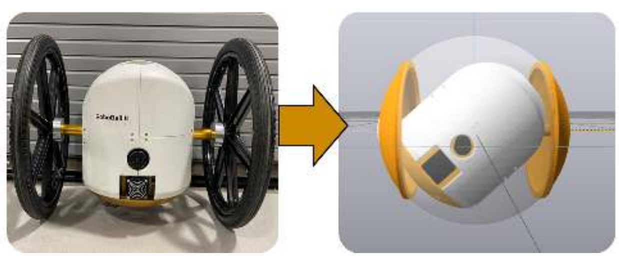
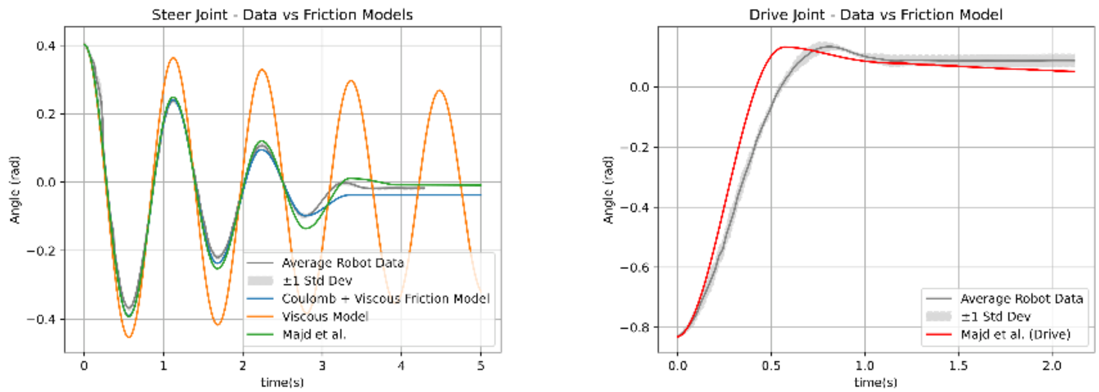


Figure 6.6: Experimental setup isolating the steer and drive joints, alongside its digital twin in MeshCat.

Of the tested models, Majd's model fit the steering direction best, so it was the only model fitted in the drive direction with a 33:1 reduction. The resulting friction models were then overlaid on the robot's data and shown in Figures 6.7A and 6.7B.



A B

Figure 6.7: Experimental data of the robot's actuation axes overlaid with model output with different friction models

This section outlined the modeling efforts to capture the dynamic effects of the joint losses of the robot. In the final tests of the chapter, where the robot rolls down a ramp, both axes will be free to move under these dissipative effects as defined by Majd. The parameters for each tuned model are given in Table 6.1.

Table 6.1: Parameters for Joint Friction Models (steer and drive axes).

Model	Parameters	Values
Viscous	c	0.104
Viscous + Coulomb	μ_s, μ_d, c	$[0.7,\ 0.65,\ 0.104]$
Majd - Steer	$f_w, f_c, \sigma, \omega_c, n$	$[0.20, 0.45, 10, 0.1, 1]$
Majd - Drive	$f_w, f_c, \sigma, \omega_c, n$	$[1.78, 1.24, 8.33, 1.36, 1.60]$

6.3 Outer Shell Contact Models

The next non-conservative aspect of the robot is its outer shell. This section details an experiment to isolate the shell dynamics from the internal pendulum using a physical clamp. Then, a detailed model that injects additional stiffness and dissipation terms as external forces to model the shell will be presented with our implementation of Drake's hydroelastic model. The models will be compared against a typical point contact assumption as a control model.

6.3.1 Experimental Shell Dynamics in Steering Direction

To initially estimate the effects of the robot, a clamp was designed to lock the steering axis of the shell, shown in Figure 6.8.

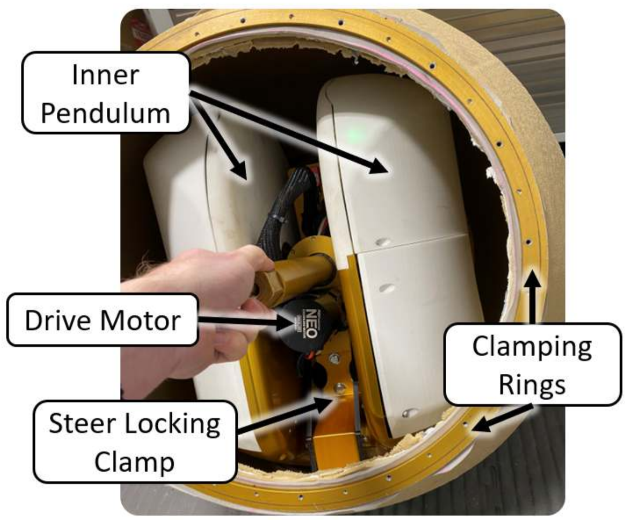


Figure 6.8: Reinforced 3D printed clamp to lock the steering.

The clamp eliminates pendulum motion, reducing the system to equivalent half-cylinder rolling dynamics. Dynamically, this constrains the steering motion of the pendulum relative to the pipe assembly such that $\theta = 0$ in Figure 6.4. The linearized equation is stated below and can be derived using typical Lagrangian methods with appropriate transforms to represent all dynamic link masses and inertia in the Pipe Assembly frame.

$$M_{eq}\ddot{\phi} + K_{eq}\phi = \tau_{app} \tag{6.4}$$

Where $m_i, r_i,$ and I_i are the mass, center of mass position, and inertia of the Pendulum, Pitch Center, and Pipe Assembly Links, respectively ($i = p,\ pc,\ s$). R is the nominal outer radius.

$$M_{eq} = (I_{eq} - 2(m_{pc}r_{pc} + m_p r_p)R)$$

$$I_{eq} = I_s + (I_{pc} + m_{pc}r_{pc}^2) + (I_p + m_p r_p^2) + m_{total}R^2$$

$$K_{eq} = (m_{pc}r_{pc} + m_p r_p)g$$

Equation (6.4) is a simple harmonic oscillator in the rolling coordinate ϕ. To investigate how well the equation models with locked steering, the robot was manually lifted to an offset angle and released. This process was repeated at various pressures, and the resulting traces were analyzed to obtain the damping ratios and natural frequencies in the steering direction. The resulting values are plotted in Figure 6.9 along with appropriate exponential and linear regressions for the damping ratio ζ and damped frequency ω_n. If the system were accurately represented by (6.4), then the system's frequency should be given by equation (6.5)

$$\omega_n^2 = \frac{K_{eq}}{M_{eq}} \tag{6.5}$$

This value is invariant to pressure, so it is shown at the bottom of Figure 6.9 as a dashed line. Numerical values from the URDF for each symbol are presented in Table 6.2 at the end of this document.

When the parsed dynamic model of the URDF is simulated in a point contact case, there are no additional contact forces applied to the rolling system ($\tau_{app} = 0$). In this case, equation (6.4) and a Drake model with locked steering under point contact are equivalent. The resulting frequency and ratios are shown in Figure 6.9, where the system aligns with the calculated frequency and there is no damping. This is not in line with the empirically measured results, indicating that the softness of the outer shell affects the dynamics in this steering direction.

6.3.2 Injected Contact Dynamics

This section will derive a model to inject additional dynamics into the model, accounting for discrepancies from the soft shell. Equation (6.4) is written in terms of the standard form of second-order ODEs. In the equation, both frequency, ($\omega_n(p)$), and damping ratio, ($\zeta(p)$), are functions of pressure, p. These relationships are derived from an exponential and linear regression of rolling data shown in the boxes of Figure 6.9.

$$\ddot{\phi} + 2\zeta(p)\omega_n(p)\dot{\phi} + \omega_n(p)^2\phi = \tau_{app} \tag{6.6}$$

By equating the coefficients of (6.4) and (6.6), we can recover relationships for additional stiffness and damping terms that can be injected into the system through applied forces to account for the dynamic effects of the soft shell as a function of pressure.

$$K_{inject}(p) = \omega_n(p)^2 M_{eq} - K_{eq} \tag{6.7}$$

$$C_{inject}(p) = 2M_{eq}\omega_n(p)\zeta(p) \tag{6.8}$$

underestimates the experimental damping values, possibly due to additional nonlinear dissipative effects from the inflatable soft shell that are not captured through the linear system incorporating pressure-dependent stiffness and damping.

$$\tau_{app} = -C_{inject}\dot{\phi} - K_{inject}\phi \tag{6.9}$$

The injected stiffness and damping move the system closer to the experimental values. The natural frequencies line up, but (6.8) underestimates the experimental damping values, possibly due to additional nonlinear dissipative effects from the inflatable soft shell that are not captured through the linear system identification.

6.3.3 Hydroelastic Contact

As an additional contact model, Drake's built-in hydroelastic contact model was implemented in this scenario. Initially developed to represent compliance in soft robotic grippers, existing tuning guidelines are primarily targeted at manipulators, and the model introduces compliance and dissipation only in the vertical direction. Consequently, the model will not contribute additional stiffness to the rolling dynamics. However, because the normal velocity flux of a sphere rotating through a cross-section is non-zero, the model can still capture some rolling-induced dissipation. The hydroelastic model requires two parameters: an elastic modulus and a dissipation coefficient. These two parameters are incorporated with the object's mesh and intersecting geometry to model the contact force between the two objects.

$$\tau_{app} = f(m_{hydro}, d_{hydro}, q) \tag{6.10}$$

The elastic modulus, m_{hydro}, represents the expected deformation (or contact "flat") of the robot and was estimated following Drake's recommended procedure, given in (6.11), where f_n is the normal force of the robot (its weight) and V is the estimated intersecting volume between the undeformed sphere and a flat surface. This contact volume is pressure-dependent and was derived from prior measurements in Chapter 2. For an initial guess of the dissipation, choose the coefficient as C_{inject} from (6.8). The results are shown in Figure 6.9, where using C_{inject} as the dissipation coefficient overestimates the damping in the hydroelastic case.

$$m_{hydro} = \frac{f_n}{V(p)} \tag{6.11}$$

$$d_{hydro} = C_{inject} \tag{6.12}$$

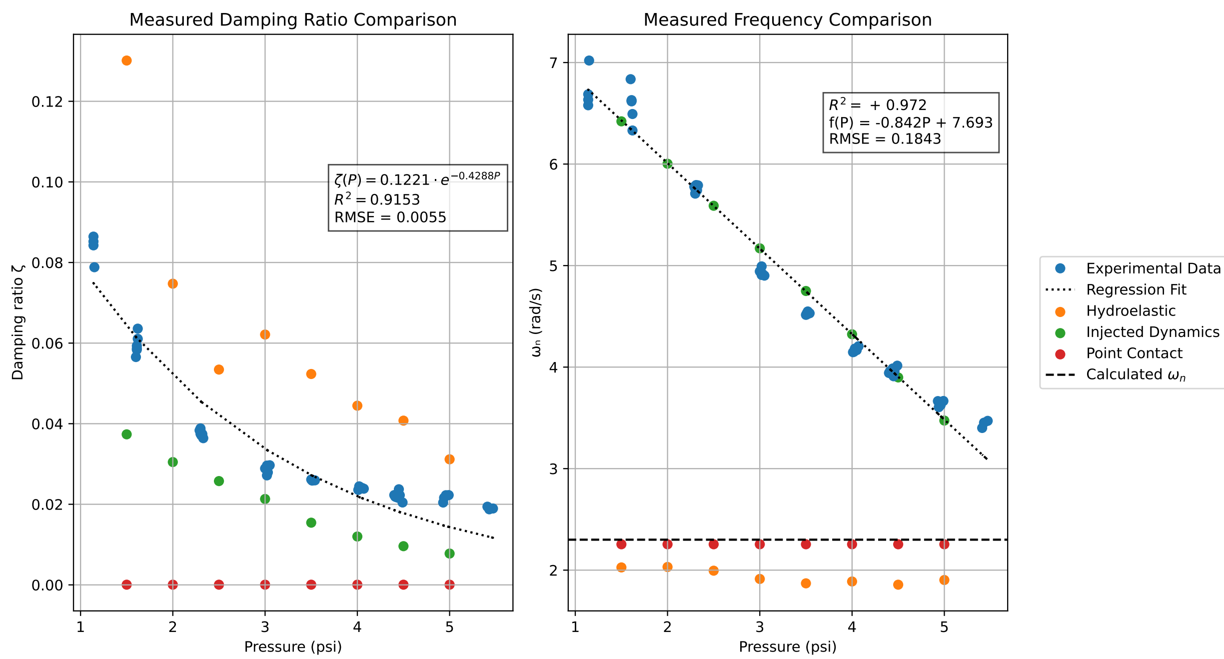


Figure 6.9: Comparison of measured damping ratio and frequency with point contact, injected dynamics, and hydroelastic models.

Through this dynamic tuning process, the shell has a significant impact on the overall system's dynamics. The model presented in (6.8) provides the best fit without additional modifications. Still, it only exists in the steering plane of the robot, whereas the hydroelastic model derives from mesh collisions and would be suitable for all rolling directions.

6.3.4 Hydroelastic Parameter Sensitivity

The values for the two hydroelastic parameters, the modulus and dissipation (m and d), were chosen to vary with pressure along the models presented in (6.11) and (6.8). This section outlines a parameter sensitivity study to determine the optimal approach for further parameter tuning.

The previously calculated values at each pressure serve as a control point, with the modulus or dissipation increased or decreased by 50% to observe the change in the simulated damping ratio. The resulting damping ratio response is shown in Figure 6.10 with associated percent changes in error in the context of the original.

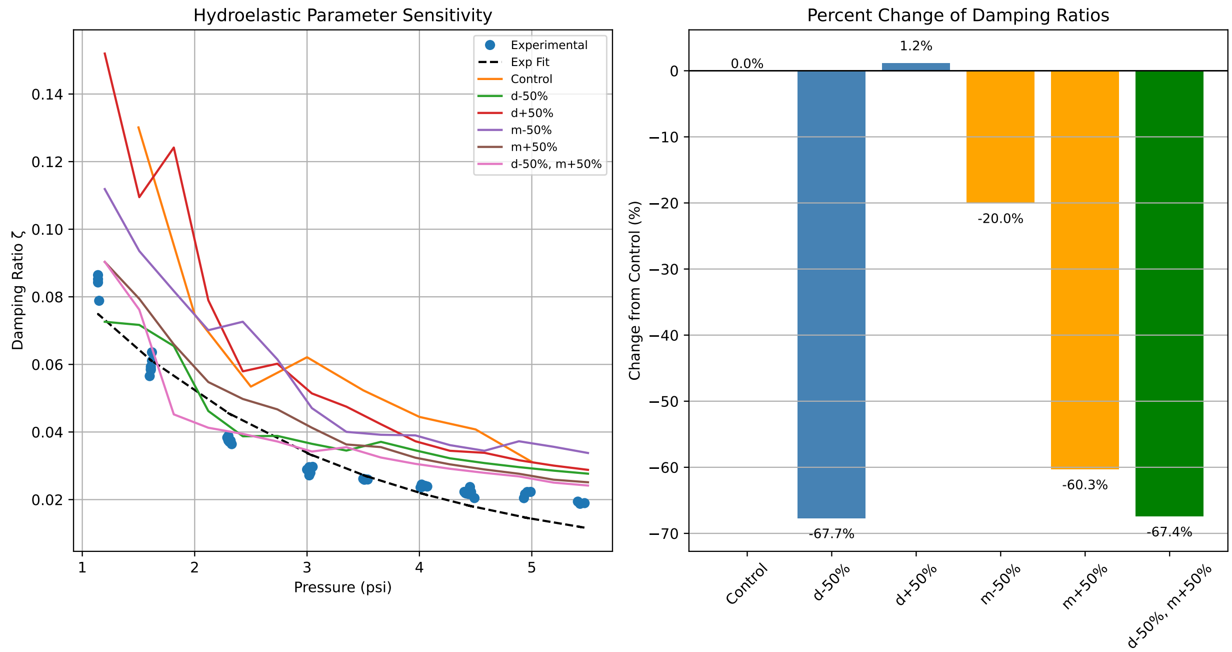


Figure 6.10: Effect of ±50%variations in hydroelastic modulus and dissipation on damping ratio predictions.

From the figure, decreasing the dissipation and increasing the modulus both decreased the percent error from the control. However, applying both together did not yield additional benefits. The hydroelastic model with a 50% reduction in dissipation is used for the analysis in the next section.

6.4 Modeling the Full Dynamics Through Uncontrolled Descent

The previous sections detailed the derivation of models for the isolated non-conservative aspects of the robot. In this section, the models for joint friction and shell dynamics are combined with the Drake rigid body engine to model the robot rolling down a ramp at different pressures. For this experiment, the pendulum is unconstrained and subjected to the frictional losses specified by the respective Majd model (Shown in Figures 6.7A and B). The only change from the steering to the drive direction is dropping the $K_{injected}$ term from (6.9) and $\dot{\phi}$ is replaced with the pipe's rolling velocity, ω_d.

In each test, the robot is inflated to a target pressure, then released from rest at a marked spot on a ramp. The robot's drive speed, as well as pendulum and pipe angles, were recorded. The experimental data were used to set the initial pipe angle and appropriate contact parameters at a certain pressure of the robot before it was released from rest. A side-by-side view of the experimental setup and virtual recreation of it is shown in Figure 6.11.

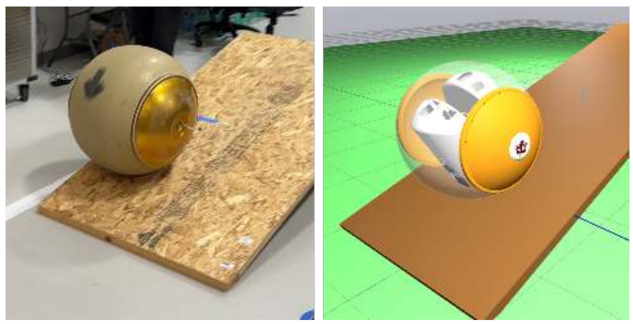
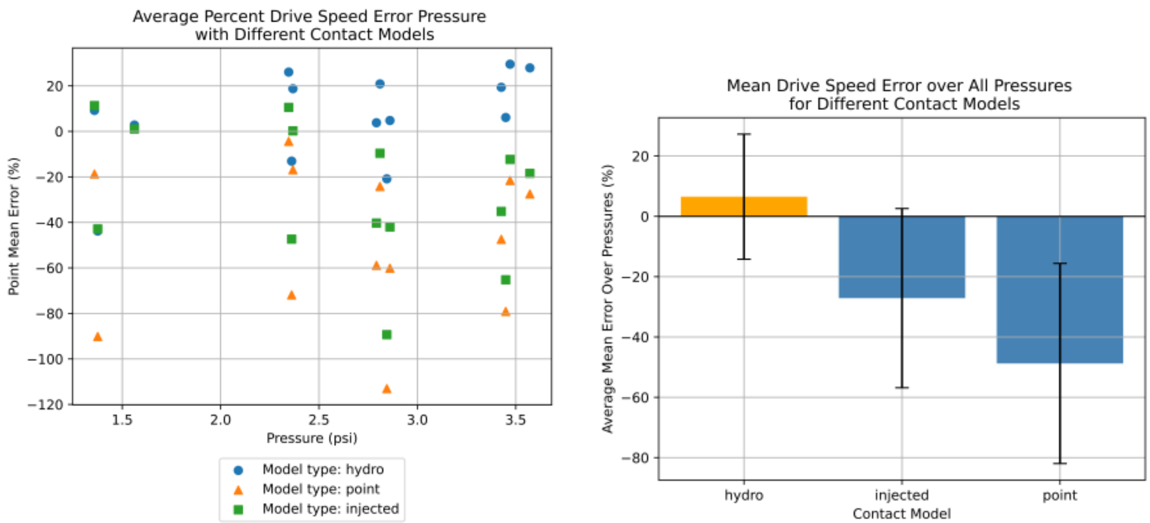


Figure 6.11: Experimental ramp descent (left) and digital twin simulation in MeshCat (right).

Only the descent phase prior to ramp exit was analyzed, excluding transition and airborne segments. The simulated velocity data is subtracted from the measured velocity data at each sample time, such that overestimated models are positive. This Point Mean Error (PME) is shown in equation 6.13 where the subscripts for the experimental and simulated drive angular velocity are e and s, respectively. This percentage error metric compares the simulated and experimental drive velocities at each time step. Figure 6.12 shows the resulting PME from each run at the tested pressure for the three different contact models.

$$PME = \frac{1}{N}\sum_{i=0}^{N}\frac{(\omega_e(i) - \omega_s(i))}{\omega_e(i)} \times 100\% \tag{6.13}$$



A B

Figure 6.12: Drive speed errors at each pressure

Figure 6.12B averages the PME from all pressures at each model for an overall metric. For this case, the tuned hydroelastic model was able to predict the experimental drive velocity more accurately on average than the injected or point contact models. While slightly underestimated, the model overdamps the system, and in line with the trend from the previous section, where reducing the dissipation from the calculated C_{inject} yields better results. However, both the injected and hydroelastic models were an improvement compared to a rigid body point contact assumption.

6.5 Discussion and Model Limitations

While the tuned hydroelastic model is accurate and exceeds the capabilities of regular point contact models, its limitations are notable. The hydroelastic representation cannot capture whole-body flexing or the additional stiffness that such deformations may introduce in rolling. The software must also set the contact parameters before running the simulation, so that this architecture cannot model changes in pressure during a simulation period.

Table 6.2: Physical parameters measured for URDF model and used in dynamic simulations.

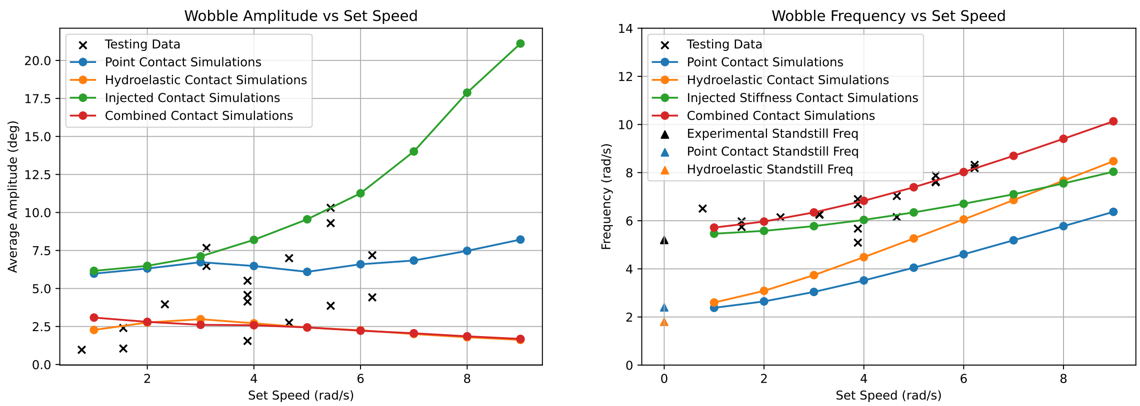
Parameter Set	Symbols	Values
Link mass (kg)	$m_{pc},\ m_p,\ m_s$	$[2.77, 22.03, 16.53]$
Link Inertias (kgm^2)	$I_{pc},\ I_p,\ I_s$	$[1.8e^{-2}, 4.4e^{-1}, 1.04]$
Distance measures (m)	$r_{pc},\ r_p,\ R$	$[0.05, 0.095, 0.305]$
Ramp Angle		16^o

6.6 Application to Wobble Experiments

The primary reason for developing the dynamic model in this chapter was to capture the wobbling effects discussed in the previous chapter. The simulation was set up to model the same locked steering experiment from Section 5.3, but with the forward dynamic and contact models derived in this chapter. The resulting frequency and amplitude responses from different contact models are shown in Figure 6.13 with the previously measured empirical data. The frequency values from Figure 6.9 at 2.7psi are also shown on the plot as triangle markers for reference at zero driving speed.

The point contact model assumes no additional dynamics imparted from the outer shell. The hydroelastic model is identical to the one tuned from equation (6.11) and (6.12). In this case, the injected dynamics only comprises the additional stiffness term, not the derived dissipation. Since the hydroelastic interpretation of dissipation was a better match than the injected dissipation, the combined model type blends the injected stiffness and hydroelastic dissipation. This overcomes the lack of rotational stiffness that the original hydroelastic model lacks.

$$\tau_{combined} = -K_{inject} + f\left(m(p), d(p)\right)_{hydro} \tag{6.14}$$



A B

Figure 6.13: Simulated Wobble tests under different contact models

From Figure 6.13B, the combined and injected stiffness models are the closest match to the measured wobble frequency data. Whereas the point and pure hydroelastic models, which do not incorporate rotational stiffness, well underestimate the frequency. This suggests that the shell's additional rotational stiffness is the reason behind the wobbling frequency. The amplitudes show another story in Figure 6.13A. All simulated models were started with an initial offset of 6 degrees. The point model kept that initial amplitude even at higher speeds. The hydroelastic and combined models exhibit a decrease in amplitude over time, resulting in a lower average amplitude throughout the studied timeframe. It is a nearly identical response as they both use the same hydroelastic dissipative effect. The injected stiffness is the only model that portrays the nutation instability, where the amplitude increases at higher speeds. However, when the stiffness is combined with injected dissipation (not shown) or hydroelastic dissipation (the combined model type), the nutation effect is mitigated. At least from studying the system through dynamics and contact mechanics alone, additional dissipation on the outer shell appears to address the nutations. Yet, the robot is best modeled with dissipation, but it still wobbles. Contact mechanics alone may not be the sole source of this behavior. The next chapter will study a different possible source of this behavior.

6.7 Chapter Conclusions

This chapter presented an empirically validated, modular dynamic modeling framework for a soft, pendulum-driven spherical robot. The approach leverages a URDF-based Articulated-Body Algorithm with interchangeable modules for joint friction and soft-shell contact dynamics. Three friction models were tuned to characterize internal joint losses. In addition, the outer shell was represented using three alternative contact models, including the hydroelastic method and a pressure-dependent injected stiffness–damping formulation. Experimental isolation of the steering dynamics enabled direct parameter tuning, and ramp descent tests demonstrated that the tuned models could also be used in the driving direction. Overall, these results indicate that an empirically tuned hydroelastic or injected dynamic contact model significantly improves prediction accuracy compared to rigid-body point contact assumptions.

The chapter closed by revisiting the wobbling dataset from the previous chapter. By combining the contact types, the model was able to capture the frequency response of the wobbling motion. However, the amplitudes of the system were only a factor of extra stiffness. Any additional dissipation appears to mitigate excessive growth in amplitude; contact mechanics alone may not be the sole source of a system's wobbling motion.

7. Nutation Instabilities in Rolling Objects

7.1 Chapter Introduction

Section 5.3 presented a mobility study that established a correlation between the robots' driving and steering directions. Chapter 6 evaluated the effect of different contact models on the wobbling frequency and amplitude. While the additional contact mechanics could match the frequency, the amplitude of the wobble was not. This chapter will take a different approach to modeling the wobbling object. Rather than modeling every aspect simultaneously with the articulated-body algorithm and contact mechanics, inspiration and alternative techniques for analysis will be drawn from other similarly wobbly objects—primarily bicycles and spacecraft.

7.1.1 Wobbling Bicycles

Bicycles and their motorized counterparts exhibit instability modes analogous to those observed in wobbling robotic systems. When left uncontrolled, a bike will wobble from its upright configuration and fall. Unlike simple rolling wheels, however, typical bicycles steer in the direction of their motion, which delays the onset of instability and manifests as the wobble. This phenomenon is a product of three parameters: the trail of the front steering assembly, the gyroscopic effect of the wheels, and the system's center of mass along its wheelbase.

Jones conducted a series of experiments aimed at constructing an unrideable bicycle to eliminate this self-steering effect [41]. The self-steering effect arises when the ground contact point of the wheel is behind the steering axis, giving the bike positive trail (Left of Figure 7.1). So, sideways motions cause the front wheel to steer into the turn. Jones built a prototype that extended the front wheel, resulting in a negative trail. This eliminated the self-steering effect, making it extremely difficult, but not impossible, to ride. In a second prototype, Jones attached a second wheel to the front of the seeing assembly. Experimental results indicated that canceling the gyroscopic contribution by spinning the secondary wheel in the opposite direction produced no significant effect on rider control or on the dynamics of falling. Conversely, when the auxiliary wheel was spun in the same direction as the primary wheel, the enhanced gyroscopic effect improved stability. These results suggested that gyroscopic effects contribute positively to stability, but their absence alone does not induce instability in a bicycle.

Subsequent studies by Kooijman et al. refined these observations using a simplified prototype [42]. Their analysis considered gyroscopic effects, steering-axis trail, and the rider's mass distribution as parameters. Results demonstrated that stability can persist even in the absence of steering-axis trail, provided that the other parameters are appropriately tuned. This establishes that bicycle stability arises from the interaction of these parameters, rather than relying on any single mechanism.

The dynamics of bicycle stability are often compared to a coin rolling on its edge. In this case, adding angular momentum asymptotically stabilizes the coin until frictional energy loss permits gravitational collapse via a wobbling motion. The same is true for bicycles.

While RoboBall shares a wobbling instability mode with the other bicycles, it differs from these analogies in several ways. Roboball rolls with an outer shell about its minimum moment of inertia, where a bike's wheel rolls about its maximum moment of inertia. When rolling forward, RoboBall's steering axis will tilt with the pendulum, but will always result in negative trail, shown on the right of Figure 7.1. RoboBall also lacks a wheelbase, which, according to Koojiman, is an essential parameter for bike-like stability.

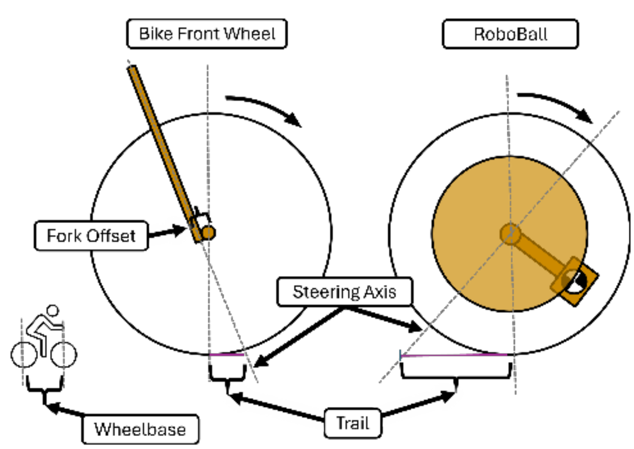


Figure 7.1: Side by side of RoboBall and Bicycle Steering Axis Trail

7.1.2 Relaxation of Rotating Asteroids and Satellites

The rotational motion of free-flying tri-axial objects is a unique and well-studied phenomenon. The most famous case is the Dzhanibekov effect, also known as the tennis racket theorem. Where an object rotating about its intermediate moment of inertia will tend to flip from one side to the other [43]. Additionally, free-flying rotating objects under some form of dissipative load will eventually settle into rotation about their major moment of inertia to minimize kinetic energy. Cigar-shaped satellites, such as Explorer I, were initially set spinning about their minor moment of inertia; however, energy dissipated by structural vibrations in its antenna caused the satellite to attempt to settle into rotations about its major inertial axis. Study of this dissipative-to-maximal-rotation effect has had a lasting impact on spin-stabilized satellite design and Eulerian mechanics theory [44]. This is also recognized by astronomers studying spinning asteroids. In that field, the use of a "relaxation angle" or the angle of the angular momentum expressed in the body frame relative to an inertia axis is used to classify asteroid spin states and study their dynamics [45]. The techniques and intuition for studying complex rotations of free-flying objects could also be applied to the study of spherical robots; however, the theory has yet to be extended to rolling objects.

7.1.3 Inertial Profiles of Robots in Literature

It's widely recognized that rotating objects with dissimilar inertias exhibit complicated motions. It would be reasonable to assume that a rotating object constrained to roll on a surface would display similar behaviors, especially if that object (or robot) is shaped like a sphere. However, engineers of spherical robots have either not addressed these effects directly or ignored them. The next couple of paragraphs will discuss several spherical robots and comment on whether they used a full 3D model or a decoupled approach. Since the inertia profile of a body is the driving factor behind the Dzhanibekov effect, the inertia tensors of their robots' outer shells will also be discussed.

Arif et al. employed a flywheel with a decoupled model to reduce oscillations and achieve higher speeds [24]; however, they did not report the shell's inertial profile. Asiri developed a full dynamic model with a control scheme based on centripetal forces, but made no observations of unexpected motion in their experiments. Their robot had an almost uniform, slightly oblate profile. Belzile used a fully coupled model and stated that the inertia was uniform, although photographs in their paper show a shaft across the sphere, which would, in fact, alter the inertia. Feng also presented a full model, but simplified it enough to reduce it to an uncoupled system. While no inertia values were provided, their kinetic energy formulation (eq. 21 in [46]) implies oblate inertias.

B. Li adopted a Lagrangian formulation with kinematic constraints embedded directly into the derivation. They briefly acknowledged a wobbling behavior during turns but did not analyze it further, focusing instead on their novel jumping mechanism. As a result, no internal inertial profiles were reported. Wang followed a decoupled approach similar to that discussed in Chapter 2 and therefore reported only one inertia. Ylikorpi introduced a soft-shell robot similar to RoboBall, featuring an inflatable outer shell. Their model considered only one plane of motion but noted that the inertia could vary with inflation pressure and shell geometry.

A summary of these studies is given in Table 7.1. For completeness, Schroll, Singhal, and RoboBall—already discussed in Chapters 5 and 6—are also included. Recall that Schroll acknowledged the nutation instability, in contrast to Singhal's competing claim. For those works that do provide inertial values, the axes have been redefined so that the middle inertia (I_2) is aligned with RoboBall's convention in Figure 7.2.

Table 7.1: Literature Model Comparison and Inertial Profiles of Outer Shells

Author	Model	Inertial Profile $[kg - m^2]$
M. Arif [24]	Uncoupled	Not Given
S. Asiri [47]	Coupled	$[2.65, 2.57, 2.65]e^{-2}$
B. Belzile [48]	Coupled	$[0.25, 0.25, 0.25]$
Z. Feng [46]	Uncoupled	non-uniform via eq. 21
B. Li [30]	Coupled	Not given
G. Schroll [14]	Coupled	$[0.25, 0.15, 0.25]$
A. Singhal [31]	Coupled	Uniform Inertia
Y. Wang [27]	Uncoupled	[N/A, 0.888, N/A]
T. Ylikorpi [16]	Uncoupled	[N/A, $(5.85 - 7.18)e^{-2}$, N/A]
RoboBall (Chapter 6)	Coupled	$[1.04, 0.64, 1.04]$

Table 7.1 shows the inconsistent treatment of possible inertial effects on these robots.

From the list of inertial profiles in the table, it is clear that possible inertial effects of the various robots' rotating outer shells have not been adequately addressed or ignored entirely. The remainder of this chapter will develop dynamics analysis techniques drawn from free-flying spacecraft but appropriately constrain them to apply to rolling objects. Additional experiments and numerical studies will be presented to develop further the relationship between a rotating inertial object and the nutation instability.

7.2 Dynamic Analysis of Rotating Bodies Supplemented with Relaxation Dynamics

This section presents two modeling paradigms: one derived from Euler's rotation equations and another from relaxation dynamics as presented by Likins in [44]. Both models will use the notation illustrated in Figure 7.2. Where the body frame axes are given by $\hat{x}_i$, and the net moments and angular velocities about the axes are given by M_i or ω_i for the i^{th} axis. Euler's equations use the individual components, while the relaxation dynamics will use the angle θ in between the angular momentum vector $\vec{\omega}$. This vector is defined in equation (7.1), and similar statements can be made for $\vec{M}$.

$$\vec{\omega} = \omega_1 \hat{x}_1 + \omega_2 \hat{x}_2 + \omega_3 \hat{x}_3 \tag{7.1}$$

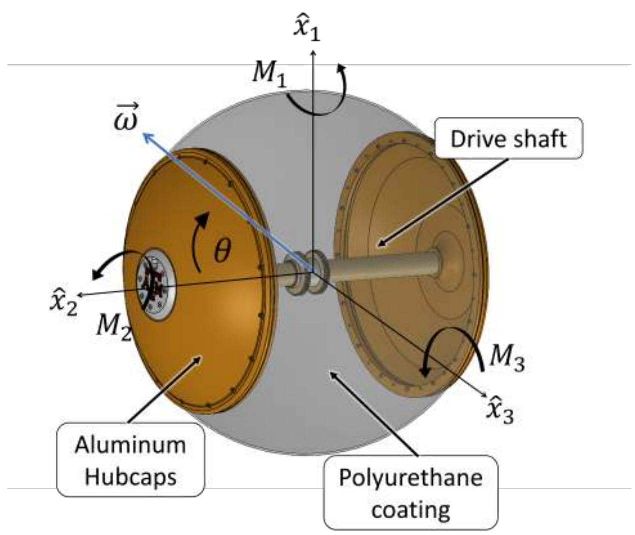


Figure 7.2: Diagram of the shell's body frame with angular velocity vector, applied moment components and relaxation angle between the velocity and minimum inertia axis $\hat{x}_2$

7.2.1 Euler Dynamics of a Rotating Body

The motion of a spinning body under the influence of external moments is described by Euler's equations, given below. Here I is the inertia tensor of the rotating body.

$$I\dot{\vec{\omega}} + \vec{\omega} \times I\vec{\omega} = \vec{M} \tag{7.2}$$

To examine the stability of free rotations with equation (7.2), assume constant rotation about ω_2 with small perturbations in ω_1 and ω_3 as δ_1 and δ_3 respectively. Ignoring $\delta_1\delta_3$ terms as negligible yields the dynamics in the form of (7.3).

$$I_1\dot{\delta}_1 + (I_3 - I_2)\omega_2\delta_3 = M_1 \tag{7.3a}$$

$$I_2\dot{\omega}_2 = M_2 \tag{7.3b}$$

$$I_3\dot{\delta}_3 + (I_2 - I_1)\omega_2\delta_1 = M_3 \tag{7.3c}$$

Ignoring M, differentiating (7.3a), then substituting (7.3b), and (7.3c) yields (7.4).

$$\begin{aligned}
I_1\ddot{\delta}_1 + (I_3 - I_2)\dot{\omega}_2\delta_3 + (I_3 - I_2)\omega_2\dot{\delta}_3 &= 0 \\
I_1\ddot{\delta}_1 + (I_3 - I_2)\omega_2\left(\frac{(I_1 - I_2)\omega_2\delta_1}{I_3}\right) &= 0 \\
I_1\ddot{\delta}_1 + \frac{(I_3 - I_2)(I_1 - I_2)}{I_3}\omega_2^2\delta_1 &= 0
\end{aligned} \tag{7.4}$$

This is in the form of a simple harmonic oscillator, which is stable for positive values of k.

$$m\ddot{x} + kx = 0 \tag{7.5}$$

If the body inertias are ordered such that I_2 is intermediate (i.e., $I_3 > I_2 > I_1$), then any small perturbations about the non-rotating axis (δ_1 or δ_3) will be unstable. This is the essence of the Intermediate Axis Theorem.

However, RoboBall's shell is not tri-axial in its inertias; rather, they are ordered by $I_1 = I_3 > I_2$. Inertial profiles of this form are commonly referred to as "oblate" objects. This reduces (7.4) to (7.6).

$$I_1\ddot{\delta} + \frac{(I_1 - I_2)^2}{I_1}\omega_2^2\delta_1 = 0 \tag{7.6}$$

In this form, the stiffness value is always positive. Unlike the previous result, any off-axis disturbance will oscillate for any rotational speed about the minor axis ω_2. This will make a free-flying object appear to wobble in place with small off-axis rotations.

Switching back to a tri-inertial scenario, consider a case where M is not ignored; instead, assume the system is under uniform damping, such as a satellite under structural dissipation. In this case, M takes the form of (7.7). Here C is a diagonal matrix of positive damping constants.

$$M = -C\omega \tag{7.7}$$

Equation (7.3) can then be written in a standard dynamic form, $\dot{x} = A(\omega)x$.

$$\begin{pmatrix} \dot{\delta}_1 \\ \dot{\omega}_2 \\ \dot{\delta}_3 \end{pmatrix} = I^{-1} \begin{bmatrix} -C & 0 & (I_2 - I_3)\omega_2 \\ 0 & -C & 0 \\ (I_1 - I_2)\omega_2 & 0 & -C \end{bmatrix} \begin{pmatrix} \delta_1 \\ \omega_2 \\ \delta_3 \end{pmatrix} \tag{7.8}$$

The characteristic equation of A is given by (7.9) with eigenvalues λ.

$$-(C + \lambda)^3 + (I_2 - I_3)(I_1 - I_2)\omega_2(C + \lambda) = 0 \tag{7.9}$$

So the eigenvalues are given by

$$\lambda_1 = -C \tag{7.10a}$$

$$\lambda_{2,3} = -C \pm |\omega_2|\sqrt{(I_2 - I_3)(I_1 - I_2)} \tag{7.10b}$$

If the inertias are distinct and ordered like before ($I_3 > I_2 > I_1$), the radical will stay real so that one of the eigenvalues could become positive for large enough speeds of ω_2. Similar to the previous analysis with the system in harmonic form for $C = 0$.

However, if the inertias are oblate ($I_3 = I_1 > I_2$), then (7.10b). takes the form of (7.11). So that the troublesome radical term becomes imaginary. Yet the real parts are still negative, so rotations about that axis are stable with some decaying harmonics.

$$\lambda_{2,3} = -C \pm i|\omega_2|\sqrt{(I_2 - I_1)^2} \tag{7.11}$$

The analysis presented here analyzed the free and damped rotations of both inertially tri-axial and oblate bodies floating in space. For an oblate body, both free and damped responses were stable, lacking the nutation instability we are after. However, this takes on a new light when the direction of the net axis of rotation is considered as the velocities decay.

7.2.2 Relaxation Dynamics of a Damped Oblate Body

Defining a relaxation angle can give insight into the evolution of a body's rotational profile over time. In astrophysics, this angle is typically defined as the angle between the body's angular momentum vector and its greatest moment of inertia [45]. Since RoboBall primarily rolls about its axis of minimum inertia, this section will define the angle with respect to that axis, rather than the maximum axes. The angle, angular velocity vector, and shell body frame are shown in Figure 7.2. When defined in this way, if the robot is rolling forward or driving straight, then $\vec{\omega}$ is aligned purely along $\hat{x}_2$, and the angle is zero. Likewise, if the shell is flipping end over end, then the vector would lie in the $\hat{x}_1 - \hat{x}_3$ plane and the relaxation angle would be 90^o. A schematic demonstrating these characteristics is shown in Figure 7.3. The angle is calculated directly with (7.12).

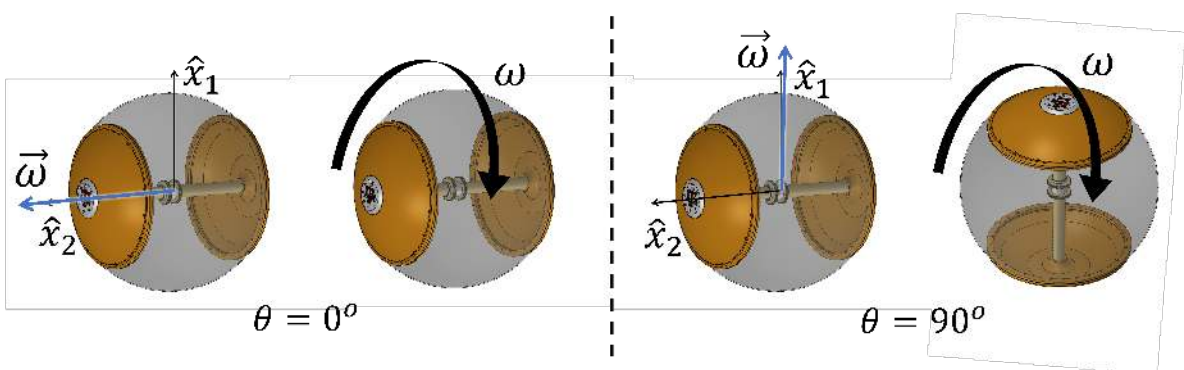


Figure 7.3: RoboBall shell in two edge case relaxation states

$$\cos(\theta) = \frac{I_2\omega_2}{||L||} \tag{7.12}$$

Where $||L||$ is the norm of the angular momentum and I_1, I_3, I_2 are the moments of inertia about the respective axis, (recall for RoboBall $I_1 = I_3$). $||L||^2$ is given by (7.13).

$$||L||^2 = \omega^T I^2 \omega = I_1^2(\omega_1^2 + \omega_3^2) + I_2^2\omega_2^2 \tag{7.13}$$

The rotational kinetic energy is given by (7.14).

$$2T = \omega^T I \omega = I_1(\omega_1^2 + \omega_3^2) + I_2\omega_2^2 \tag{7.14}$$

The relaxation angle θ can be expressed in terms of T and $||L||$ following an identical procedure shown by Likins in Section IV of [44]. The resulting relationship is expressed in (7.15).

$$\cos^2(\theta) = \frac{I_2(2I_1T - ||L||^2)}{||L||^2(I_1 - I_2)} \tag{7.15}$$

If the change in momentum is small, then (7.15) can be differentiated directly to (7.16).

$$\dot{\theta} = \frac{2}{\sin(2\theta)} \frac{I_1 I_2}{||L||^2(I_2 - I_1)} \dot{T} \tag{7.16}$$

This equation is the *relaxation dynamics* of a rotating body. It is a tool to study the evolution of a rotating oblate body's rotational profile over time.

The behavior of equation (7.16) can be studied by separating it into three terms and determining their signs: the sinusoidal fraction, the inertia ratio, and the rate of change of kinetic energy. The first part, the sinusoid ratio, is positive within the quadrant between $\hat{x}_2$ and the $\hat{x}_1 - \hat{x}_3$ plane from Figure 7.2.

$$\frac{2}{\sin(2\theta)} > 0 \text{ for } \theta \in (0, 90^o)$$

For the inertia ratio:

$$\frac{I_1 I_2}{||L||^2(I_2 - I_1)}$$

As a norm, $||L||^2$ is always positive. Therefore, the sign of this term depends on the ordering of the inertias. For RoboBall, $I_1 > I_2$, so this term is negative. That leaves the stability of (7.16) down to the rate of kinetic energy change, $\dot{T}$. This can be expressed in terms of the applied moment to the body using equations (7.2) and (7.14).

$$\begin{aligned} 2\dot{T} &= \dot{\omega}^T I\omega + \omega^T I\dot{\omega} \\ \dot{T} &= \omega^T I\dot{\omega} \\ &= \omega^T(M - \omega \times (I\omega)) \\ \dot{T} &= \omega^T M \end{aligned} \tag{7.17}$$

As with the previous sections, we can apply uniform damping to the system to mimic structural dissipation of a satellite.

$$M = -C\omega \tag{7.18}$$

So the relaxation dynamics become

$$\dot{\theta} = \frac{2}{\sin(2\theta)} \frac{I_1 I_2}{||L||^2(I_2 - I_1)}(-\omega^T C\omega) \tag{7.19}$$

The term $-\omega^T C\omega$ is negative definite for all non-zero ω. The positive sine term, negative inertia ratio, and negative dissipation terms net a positive sign for $\dot{\theta}$. Therefore, a rotating body under dissipation will experience a reduction in angular magnitude, but a shift in net rotational direction towards the $\hat{x}_1 - \hat{x}_2$ plane. As RoboBall's shell rotates under some dissipative load, it will shift to flip hubcap-to-hubcap.

7.2.3 Comparing Responses of Rotation and Relaxation Dynamics

Figures 7.4 and 7.5 show results from numerically integrating (7.2) for both free and uniformly damped rotations, as defined by (7.18). The middle and right of the figures contain the calculation of the relaxation angle with equation (7.12) and rotational kinetic energy as defined by (7.14).

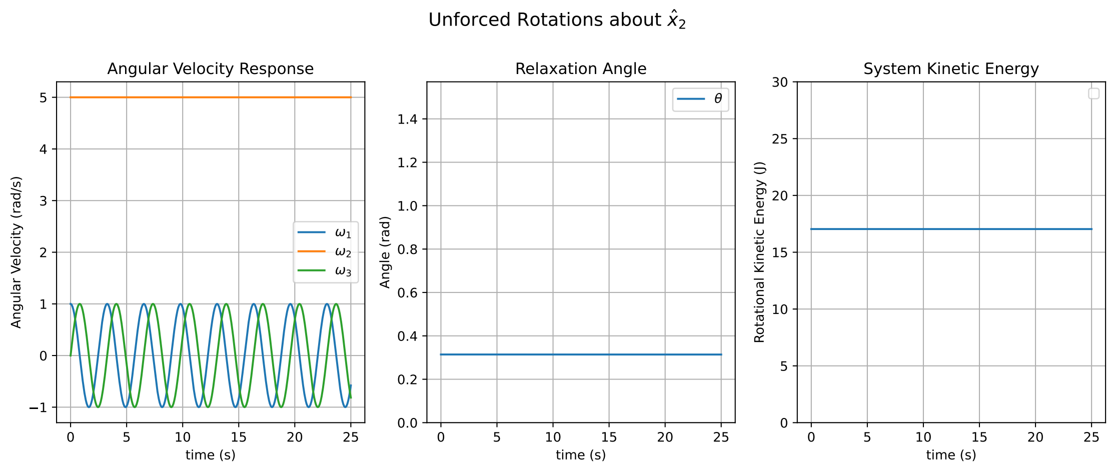


Figure 7.4: Free rotations of an oblate body about $\hat{x}_2$

Figure 7.4 displays behaviors for a floating and rotating oblate body discussed in the previous sections. One of the eigenvalues is zero under zero damping, while the others exhibit sinusoidal behavior. The relaxation angle is constant because there is no change in energy but not zero since not all of the angular momentum is aligned with ω_2.

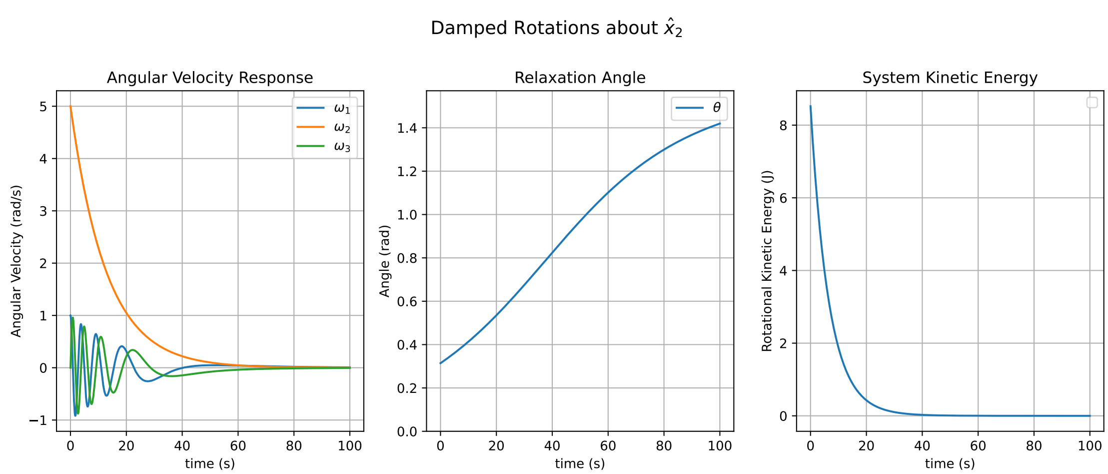


Figure 7.5: Damped rotations of the shell about the drive axis

Figure 7.4 shows the same system as above, but now uniformly damped in all axes. The system is damped, so the velocities and the kinetic energy decay. However, the relaxation angle increases, indicating that as it slows, the body will eventually transition from rotating about the drive axis to flipping from hubcap to hubcap before coming to a stop. This behavior is similar to what is observed in rolling objects undergoing a nutation instability.

Since the momentum angle encodes a direction, this effect could also be shown, although less elegantly, by defining a set of orientation states through eigenvalue analysis. This approach avoids the complexities with Euler-angle singularities or quaternion math.

7.3 Relaxation Dynamics for Constrained Bodies

The previous section studied the behavior of free-flying objects in two cases: free rotations and under uniform damping. Since RoboBall is a rolling dynamics system, the theory will need to be extended. This section will bridge the gap between the previous section by applying appropriate constraints for a sphere rolling on an inclined surface.

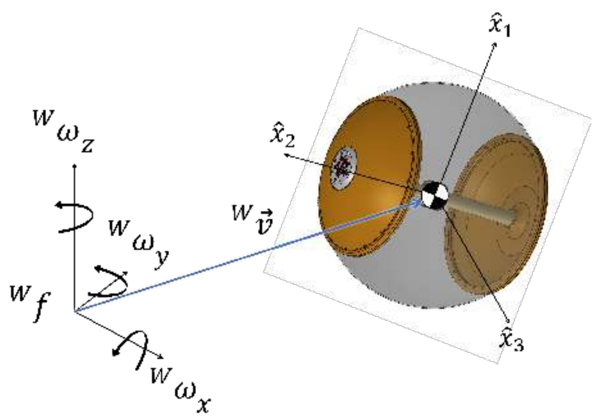


Figure 7.6: The robot shell located in the world frame

Rolling constraint and contact Jacobian

Figure 7.6 illustrates the motion of the robot's shell within the world frame. For the remainder of this section, values expressed in the world or inertial frame will be preceded with a superscript

${}^{w}(\cdot)$. Values expressed in the body frame will have no superscript to remain consistent with the previous sections. When a sphere rolls without slipping on a plane, the translational velocity ${}^{w}v$ is related to its rotational velocity in the world frame by equation (7.20), where R is the nominal outer shell radius.

$$
\begin{aligned}
{}^{w}v_x &= R\,{}^{w}\omega_y \\
{}^{w}v_y &= -R\,{}^{w}\omega_x \\
{}^{w}v_z &= 0
\end{aligned}
$$

or more compactly stated as

$$ {}^{w}v = \bar{R}\,{}^{w}\omega \tag{7.20} $$

where

$$ \bar{R} = \begin{bmatrix} 0 & R & 0 \\ -R & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, $$

Defining ${}^{w}R_b$ as a transform from the body frame to the world frame, express (7.20) in the form of (7.21) with respect to the body angular velocities.

$$ {}^{w}v = \bar{R}\,{}^{w}R_b\,\omega \tag{7.21} $$

Define the generalized velocity vector

$$ \dot{q} = \begin{bmatrix} {}^{w}v \\ \omega \end{bmatrix}. \tag{7.22} $$

The constraints can be written compactly as a contact Jacobian as

$$J(q)\dot{q} = 0, \qquad J(q) = \begin{bmatrix} \text{eye}_3 & -\bar{R}^{\,w}R_b \end{bmatrix}. \tag{7.23}$$

In (7.23), eye_3 represents the identity matrix of size three. This notation is introduced to keep it distinct from the body inertia tensor I. Its transpose

$$J(q)^T = \begin{bmatrix} \text{eye}_3 \\ -(\bar{R}^{\,w}R_b)^T \end{bmatrix} \tag{7.24}$$

acts as the Jacobian mapping Lagrange multipliers, λ, into the system. These multipliers represent the frictional forces that hold the constraints in (7.20) true.

Constrained dynamics

The equations of motion with constraint forces take the form below with notation from the appendix of [49]

$$A(q)\ddot{q} + H(q, \dot{q}) = F + J^T\lambda, \tag{7.25}$$

with a block-diagonal mass matrix

$$A = \begin{bmatrix} m\,\text{eye}_3 & 0 \\ 0 & I \end{bmatrix}, \tag{7.26}$$

where m is the mass and I is the inertia tensor in the body frame of the robot's shell.

Splitting into linear and angular components in (7.27). Here m is the shell mass, F_v are applied linear forces in the world frame, such as gravity, with other angular terms consistent with (7.2).

$$m^w\dot{v} = F_v + \lambda \tag{7.27a}$$

$$I\dot{\omega} + \omega \times (I\omega) = M - (\bar{R}^{\,w}R_b)^T\lambda. \tag{7.27b}$$

Solving for the contact force

From the linear block, solve for λ directly with:

$$\lambda = m\dot{v} - F_v \tag{7.28}$$

Substituting into the angular block gives the reduced rotational dynamics:

$$I\dot{\omega} + \omega \times (I\omega) = M - (\bar{R}\,{}^{w}R_b)^T\big(m^{w}\dot{v} - F_v\big) \tag{7.29}$$

Expression for angular kinetic energy rate

Redefine the kinetic energy rate following a similar procedure from equation (7.17).

$$\begin{aligned}
2\dot{T} &= \dot{\omega}^T I\omega + \omega^T I\dot{\omega} \\
\dot{T} &= \omega^T I\dot{\omega} \\
&= \omega^T(M - \omega \times (I\omega) - (\bar{R}\,{}^{w}R_b)^T\big(m^{w}\dot{v} - F_v\big)) \\
\dot{T} &= \omega^T M - \omega^T(\bar{R}\,{}^{w}R_b)^T\big(m^{w}\dot{v} - F_v\big)
\end{aligned} \tag{7.30}$$

recall from the rolling constraint,

$${}^{w}v = \bar{R}\,{}^{w}R_b\,\omega,$$

we obtain

$${}^{w}v^{T\,w}\dot{v} = \omega^T(\bar{R}\,{}^{w}R_b)^{T\,w}\dot{v}.$$

Therefore,

$$-\omega^T(\bar{R}\,{}^{w}R_b)^T m^{w}\dot{v} = -m^{w}v^{T\,w}\dot{v} = -\frac{d}{dt}\left(\tfrac{1}{2}m^{w}v^{T\,w}v\right).$$

That is,

$$-\omega^T(\bar{R}\,{}^wR_b)^T m\,{}^w\dot{v} = -\dot{T}_{\text{trans}} \tag{7.31}$$

where

$$T_{\text{trans}} = \tfrac{1}{2}mv^Tv \tag{7.32}$$

is the translational kinetic energy.

In this case (7.30) can be simplified to (7.33)

$$\dot{T} = \omega^T M - \dot{T}_{\text{trans}} + \omega^T(\bar{R}\,{}^wR_b)^T F_v \tag{7.33}$$

If the system is rolling unpowered down a slope and the global coordinate frame and initial body frame aligned along the angle of the slope, as shown in Figure 7.7. Then ${}^wR_b = eye_3$ and F_v can represent gravity acting on the system.

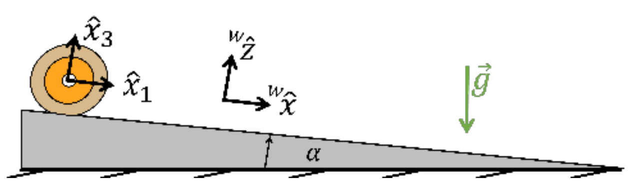


Figure 7.7: World frame aligned with object rolling down slope

$${}^wF_v = \begin{pmatrix} mg\sin(\alpha) \\ 0 \\ -mg\cos(\alpha) \end{pmatrix} \tag{7.34}$$

the final term in (7.33) can be simplified to

$$\omega^T(\bar{R}\,{}^wR_b)^T F_v = \omega_2 mgR\sin(\alpha) \tag{7.35}$$

This is the modified energy rate to be used in the relaxation dynamics relation for rolling systems. The relaxation dynamics for an oblate object rolling down a slope with uniform dissipative contact are stated below.

$$\dot{\theta} = \frac{2}{\sin(2\theta)} \frac{I_1 I_2}{||L||^2 (I_2 - I_1)} (-\omega^T C \omega - \dot{T}_{\text{trans}} + \omega_2 m g R \sin(\alpha)) \tag{7.36}$$

From equation (7.36), all variables in the final gravity term are positive and will stay positive as ω_2 increases down the slope. However, as the object accelerates down the ramp, $\dot{T}_{trans}$ will also be positive along with the quadratic damping term. At fast enough speeds, $\dot{T}_{trans}$ and the quadratic damping term will overpower the linear-in-velocity gravity term, resulting in a negative sign for $\dot{T}$. In this case, the system will experience the relaxation effect and flip hubcap-to-hubcap.

This is counterintuitive to our free-flying object scenario. If the relaxation dynamics from (7.16) are considered without accounting for the translational coupling between the angular and translational components, intuition would assume that the net angular kinetic energy would increase as the system rolls down the slope, stabilizing the system. The next section will support (7.36) with an experiment with the robot shell.

7.4 Shell Rolling Experiments

To verify (7.36), a series of experiments was conducted with only the robot's shell. A VN-100 IMU was mounted inside RoboBall's hubcaps to measure $\vec{\omega}$. This additional weight was balanced with weights to keep the inertial profile as consistent as possible. A schematic of this setup is shown in Figure 7.8. A bare shaft was used in place of the pendulum to keep the shell's shape.

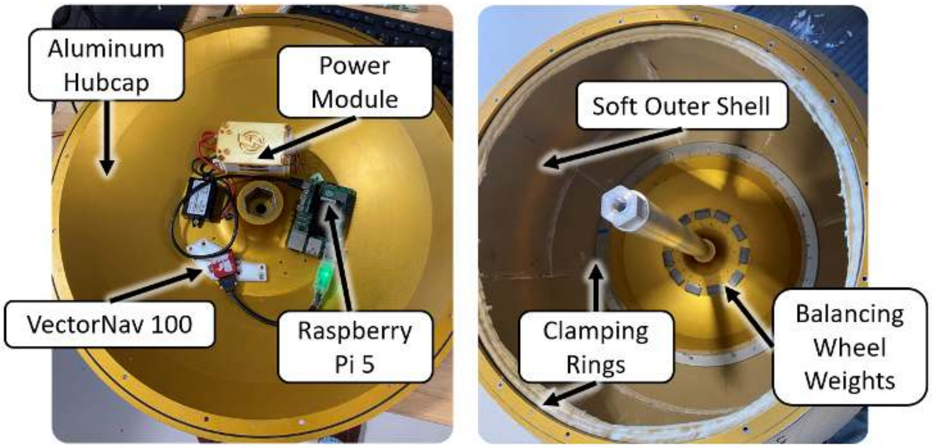


Figure 7.8: Instrumentation of Robot Shell

This empty shell was then placed at the top of a slightly sloped concrete slab and allowed to roll down. This allowed for multiple runs with a consistent set of initial conditions since the shell alone has no actuation capabilities. Figure 7.9 shows screenshots of the shell undergoing the nutation instability. As it rolls under the influence of gravity, it gradually relaxes, flipping hubcap over hubcap.

Figure 7.9: Screenshots of the shell experiencing a nutation instability

Data from one of the runs is shown on the left of Figure 7.10 with the relaxation angle and kinetic energy calculated in post-processing. Recall from (7.16) that the relaxation angle is a function of the angular momentum of the system, so small irregularities from the concrete and shell will affect the net energy of the system in a chaotic way. A third-order Savitzky-Golay filter was employed to remove most of the noise in the angular velocity readings.

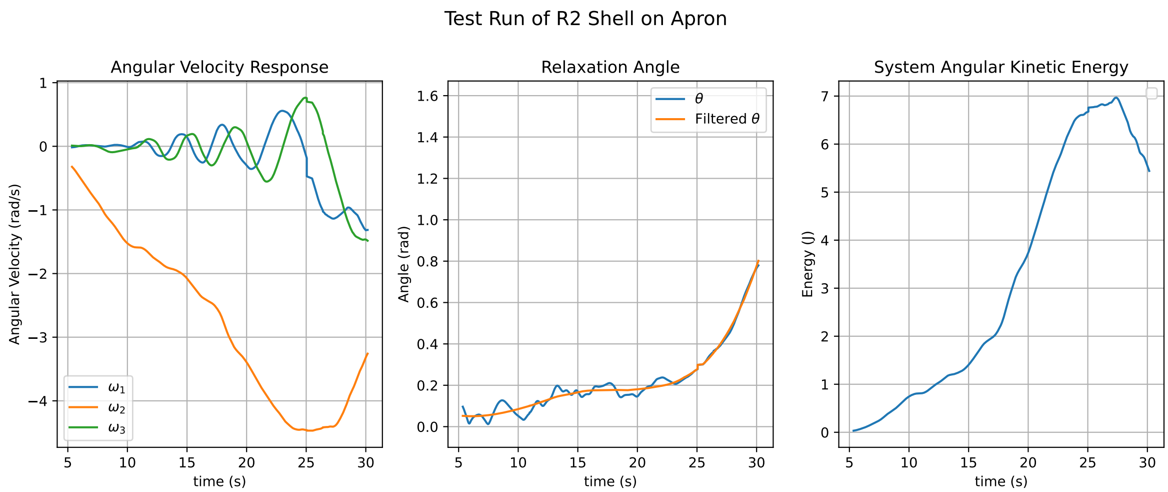


Figure 7.10: Data from a single run of the shell rolling down a slope

To determine if pressure has a significant effect on the relaxation rate, trails were conducted two main pressure levels. The filtered relaxation angles were used to calculate an average rate of relaxation to determine if system pressure had any significant effect on the relaxation dynamics. In this testing case, there appears to be no significant relationship between the rate of relaxation and pressure, shown by the linear regression in Figure 7.11.

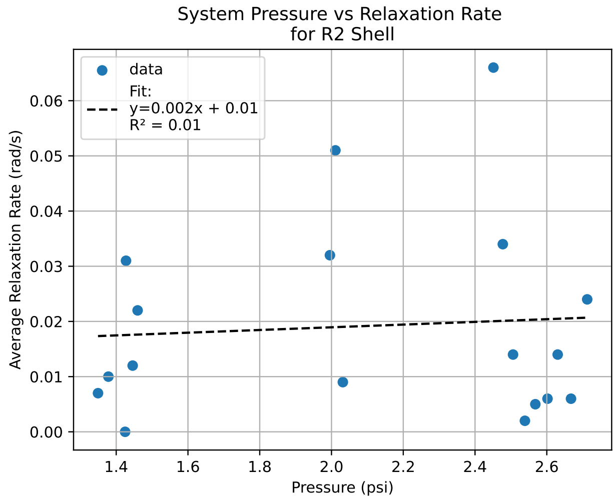


Figure 7.11: Relaxation Rates with R2 Shell at different Pressures

7.5 Simulation Studies

The modeling environment developed in the previous chapter can help highlight other significant aspects of the relaxation model. This section will highlight different contact model types, caveats with initial conditions, and the effects of flipping inertias. For this numerical study, the URDF described in Figure 6.3 was edited to only describe the pipe assembly and bedliner components of the robot, removing the pendulum and pitch center. This shell was then set to a small offset pipe angle and allowed to roll freely down a simulated ramp.

For the simulations, the small initial pipe angle is necessary to induce the relaxation effect. This is a holdover from Eulers rotational equations where pure rotations about every point is an equilibrium point. Sincere there are no irregularities in the shell or ground in the simulation where will be nothing to bump it off the equilibrium manifold and cause the system to relax, as shown in Figure 7.12.

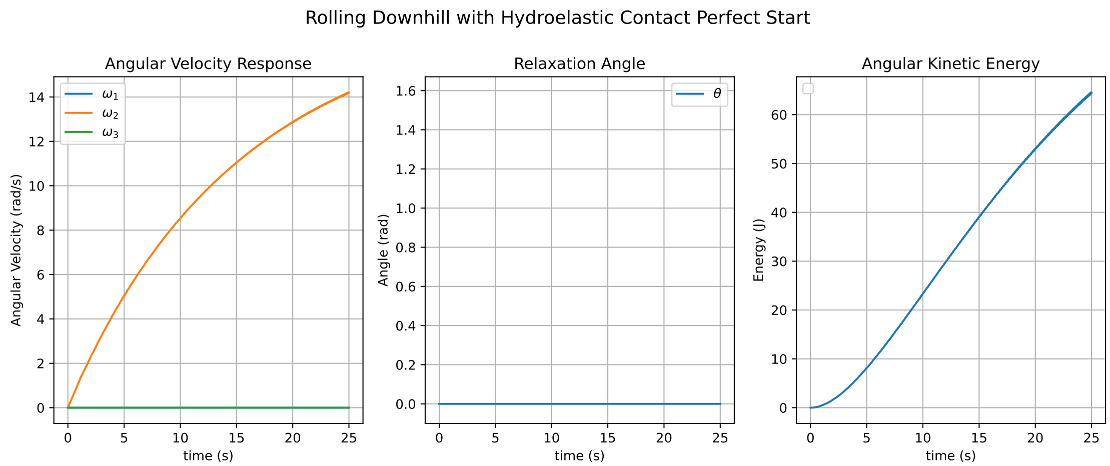


Figure 7.12: Simulated Response and Relaxation Angle of a Robot Shell Rolling down a slope with perfect initial conditions

The same setup was repeated with the point and hydroelastic contact models tuned in Chapter 6. Figure 7.13 and 7.14 show the results of these simulations. Recall from Chapter 6 that the hydroelastic contact model simply adds pressure-dependent damping to the system. Similar to the damping given to the system from equation (7.18). The injected dynamics does not exist with no pendulum so it is not included here. In the context of the relaxation dynamics, this would only affect the $\omega^T M$ term. Since the shell is constantly rolling down a slope, $\dot{T}_{\text{trans}}$ will always be positive as long as the rolling constraints hold, adding in the dissipative term only pushes the damped case to relax sooner.

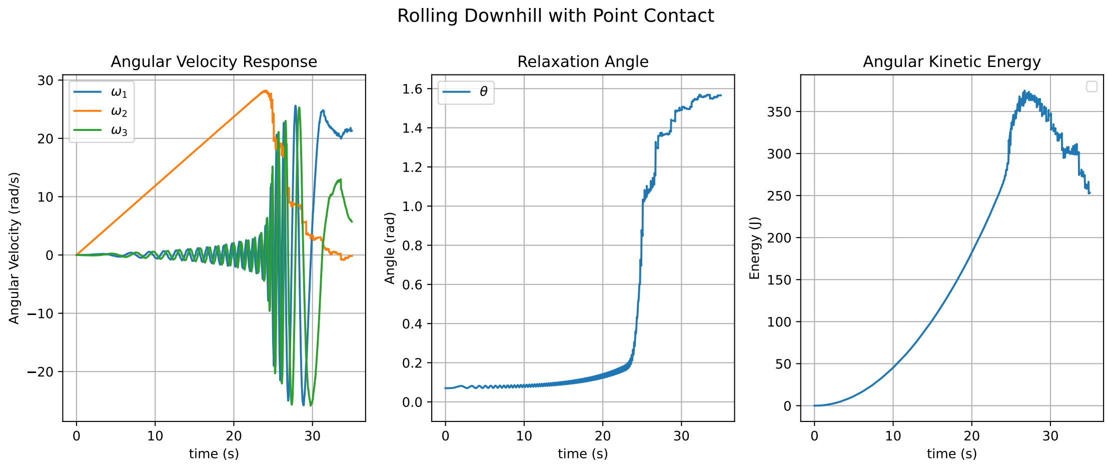


Figure 7.13: Simulated Response and Relaxation angle of Robot Shell Rolling down a Slope Modeled with a Point contact model

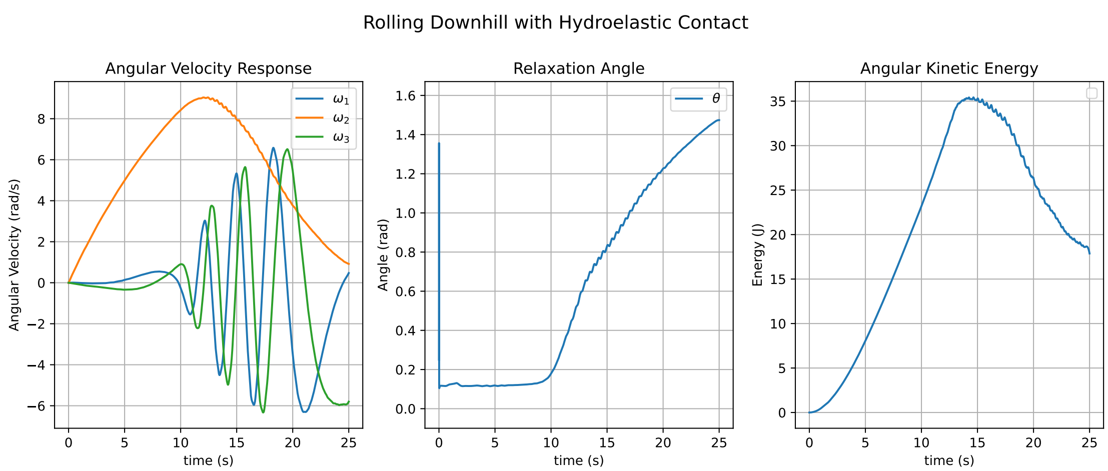


Figure 7.14: Simulated Response and Relaxation Angle of a Robot Shell Rolling down a slope modeled with a hydroelastic contact model

If a robot designer wished to stabilize the system, flipping the inertia profile of the system would flip the sign of (7.36). This would recover the gyroscopic stabilization effect of a bike and rolling coin. An example simulation with the values of I_1 and I_2 flipped is shown in Figure 7.15 with the relaxation angle stabilizing as it rolls down a slope.

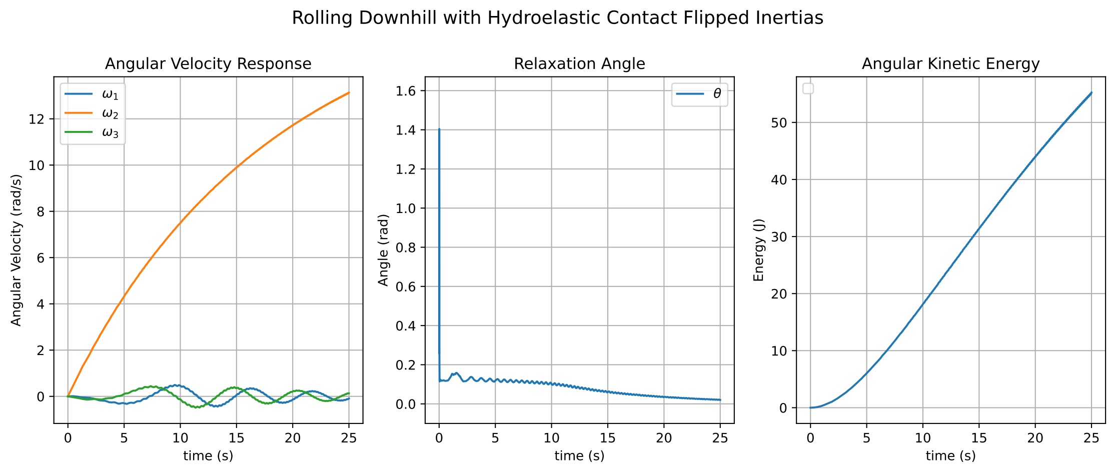


Figure 7.15: Simulated Response and Relaxation Angle of a Robot Shell Rolling down a slope modeled with a flipped inertial profile

7.6 Rolling Dynamics with a Pendulum

The previous section derived the relaxation dynamics for a sphere rolling on an inclined plane. This section will explore what is necessary to extend the theory to a rolling sphere with a pendulum. The model will then be studied in the context of the wobble tests from Chapter 5.

Equation (7.36) gave the relaxation dynamics in terms of a constrained rolling oblate sphere. To apply the concept to an actuated spherical robot, an appropriate selection for M must be made to reflect the actuation mechanism into the shell's frame. The pendulum dynamics can be isolated from the dissipative effects of the outer shell by substituting in $M = M_{\text{pend}} - C\omega$ into (7.37). Here M_{pend} is the dynamics of the pendulum expressed in the shell frame, hence no superscript.

$$\dot{\theta} = \frac{2}{\sin(2\theta)} \frac{I_1 I_2}{||L||^2 (I_2 - I_1)} \left(\omega^T M_{\text{pend}} - \omega^T C\omega - \dot{T}_{\text{trans}} \right) \tag{7.37}$$

To determine the stability of the system, the sign of $\omega^T M_{\text{pend}}$ must be assessed. This is not a trivial exercise since the forces of the pendulum depend on its pose with respect to gravity in the world frame and the drive and steer angles (${}^{w}R_b$, θ_d, and θ_s respectively). However, it is possible to choose M_{pend} such that the system is stable at high speeds, as shown in [10] and [27]. Irresponsible choices can also drive the system unstable; therefore, the sign of $\omega^T M_{\text{pend}}$ is arbitrary and can be both positive and negative.

$$M_{\text{pend}} = f({}^{w}R_b, \theta_d, \theta_s) \tag{7.38}$$

7.6.1 Revisiting the Mobility Studies

Section 5.3 described a positive relationship between a wobble amplitude and frequency according to the tracked speed of the system. That data was calculated by isolating the amplitude and period of the response peaks to determine the vibrational amplitude and frequency for the system with its steering axis locked.

The robot also had an IMU mounted on its pitch center that captured the off-axis velocities of the shell ($\omega_{1,3}$) during those tests. The driven speed of the shell (ω_2) is calculated from combining the pitching speed of the pendulum with the values measured from the drive motor encoders. With these values, we can plot the relaxation angle of the system continuously over time, rather than the amplitudes and frequencies.

The relaxation angle from five tests at increasing speeds is shown in the middle of Figure 7.16.

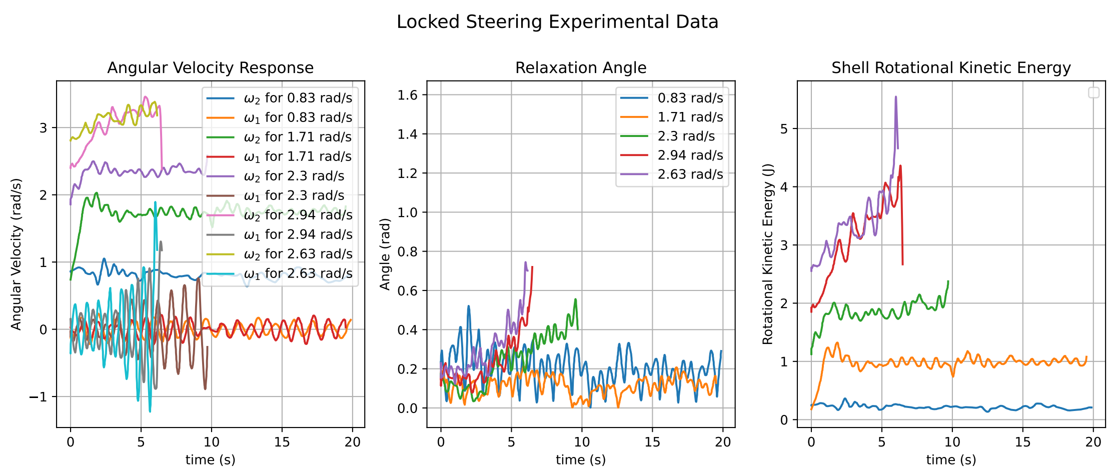


Figure 7.16: Locked Steering Tests Revisited with Relaxation Angles

The angle remains relatively bounded at slower speeds, but at faster speeds, past around 2 rad/s, the average angle starts to increase, a sign of the nutation instability as previously discussed. For this scenario, slower speeds allow the locked pendulum to balance out the dissipative terms in (7.37) and stabilize the system, but only until a certain speed. Then the dissipative effects overpower the pendulum, and the instability takes over.

This logic can even extend to the water testing in Section 5.1. Initially, the robot's linear speeds were measured in water to find the limiting factor for its performance in water. We could not compare it to the wobbling case on land because there were no amplitudes or vibrational peaks to study. By recasting the data into a relaxation angle, we can draw observations on why the robot on water was able to reach higher rotational speeds than currently possible on land. The respective parameters are plotted in Figure 7.17.

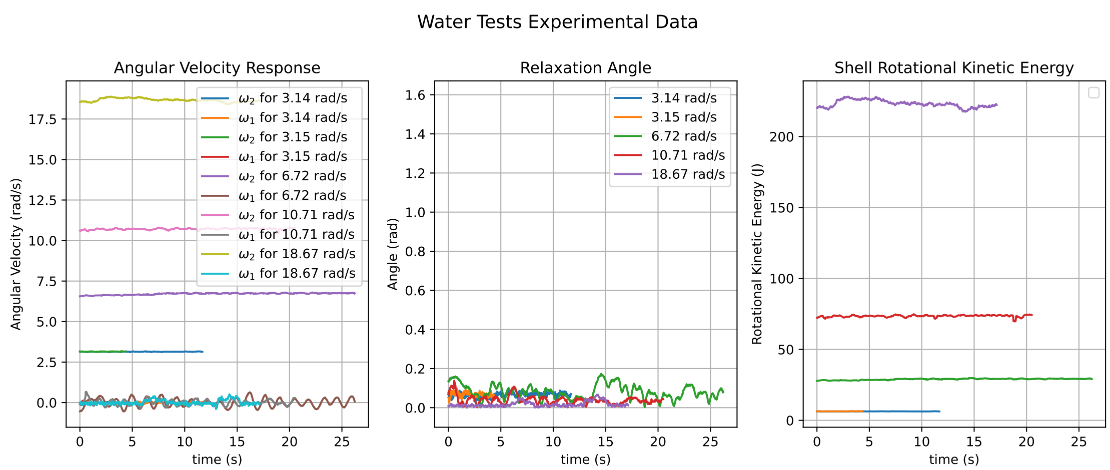


Figure 7.17: Relaxation Parameters in Water Mobility Tests

From the figure, none of the cases appear to relax over the tested timeframe. This is probably due to the breakdown of the kinematic constraints that transmit the shell's rotational motion into forward motion. So that the translational energy sink $\dot{T}_{\text{trans}}$ is eliminated from the dynamics. The motors and pendulum then even out the dissipative terms introduced by the water. The most likely reason the system can stabilize on the water with ease, even in the presence of the dissipative effect of the water.

7.7 Comments On The Relaxation Angle as a Feedback Parameter

While the relaxation angle can be a useful tool for studying the dynamics of a system, inevitable tradeoffs arise with its potential application in a control scheme. The first is its singularity when not moving. While the angle avoids position-based singularities apparent in Euler angle rotation representation systems, it is an inverse function of the norm of the angular momentum. As a result, the parameter becomes ill-conditioned at slower speeds, but it could be of use in higher-speed regimes.

$$\theta \propto \frac{1}{||L||}$$

The second drawback is the dependence of the parameters on noise. Inspecting figures 7.10 and 7.16 show the noise in the computed angle, even after the gyroscope readings have been filtered. Astrodynamicists using the angle to study asteroids, like M. Efroimsky in [45], deal with this by working with the time derivative of the squared sine of the angle averaged over the precession period, shown in equation (7.39). This takes the place of the Savitzky-Golay filter used in Figure 7.10. However, a rolling object appears to relax much faster than asteroids, so waiting every period for a datapoint may not be fast enough for stabilizing control logic. Additionally, a rolling object relaxes with small periodic amplitudes on a curved response that are difficult to target with filters for calculating the period, as observed in Figure 7.14.

$$\frac{d}{dt}\langle \sin^2(\theta) \rangle \tag{7.39}$$

7.8 Chapter Conclusions

This chapter extends the theory of relaxation dynamics, well established in satellite and asteroid mechanics, to the domain of rolling spherical robots. By analyzing oblate inertial profiles under rolling constraints, we showed that the wobbling and eventual flipping observed in spherical shells are natural consequences of the relaxation effect. The derived equations, supported by experimental trials with varying internal configurations and constraint conditions, demonstrate that translational rolling constraints play a dominant role in driving this instability by acting as an additional dissipation mechanism.

Looking forward, this connection between classical rotational dynamics and constrained robotic motion opens the door to new control approaches. These techniques could be adapted to actively regulate momentum alignment in rolling robots. More broadly, the relaxation framework offers a compact tool for analyzing and predicting stability in mobile rolling platforms where shape, inertia, and ground interaction are tightly coupled.

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